



3042 SIC Journal - sic

BSCit (Ramniranjan Jhunhunwala College)



Scan to open on Studocu



**RAMNIRANJAN JHUNJHUNWALA COLLEGE GHATKOPAR (W),
MUMBAI - 400 086**

DEPARTMENT OF INFORMATION TECHNOLOGY

2022 -2023

**T.Y. B. SC.(I.T.) SEM VI
RJSUITP602
SECURITY IN COMPUTING PRACTICAL**

Name. Utkarsh Pandey.

Roll No. 3042.

Utkarsh Pandey

INDEX

Sr. No.	List of Practical	Date
1.	Configure Cisco Routers for Syslog, NTP, and SSH Operations	08-12-22
2.	Packet Tracer - Configure AAA Authentication on Cisco Routers	12-12-22
3.	Configuring Extended ALs Scenario-1	15-12-22
4.	Configuring Extended ALs Scenario-2	
5.	Configure IP ACLs to Mitigate Attacks	15-12-22
6.	Configuring IPv6 ACLs	04-01-23
7.	Configure Zone based policy firewall	11-01-23
8	Configure IOS (IPS)	25-01-23
9	Layer 2 security	01-02-23
10	Configure and verify site to site IPsec VPN using CLI	08-02-23

SECURITY IN COMPUTING PRACTICALS

Name : Utkarsh PandeyRoll No : 3042

Pract.No	Title	Date	Sign
1	Configure Cisco Routers for Syslog, NTP, and SSH Operations a. Configure OSPF MD5 Authentication b. Configure NTP c. Configure Routers to Log Messages to the Syslog Server d. Configure R3 to Support SSH Connections	8/12/22	<i>Hareesh</i>
2	Configure AAA Authentication on Cisco Routers a. Configure Local AAA Authentication for Console Access on R1 b. Configure Local AAA Authentication for vty Lines on R1 c. Configure Server-Based AAA Authentication Using TACACS+ on R2 d. Configure Server-Based AAA Authentication Using RADIUS on R3	12/12/22	<i>Hareesh</i>
3	Configuring Extended ACLs - Scenario 1 a. Configure, Apply and Verify an Extended Numbered ACL b. Configure, Apply and Verify an Extended Named ACL	15/12/22	<i>Hareesh</i>
4	Configuring Extended ACLs - Scenario 2 a. Configure, Apply and Verify an Extended Numbered ACL	15/12/22	<i>Hareesh</i>
5	Configure IP ACLs to Mitigate Attacks a. Verify Basic Network Connectivity b. Secure Access to Routers c. Create a Numbered IP ACL 120 on R1 d. Modify an Existing ACL on R1 e. Create a Numbered IP ACL 110 on R3 f. Create a Numbered IP ACL 100 on R3	04/01/23	<i>Hareesh</i>
6	Configuring IPv6 ACLs a. Configure, Apply, and Verify an IPv6 ACL b. Configure, Apply, and Verify a Second IPv6 ACL	11/01/23	<i>Hareesh</i>
7	Configuring a Zone-Based Policy Firewall (ZPF) a. Verify Basic Network Connectivity b. Create the Firewall Zones on R3 c. Identify Traffic Using a Class-Map d. Specify Firewall Policies e. Apply Firewall Policies f. Test Firewall Functionality from IN-ZONE to OUT-ZONE g. Test Firewall Functionality from OUT-ZONE to	18/01/23	<i>Hareesh</i> 19/1/23

SECURITY IN COMPUTING PRACTICALS

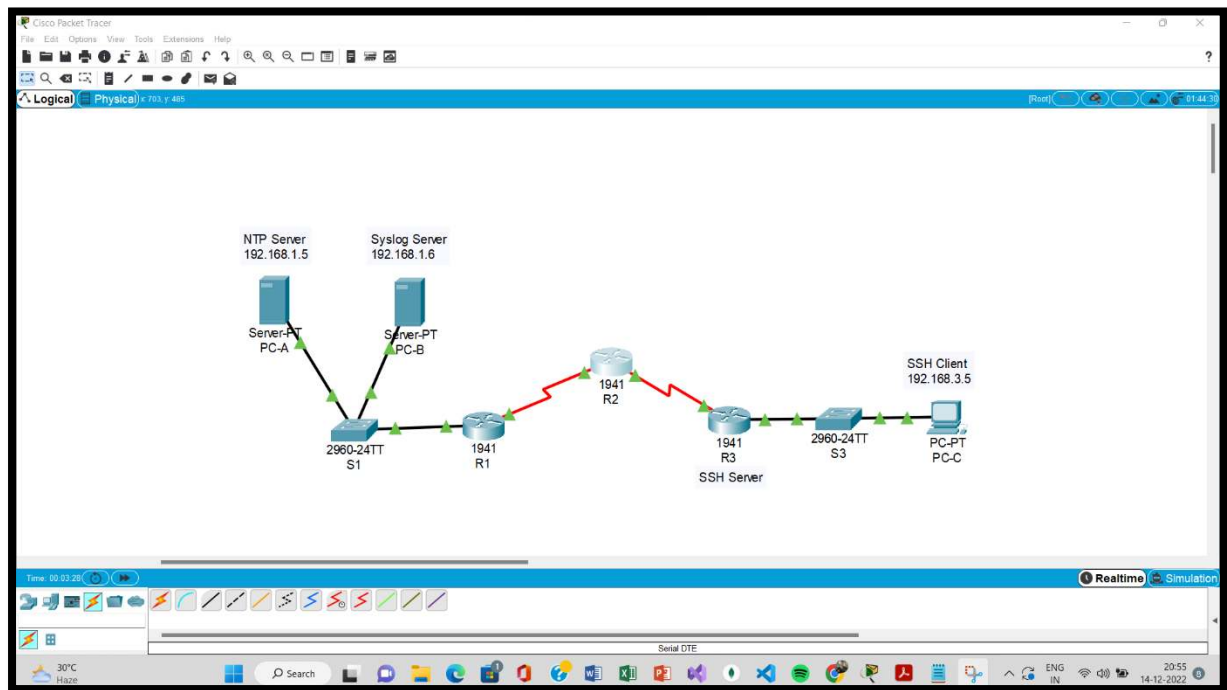
Name : Utkarsh PandeyRoll No : 3042

	IN-ZONE		
8	Configure IOS Intrusion Prevention System (IPS) Using the CLI a. Enable IOS IPS b. Configure logging. c. Modify the Signature d. Verify IPS.	25/01/23	<i>Utkarsh</i> 25/1/23
9	Layer 2 Security a. Configure Root Bridge b. Protect Against STP Attacks c. Configure Port Security and Disable Unused Ports	01/02/23	<i>Utkarsh</i> 11/2/23
10	Configure and Verify a Site-to-Site IPsec VPN Using CLI a. Configure R1 to support a site-to-site IPsec VPN with R3.	08/02/23	<i>Utkarsh</i> 08/2/23

Practical 1

Q1} Packet Tracer - Configure Cisco Routers for Syslog, NTP, and SSH Operations.

Topology:



Addressing Table:

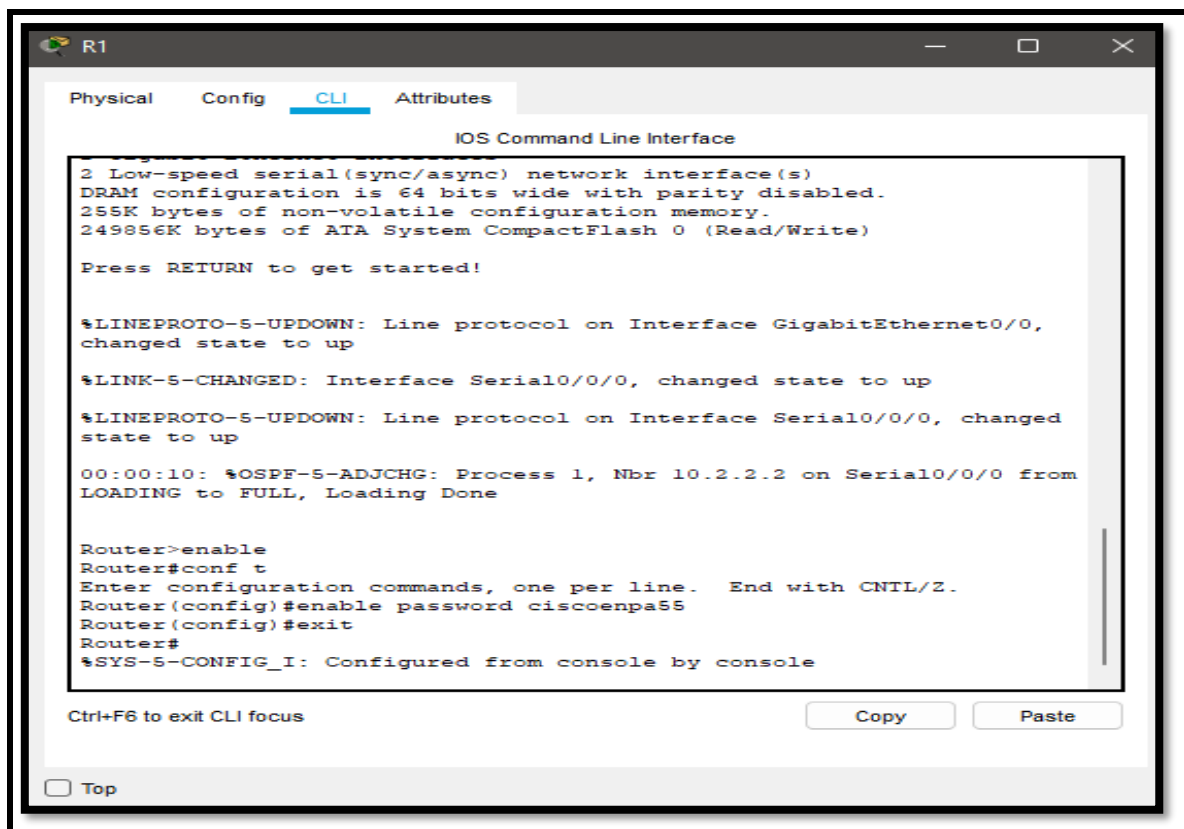
Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.5	255.255.255.0	192.168.1.1	S1 F0/6
PC-B	NIC	192.168.1.6	255.255.255.0	192.168.1.1	S2 F0/18
PC-C	NIC	192.168.3.5	255.255.255.0	192.168.3.1	S3 F0/18

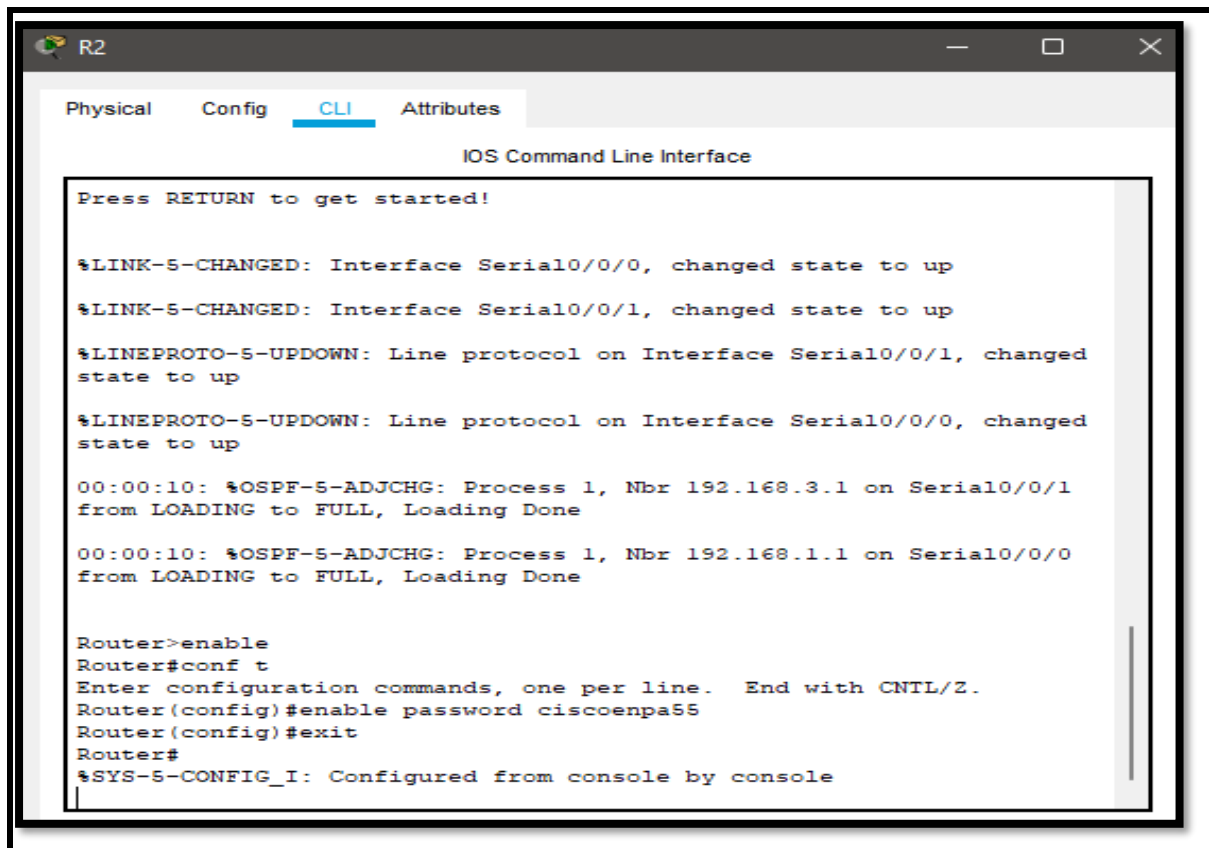
The routers have been pre-configured with the following passwords:

- Enable password: ciscoenpa55
- Password for vty lines: ciscovtypa55

To set Router enable password

- enable
- conf t
- enable password ciscoenpa55
- exit





The screenshot shows the CLI window for router R2. The 'CLI' tab is selected. The interface displays the following text:

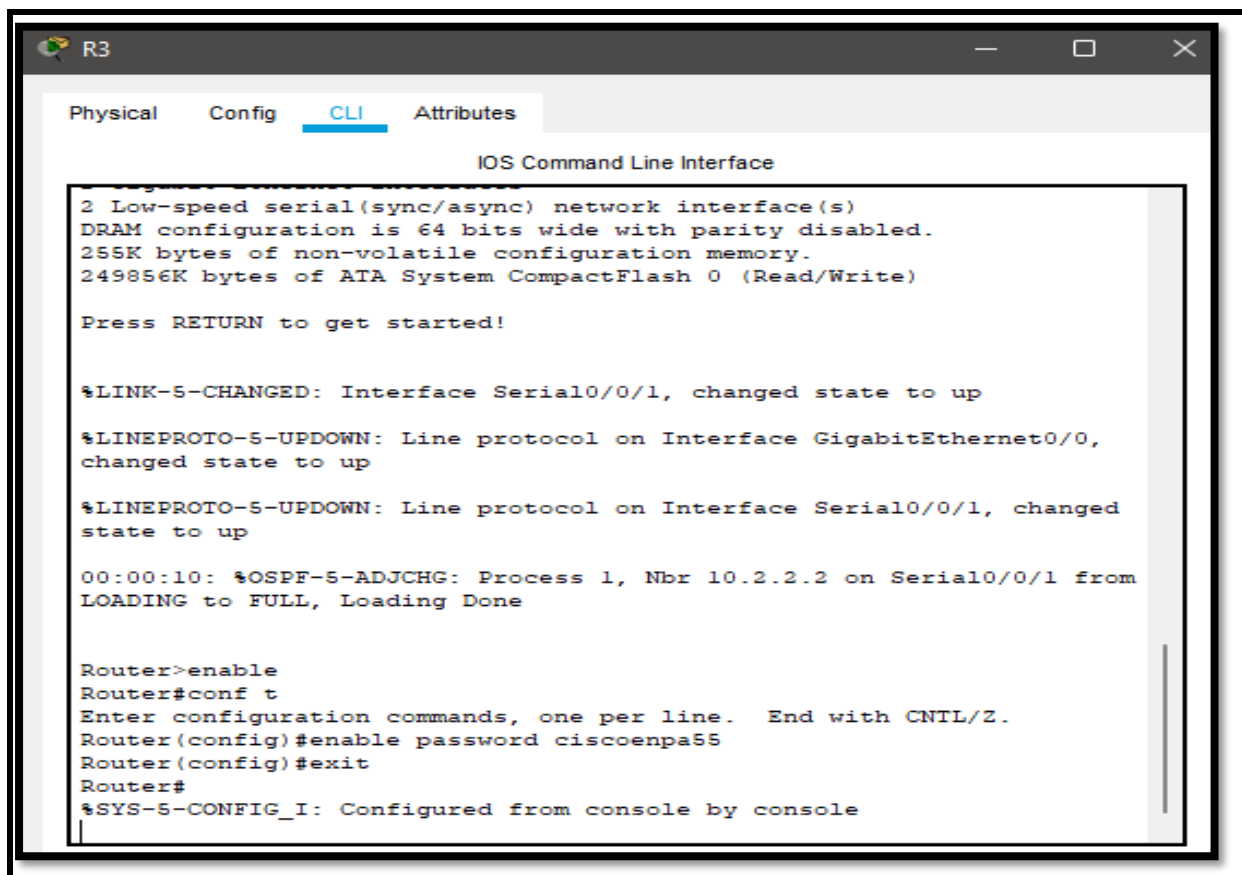
```
Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed
state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed
state to up

00:00:10: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on Serial0/0/1
from LOADING to FULL, Loading Done

00:00:10: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial0/0/0
from LOADING to FULL, Loading Done

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#enable password ciscoenpa55
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```



The screenshot shows the CLI window for router R3. The 'CLI' tab is selected. The interface displays the following text:

```
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed
state to up

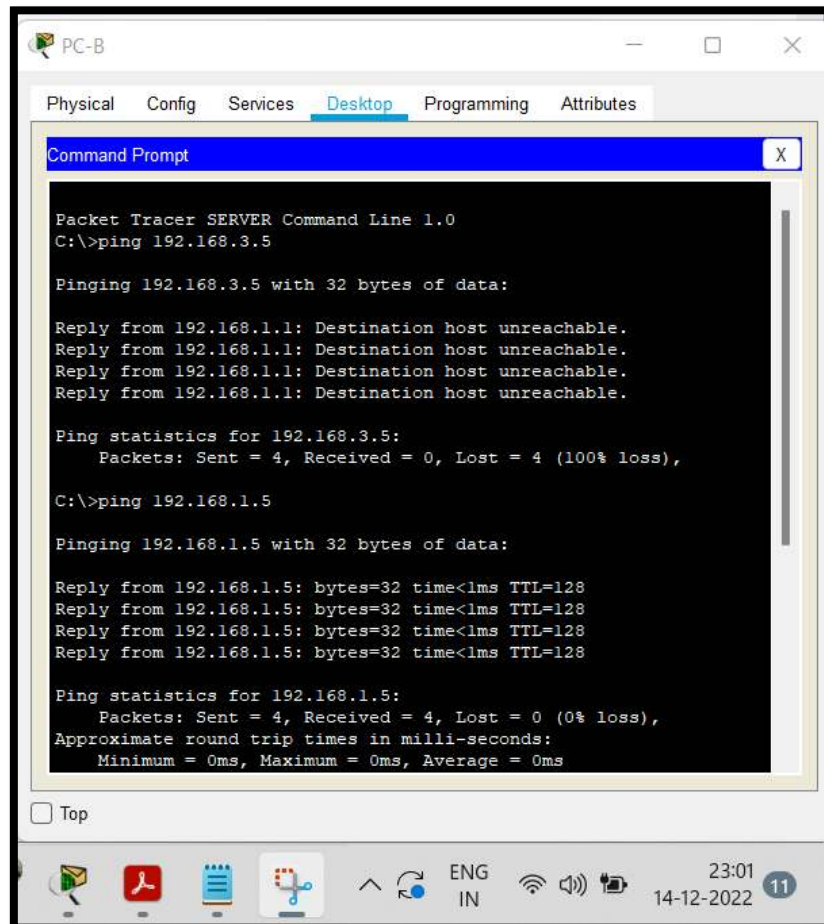
00:00:10: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/1 from
LOADING to FULL, Loading Done

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#enable password ciscoenpa55
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```


Part 1: Configure OSPF MD5 Authentication

Step 1: Test connectivity. All devices should be able to ping all other IP addresses.

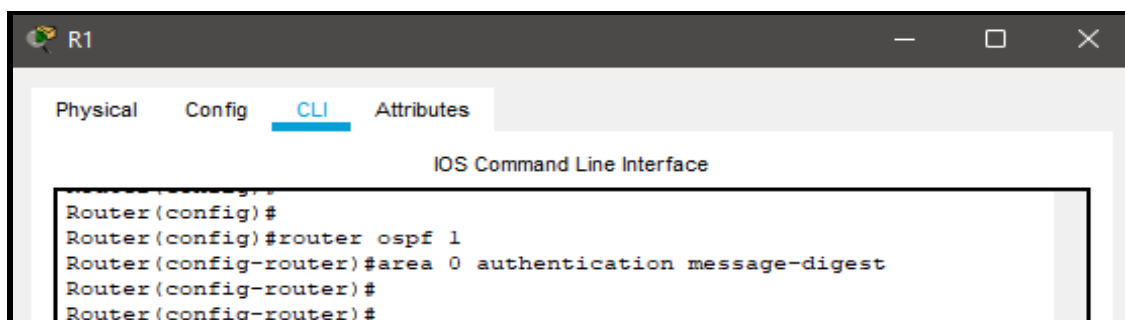
PC-A: C:/>ping 192.168.3.5



Step 2: Configure OSPF MD5 authentication for all the routers in area 0.

R1(config)# router ospf 1

R1(config-router)# area 0 authentication message-digest



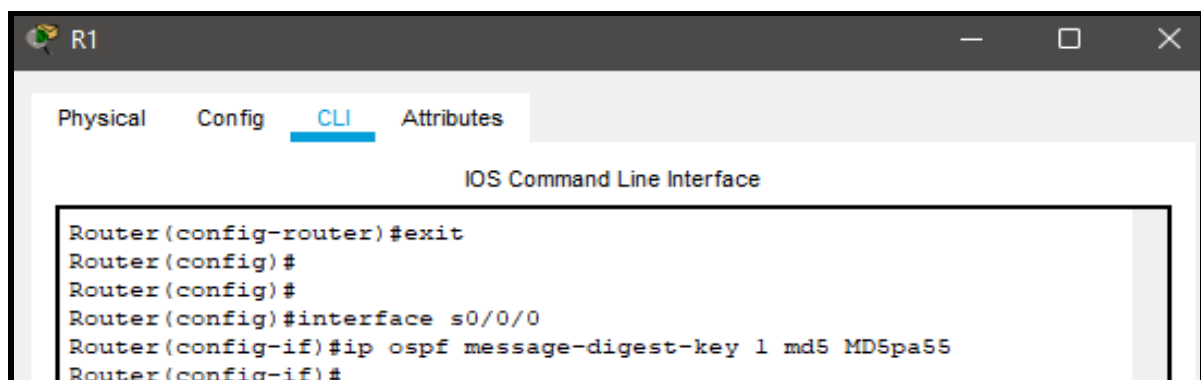
R2(config)# router ospf 1**R2(config-router)# area 0 authentication message-digest**A screenshot of the R2 router configuration window. The window has tabs for Physical, Config, CLI (selected), and Attributes. The CLI tab shows the IOS Command Line Interface with the following commands entered:

```
Router(config)#  
Router(config)#  
Router(config)#router ospf 1  
Router(config-router)#area 0 authentication message-digest  
Router(config-router)#
```

R3(config)# router ospf 1**R3(config-router)# area 0 authentication message-digest**A screenshot of the R3 router configuration window. The window has tabs for Physical, Config, CLI (selected), and Attributes. The CLI tab shows the IOS Command Line Interface with the following commands entered:

```
Router(config)#  
Router(config)#  
Router(config)#router ospf 1  
Router(config-router)#area 0 authentication message-digest  
Router(config-router)#
```

Step 3: Configure the MD5 key for all the routers in area 0. Configure an MD5 key on the serial interfaces on R1, R2 and R3. Use the password MD5pa55 for key 1

R1(config)# interface s0/0/0**R1(config-if)# ip ospf message-digest-key 1 md5 MD5pa55**A screenshot of the R1 router configuration window. The window has tabs for Physical, Config, CLI (selected), and Attributes. The CLI tab shows the IOS Command Line Interface with the following commands entered:

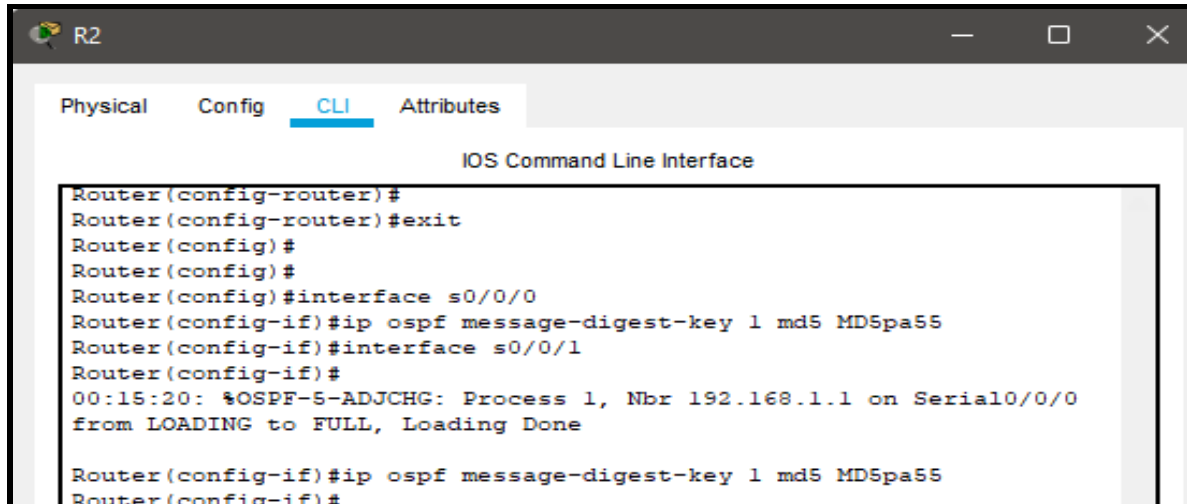
```
Router(config-router)#exit  
Router(config)#  
Router(config)#  
Router(config)#interface s0/0/0  
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55  
Router(config-if)#
```

R2(config)# interface s0/0/0

R2(config-if)# ip ospf message-digest-key 1 md5 MD5pa55

R2(config-if)# interface s0/0/1

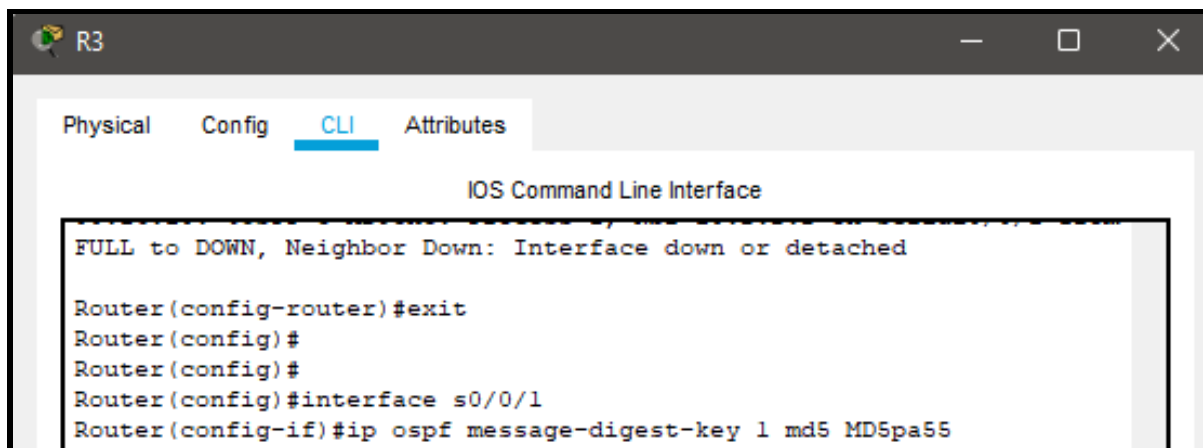
R2(config-if)# ip ospf message-digest-key 1 md5 MD5pa55



```
R2
Physical Config CLI Attributes
IOS Command Line Interface
Router(config-router)#
Router(config-router)#exit
Router(config)#
Router(config)#
Router(config)#interface s0/0/0
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
Router(config-if)#interface s0/0/1
Router(config-if)#
00:15:20: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial0/0/0
from LOADING to FULL, Loading Done
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
Router(config-if)#
```

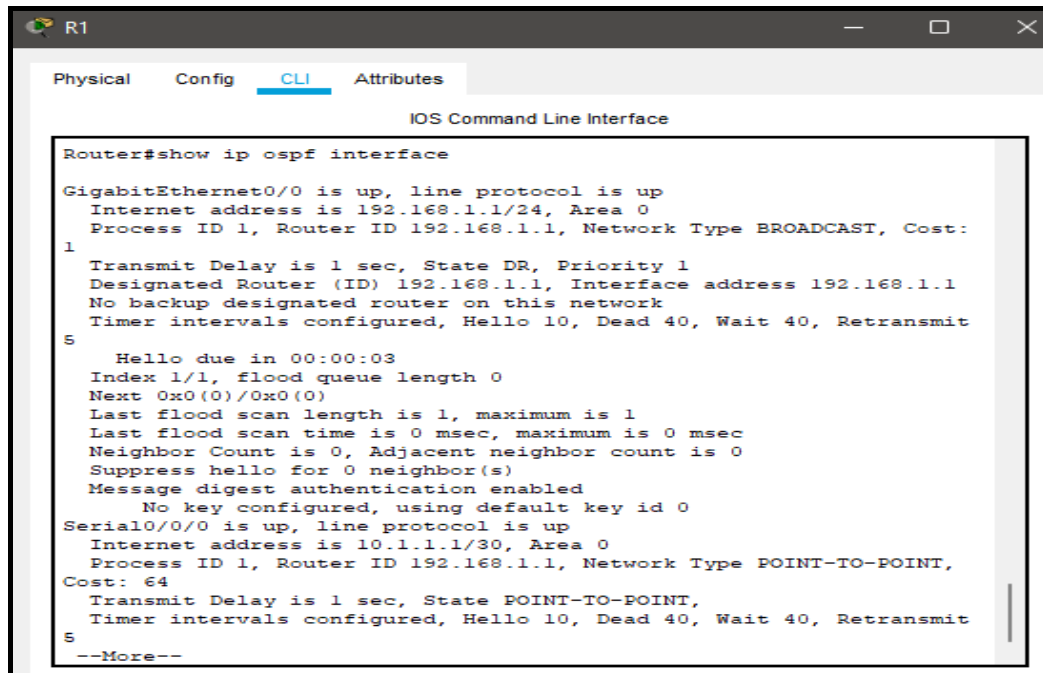
R3(config)# interface s0/0/1

R3(config-if)# ip ospf message-digest-key 1 md5 MD5pa55



```
R3
Physical Config CLI Attributes
IOS Command Line Interface
FULL to DOWN, Neighbor Down: Interface down or detached
Router(config-router)#exit
Router(config)#
Router(config)#
Router(config)#interface s0/0/1
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
```

Step 4: Verify configurations. Verify the MD5 authentication configurations using the commands show ip ospf interface Verify end-to-end connectivity.



```

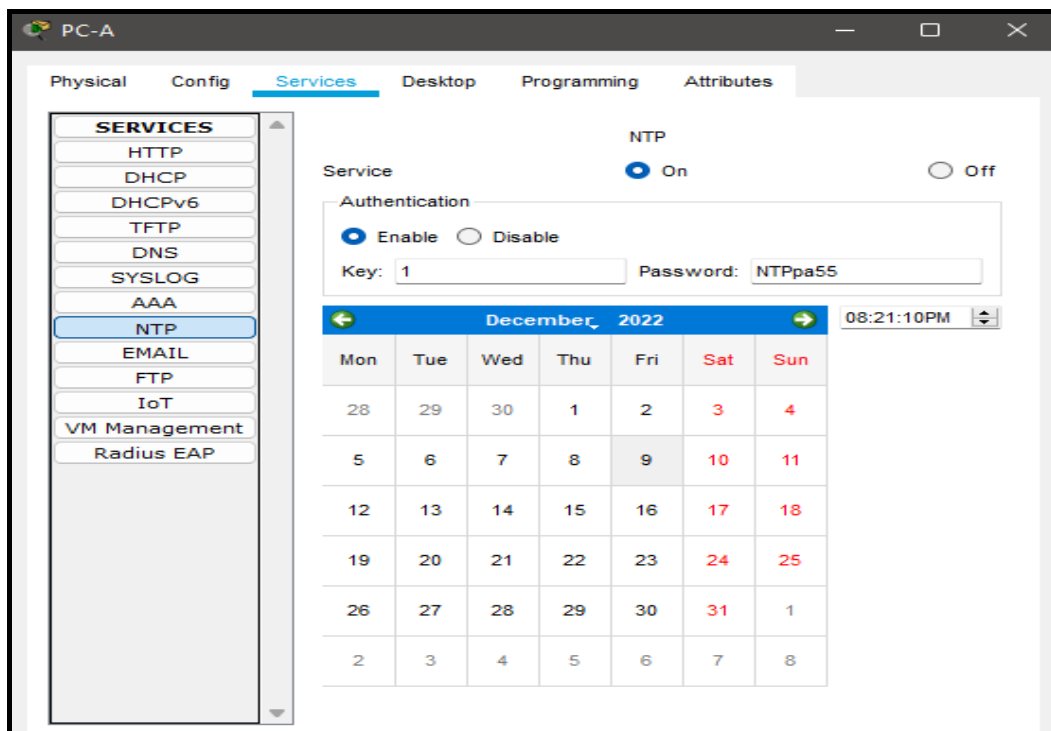
R1
Physical Config CLI Attributes
IOS Command Line Interface

Router#show ip ospf interface

GigabitEthernet0/0 is up, line protocol is up
Internet address is 192.168.1.1/24, Area 0
Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost:
1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 192.168.1.1, Interface address 192.168.1.1
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit
S
Hello due in 00:00:03
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)
Message digest authentication enabled
No key configured, using default key id 0
Serial0/0/0 is up, line protocol is up
Internet address is 10.1.1.1/30, Area 0
Process ID 1, Router ID 192.168.1.1, Network Type POINT-TO-POINT,
Cost: 64
Transmit Delay is 1 sec, State POINT-TO-POINT,
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit
S
--More--
  
```

Part 2: Configure NTP

Step 1: Enable NTP authentication on PC-A



PC-A

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP**
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

NTP

Service ☒ On ☐ Off

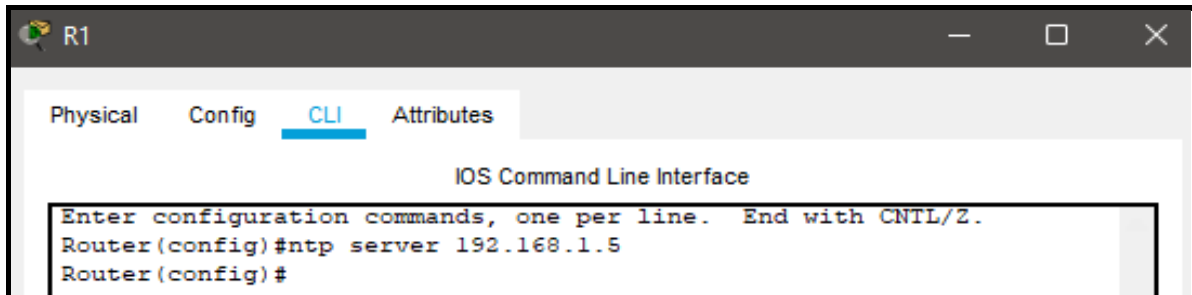
Authentication

☒ Enable ☐ Disable

Key: 1 Password: NTPpa55

December 2022 08:21:10PM

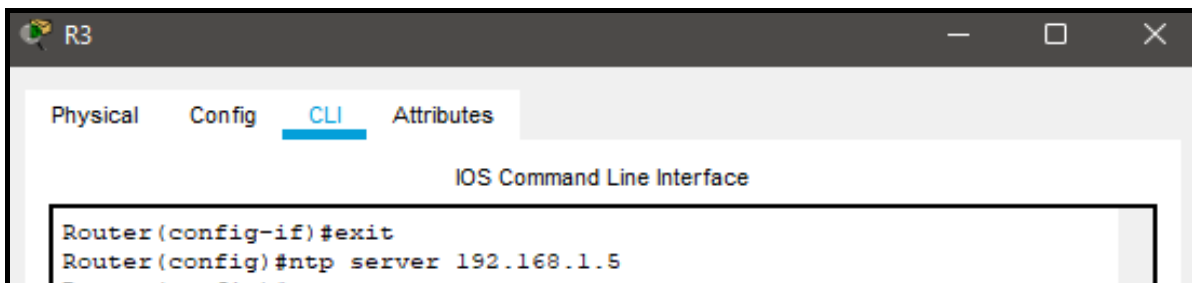
Mon	Tue	Wed	Thu	Fri	Sat	Sun
28	29	30	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1
2	3	4	5	6	7	8

Step 2: Configure R1, R2, and R3 as NTP clients**R1(config)# ntp server 192.168.1.5**A screenshot of the R1 router's CLI interface. The window title is 'R1'. At the top, there are tabs for 'Physical', 'Config', 'CLI' (which is selected and highlighted in blue), and 'Attributes'. Below the tabs, the text 'IOS Command Line Interface' is displayed. The main area shows the command prompt 'Router(config)#' and the command 'ntp server 192.168.1.5' being entered. The prompt returns to 'Router(config)#' after the command is executed.

```
R1
Physical Config CLI Attributes
IOS Command Line Interface
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ntp server 192.168.1.5
Router(config)#
```

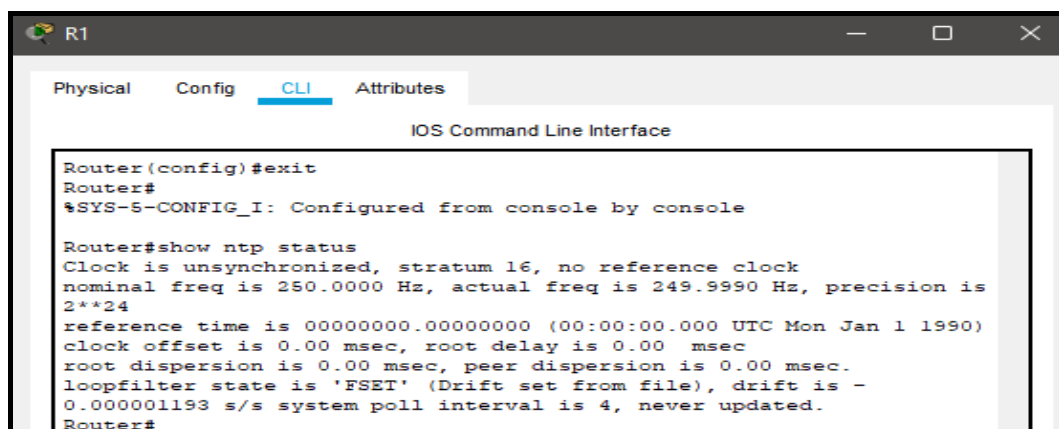
R2(config)# ntp server 192.168.1.5A screenshot of the R2 router's CLI interface. The window title is 'R2'. At the top, there are tabs for 'Physical', 'Config', 'CLI' (which is selected and highlighted in blue), and 'Attributes'. Below the tabs, the text 'IOS Command Line Interface' is displayed. The main area shows the command prompt 'Router(config-if)#', followed by the command 'exit' to return to 'Router(config)#', and then the command 'ntp server 192.168.1.5'. The prompt returns to 'Router(config)#' after the command is executed.

```
R2
Physical Config CLI Attributes
IOS Command Line Interface
Router(config-if)#exit
Router(config)#ntp server 192.168.1.5
Router(config)#
```

R3(config)# ntp server 192.168.1.5A screenshot of the R3 router's CLI interface. The window title is 'R3'. At the top, there are tabs for 'Physical', 'Config', 'CLI' (which is selected and highlighted in blue), and 'Attributes'. Below the tabs, the text 'IOS Command Line Interface' is displayed. The main area shows the command prompt 'Router(config-if)#', followed by the command 'exit' to return to 'Router(config)#', and then the command 'ntp server 192.168.1.5'. The prompt returns to 'Router(config)#' after the command is executed.

```
R3
Physical Config CLI Attributes
IOS Command Line Interface
Router(config-if)#exit
Router(config)#ntp server 192.168.1.5
Router(config)#
```

Verify client configuration using the command show ntp status.

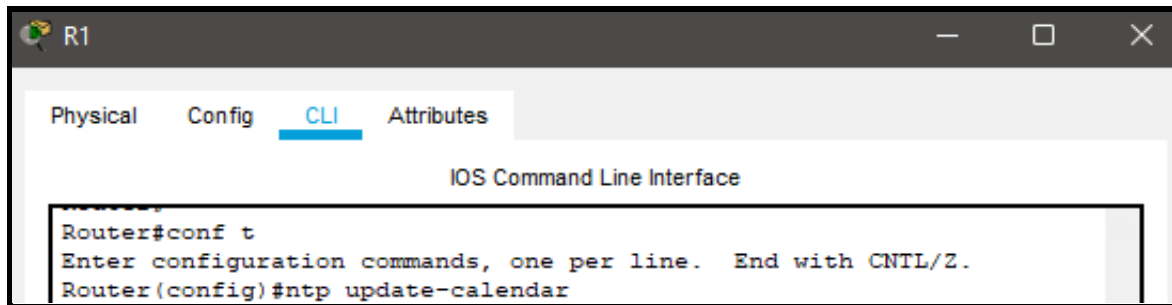
A screenshot of the R1 router's CLI interface showing the output of the 'show ntp status' command. The window title is 'R1'. At the top, there are tabs for 'Physical', 'Config', 'CLI' (which is selected and highlighted in blue), and 'Attributes'. Below the tabs, the text 'IOS Command Line Interface' is displayed. The main area shows the command prompt 'Router#', followed by the command 'show ntp status'. The output displays various NTP parameters including clock synchronization status, frequency, reference time, and drift rate.

```
R1
Physical Config CLI Attributes
IOS Command Line Interface
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ntp status
Clock is unsynchronized, stratum 16, no reference clock
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is
2**24
reference time is 00000000.00000000 (00:00:00.000 UTC Mon Jan 1 1990)
clock offset is 0.00 msec, root delay is 0.00 msec
root dispersion is 0.00 msec, peer dispersion is 0.00 msec.
loopfilter state is 'FSET' (Drift set from file), drift is -
0.000001193 s/s system poll interval is 4, never updated.
Router#
```

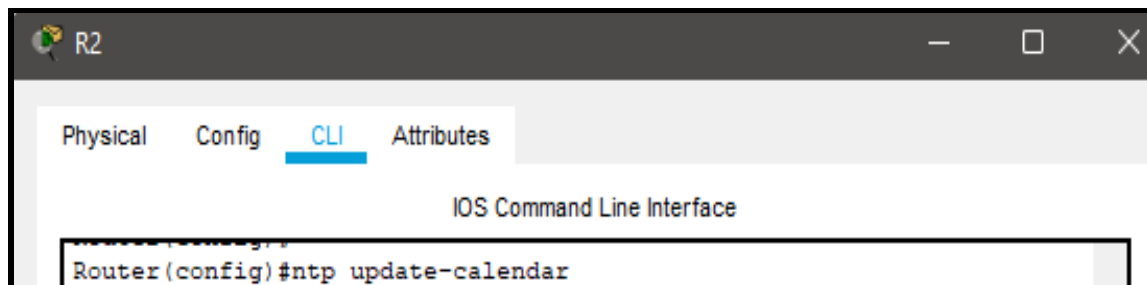
Step 3: Configure routers to update hardware clock. Configure R1, R2, and R3 to periodically update the hardware clock with the time learned from NTP.

R1(config)# ntp update-calendar



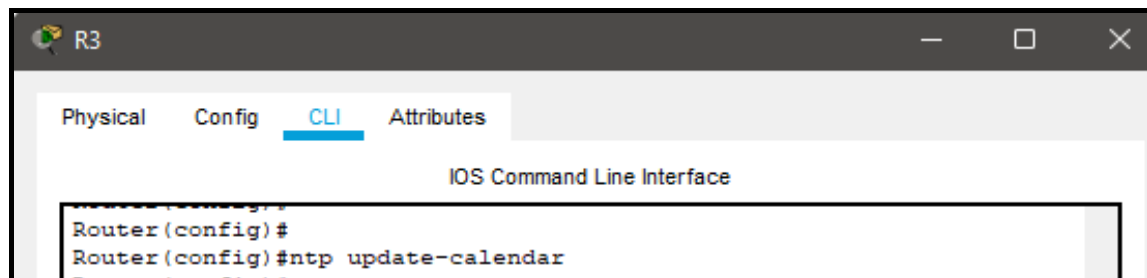
```
R1
Physical Config CLI Attributes
IOS Command Line Interface
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ntp update-calendar
```

R2(config)# ntp update-calendar



```
R2
Physical Config CLI Attributes
IOS Command Line Interface
Router(config)#ntp update-calendar
```

R3(config)# ntp update-calendar

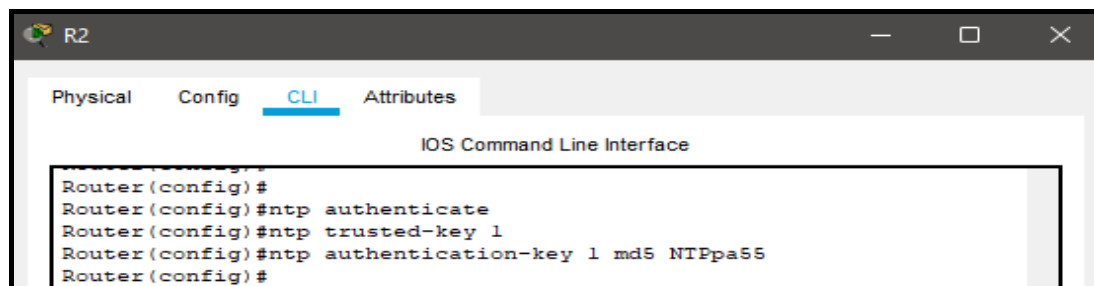
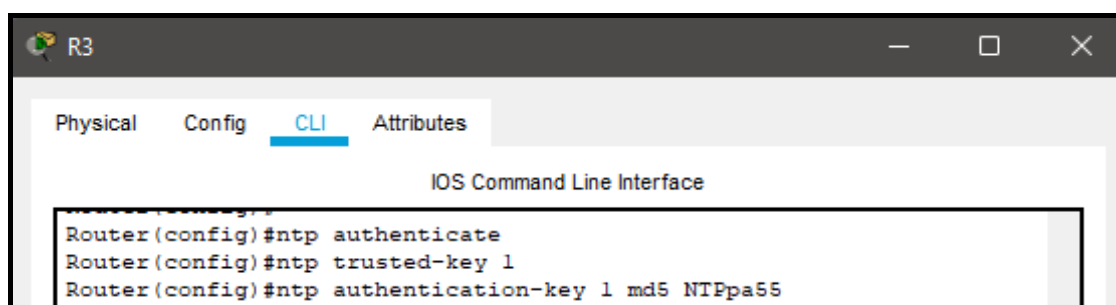


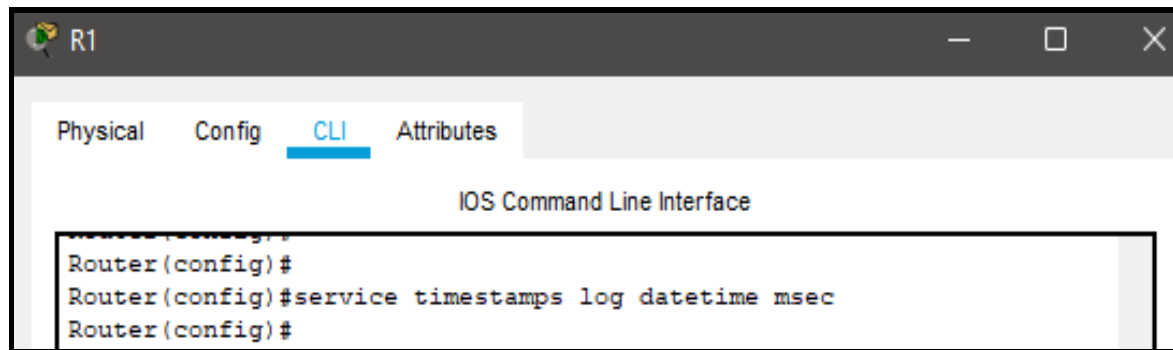
```
R3
Physical Config CLI Attributes
IOS Command Line Interface
Router(config)#
Router(config)#ntp update-calendar
```

Exit global configuration and verify that the hardware clock was updated using the command show clock.

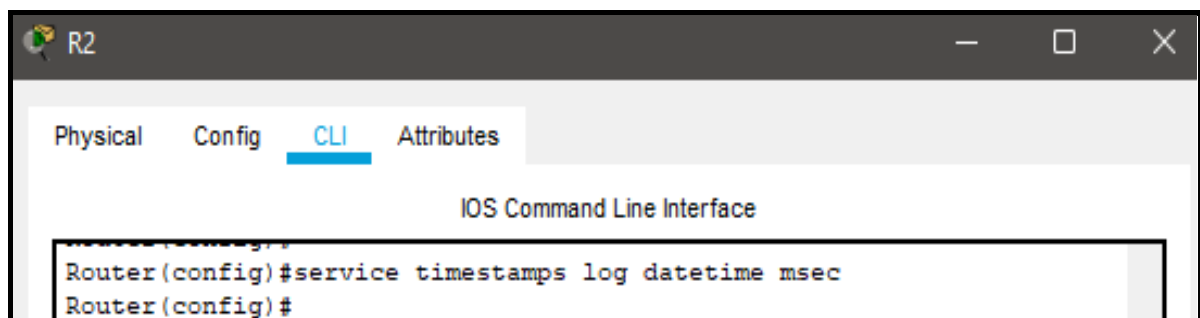


```
R1
Physical Config CLI Attributes
IOS Command Line Interface
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show clock
20:34:15.452 UTC Fri Dec 9 2022
Router#
```

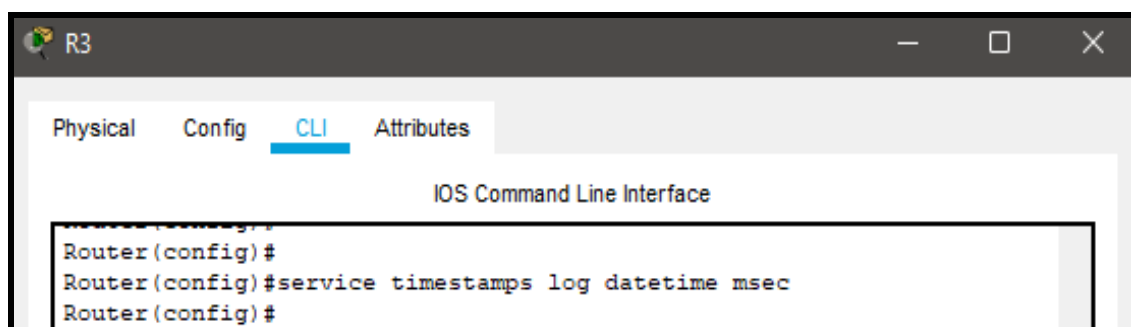
Step 4: Configure NTP authentication on the routers.**R1(config)# ntp authenticate****R1(config)# ntp trusted-key 1****R1(config)# ntp authentication-key 1 md5 NTPpa55**A screenshot of the R1 router's CLI interface. The window title is 'R1'. It has tabs for 'Physical', 'Config', 'CLI' (selected), and 'Attributes'. Below the tabs is the text 'IOS Command Line Interface'. The command history shows: 'Router(config)#', 'Router(config)#ntp authenticate', 'Router(config)#ntp trusted-key 1', 'Router(config)#ntp authentication-key 1 md5 NTPpa55', and 'Router(config)#'.**R2(config)# ntp authenticate****R2(config)# ntp trusted-key 1****R2(config)# ntp authentication-key 1 md5 NTPpa55**A screenshot of the R2 router's CLI interface. The window title is 'R2'. It has tabs for 'Physical', 'Config', 'CLI' (selected), and 'Attributes'. Below the tabs is the text 'IOS Command Line Interface'. The command history shows: 'Router(config)#', 'Router(config)#ntp authenticate', 'Router(config)#ntp trusted-key 1', 'Router(config)#ntp authentication-key 1 md5 NTPpa55', and 'Router(config)#'.**R3(config)# ntp authenticate****R3(config)# ntp trusted-key 1****R3(config)# ntp authentication-key 1 md5 NTPpa55**A screenshot of the R3 router's CLI interface. The window title is 'R3'. It has tabs for 'Physical', 'Config', 'CLI' (selected), and 'Attributes'. Below the tabs is the text 'IOS Command Line Interface'. The command history shows: 'Router(config)#ntp authenticate', 'Router(config)#ntp trusted-key 1', and 'Router(config)#ntp authentication-key 1 md5 NTPpa55'.Step:5 Configure routers to timestamp log messages.**R1(config)# service timestamps log datetime msec**



R2(config)# service timestamps log datetime msec



R3(config)# service timestamps log datetime msec



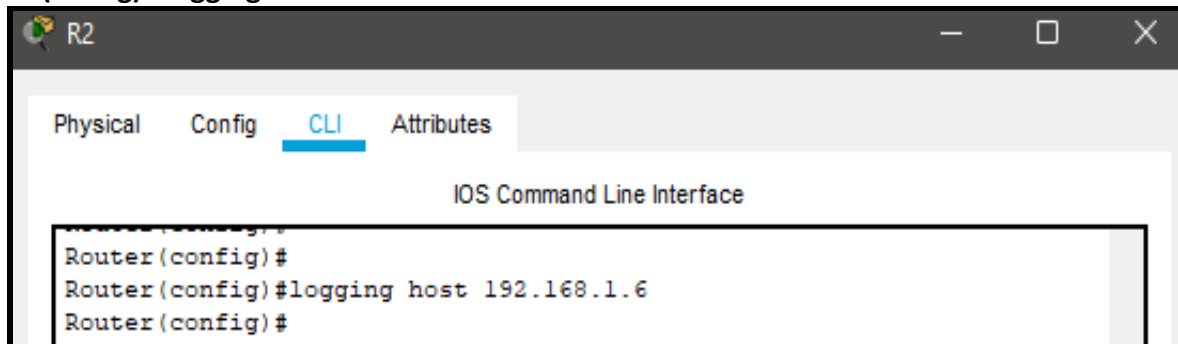
Part 3: Configure Routers to Log Messages to the Syslog Server

Step:1 Configure the routers to identify the remote host (Syslog Server) that will receive logging messages

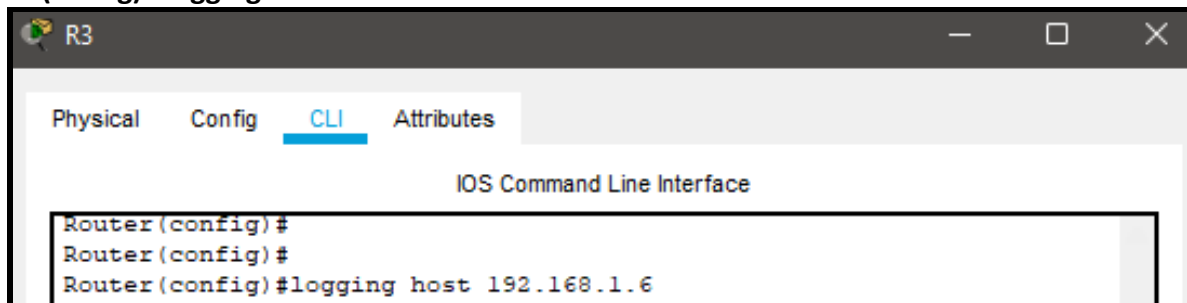
R1(config)# logging host 192.168.1.6



R2(config)# logging host 192.168.1.6

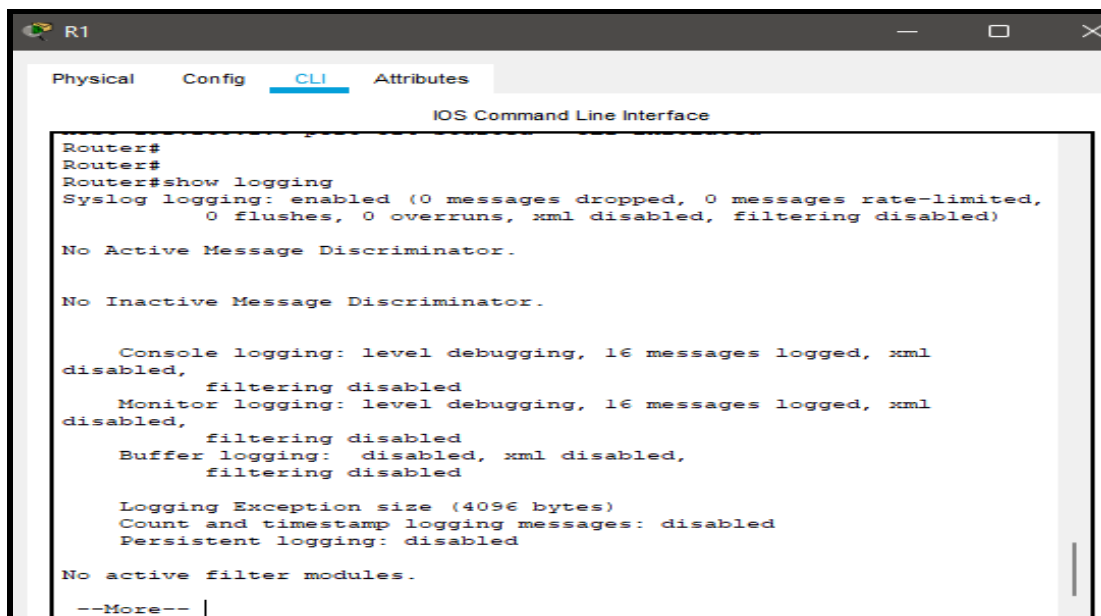


R3(config)# logging host 192.168.1.6



Step:2 Verify logging configuration

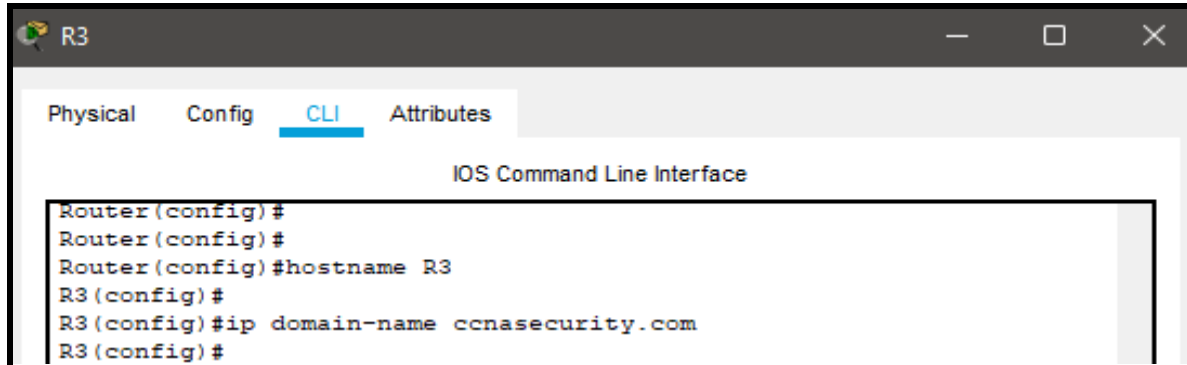
Use the command show logging to verify logging has been enabled.



Part 4: Configure R3 to Support SSH Connections

Step 1: Configure a domain name on R3.

R3(config)# ip domain-name ccnasecurity.com

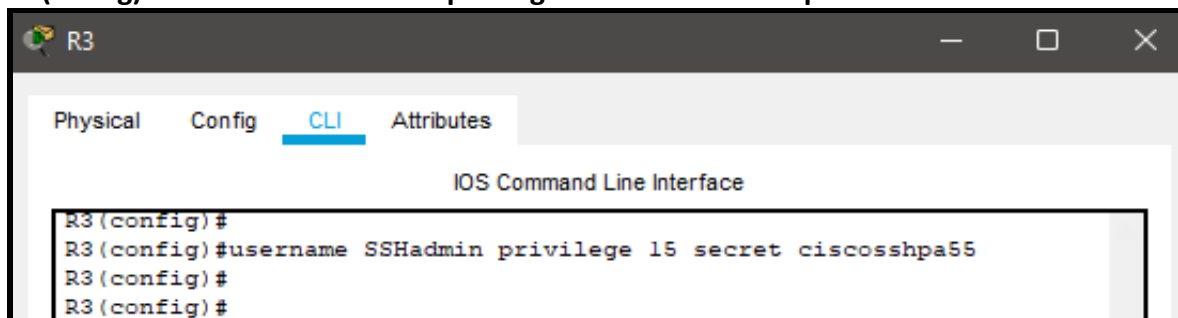
A screenshot of the R3 router configuration window. The window has tabs for Physical, Config, CLI, and Attributes, with CLI selected. The title bar says 'R3'. Below the tabs, it says 'IOS Command Line Interface'. The command history shows: Router(config)#, Router(config)#, Router(config)#hostname R3, R3(config)#, R3(config)#ip domain-name ccnasecurity.com, and R3(config)#.

```
R3
Physical Config CLI Attributes
IOS Command Line Interface
Router(config)#
Router(config)#
Router(config)#hostname R3
R3(config)#
R3(config)#ip domain-name ccnasecurity.com
R3(config)#
```

Step 2: Configure users for login to the SSH server on R3.

Create a user ID of SSHadmin with the highest possible privilege level and a secret password of ciscosshpa55.

R3(config)# username SSHadmin privilege 15 secret ciscosshpa55

A screenshot of the R3 router configuration window. The window has tabs for Physical, Config, CLI, and Attributes, with CLI selected. The title bar says 'R3'. Below the tabs, it says 'IOS Command Line Interface'. The command history shows: R3(config)#, R3(config)#username SSHadmin privilege 15 secret ciscosshpa55, R3(config)#, and R3(config)#.

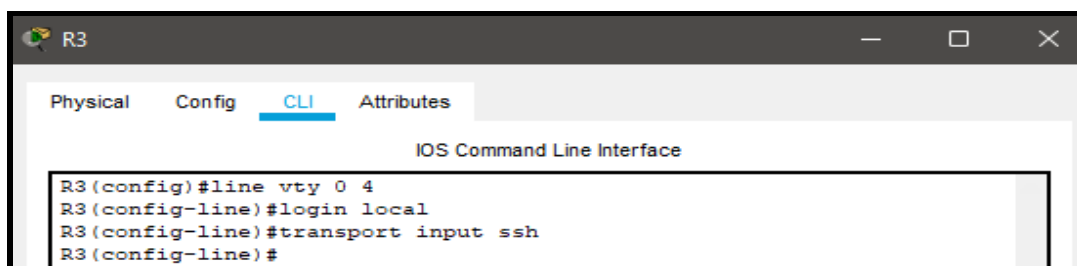
```
R3
Physical Config CLI Attributes
IOS Command Line Interface
R3(config)#
R3(config)#username SSHadmin privilege 15 secret ciscosshpa55
R3(config)#
R3(config)#
```

Step 3: Configure the incoming vty lines on R3 Use the local user accounts for mandatory login and validation. Accept only SSH connections.

R3(config)# line vty 0 4

R3(config-line)# login local

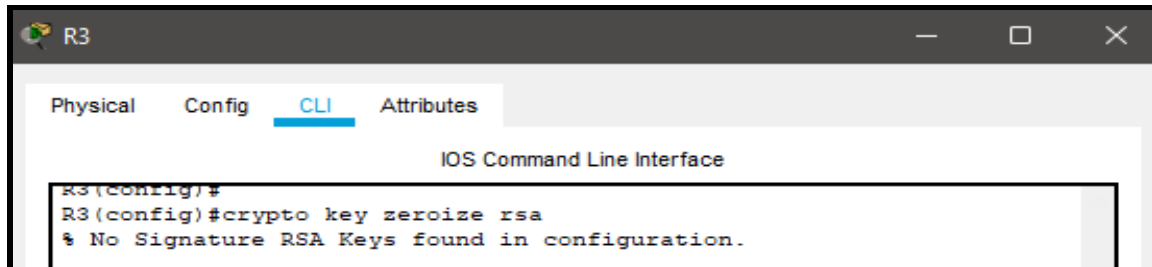
R3(config-line)# transport input ssh

A screenshot of the R3 router configuration window. The window has tabs for Physical, Config, CLI, and Attributes, with CLI selected. The title bar says 'R3'. Below the tabs, it says 'IOS Command Line Interface'. The command history shows: R3(config)#line vty 0 4, R3(config-line)#login local, R3(config-line)#transport input ssh, and R3(config-line)#.

```
R3
Physical Config CLI Attributes
IOS Command Line Interface
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#transport input ssh
R3(config-line)#
```

Step 4: Erase existing key pairs on R3 RSA key pairs should be erased on the router.

R3(config)# crypto key zeroize rsa

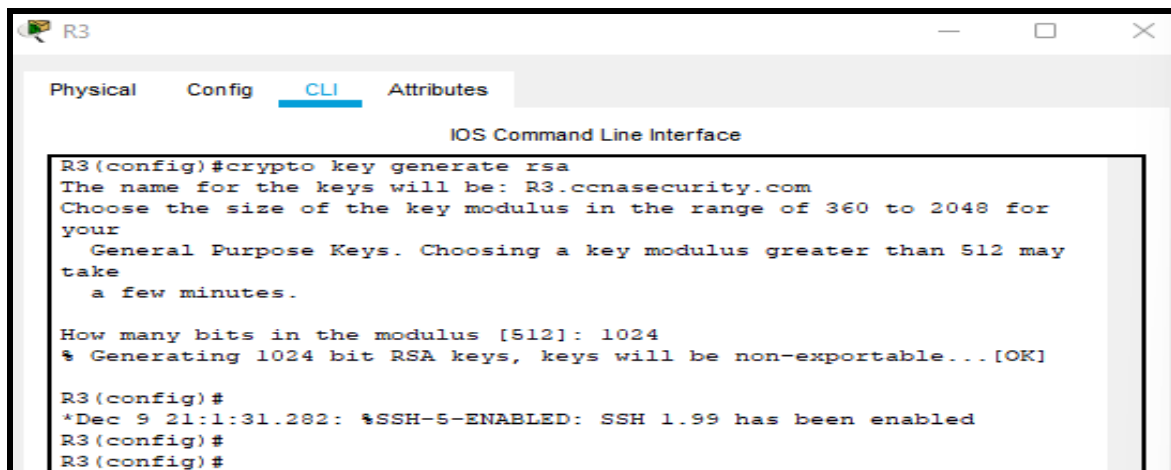


The screenshot shows a terminal window titled 'R3' with tabs for Physical, Config, CLI (selected), and Attributes. The CLI tab displays the 'IOS Command Line Interface' with the following text:

```
R3(config)#
R3(config)#crypto key zeroize rsa
% No Signature RSA Keys found in configuration.
```

Step 5: Generate the RSA encryption key pair for R3.

R3(config)# crypto key generate rsa



The screenshot shows a terminal window titled 'R3' with tabs for Physical, Config, CLI (selected), and Attributes. The CLI tab displays the 'IOS Command Line Interface' with the following text:

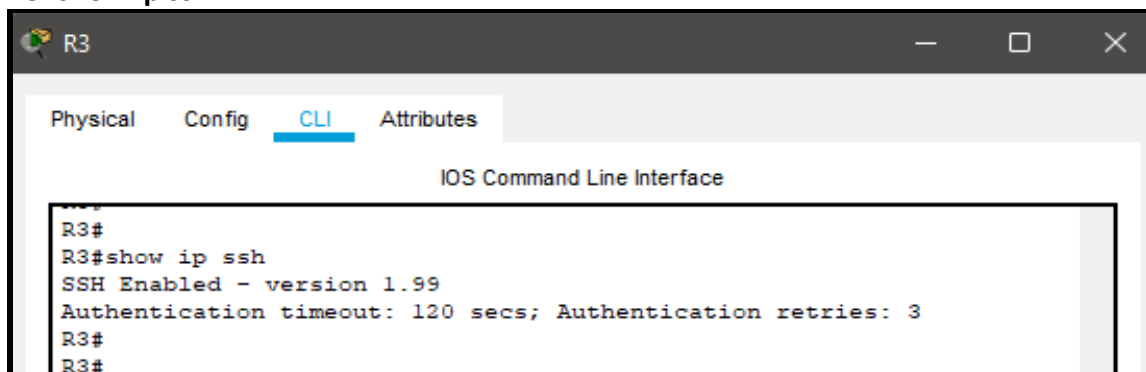
```
R3(config)#crypto key generate rsa
The name for the keys will be: R3.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 2048 for
your
  General Purpose Keys. Choosing a key modulus greater than 512 may
take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R3(config)#
*Dec 9 21:1:31.282: %SSH-5-ENABLED: SSH 1.99 has been enabled
R3(config)#
R3(config)#
```

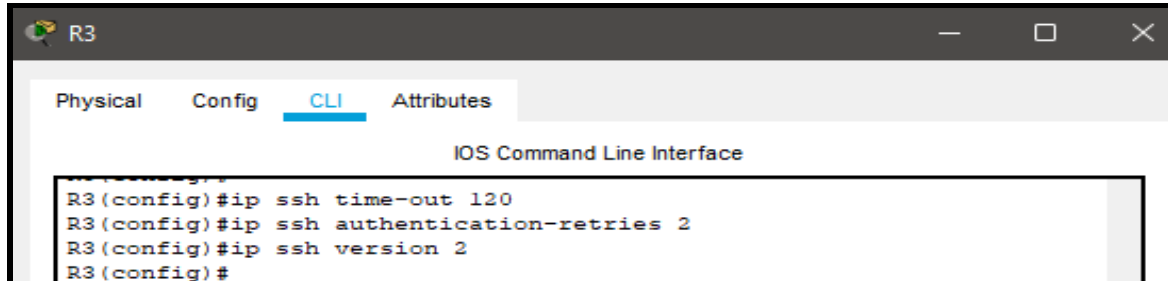
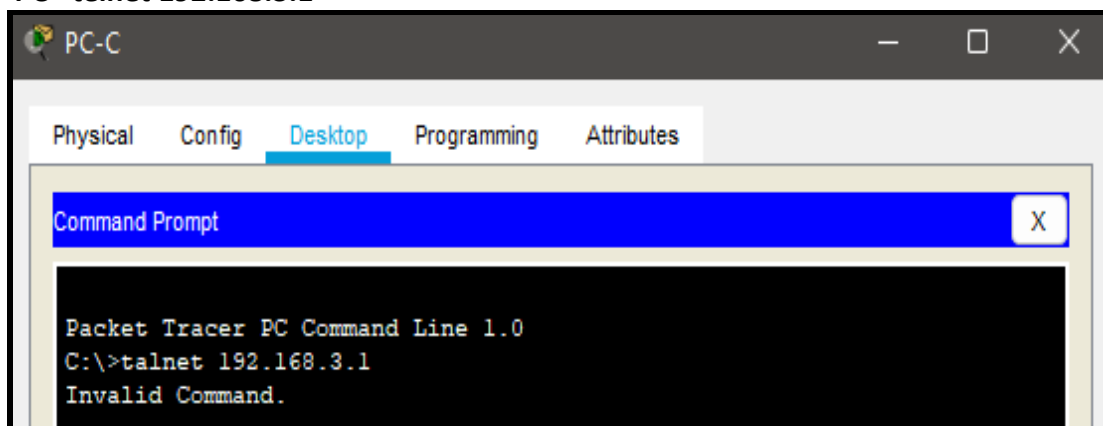
Step 6: Verify the SSH configuration.

R3#show ip ssh

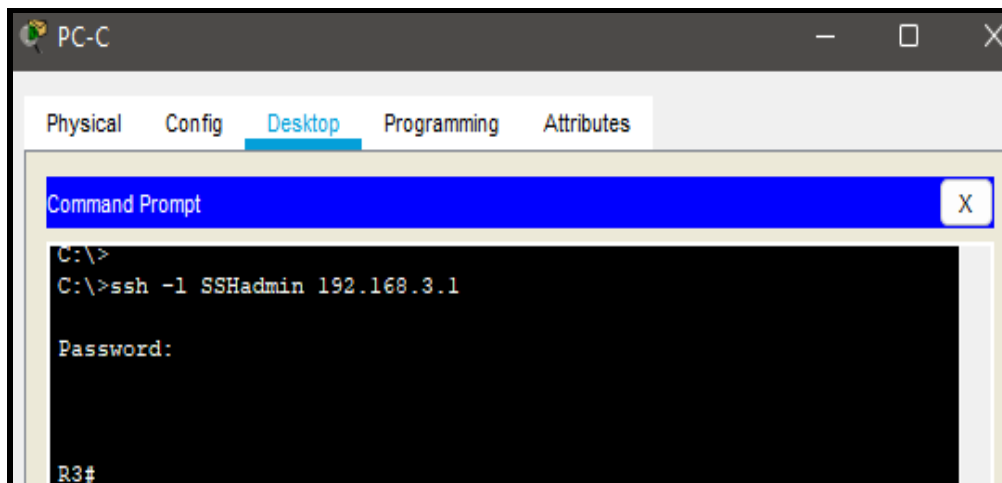


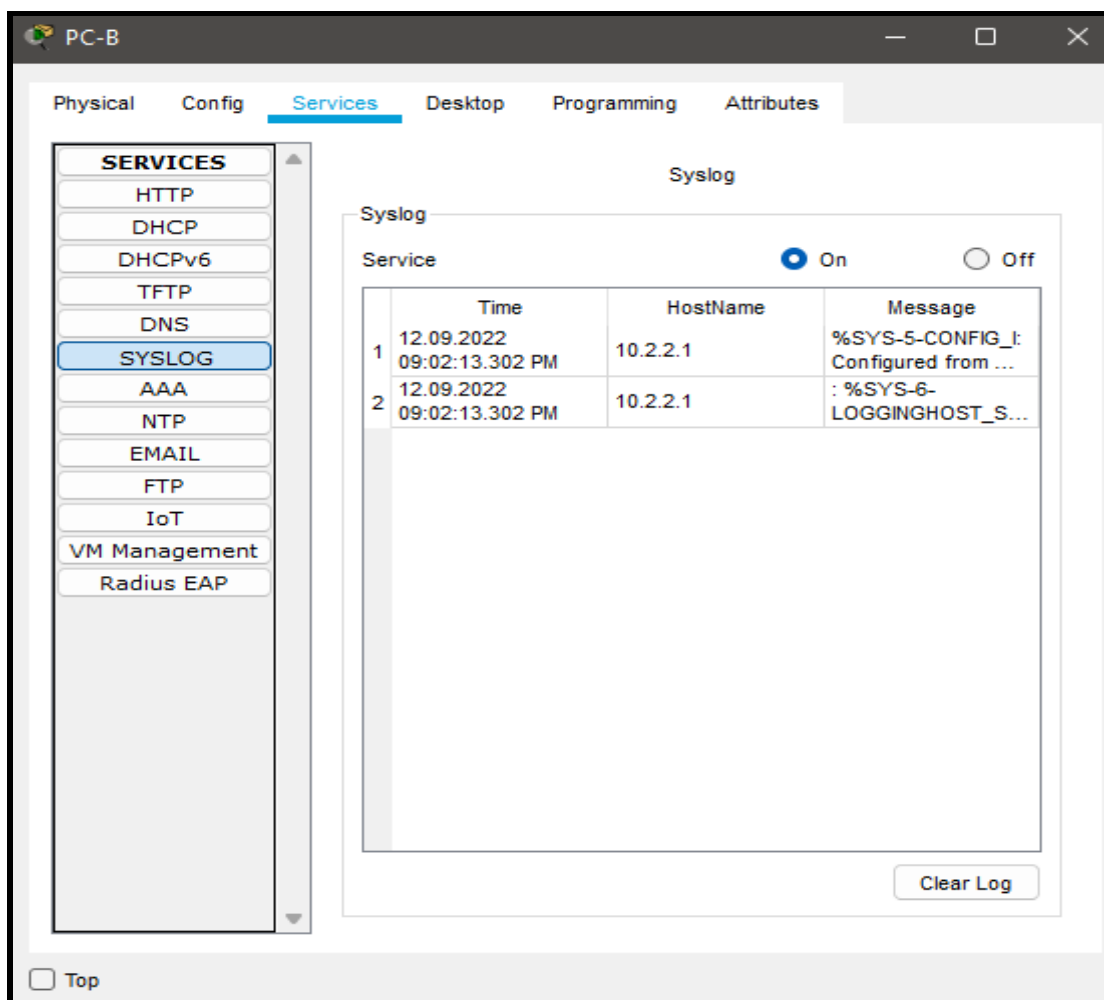
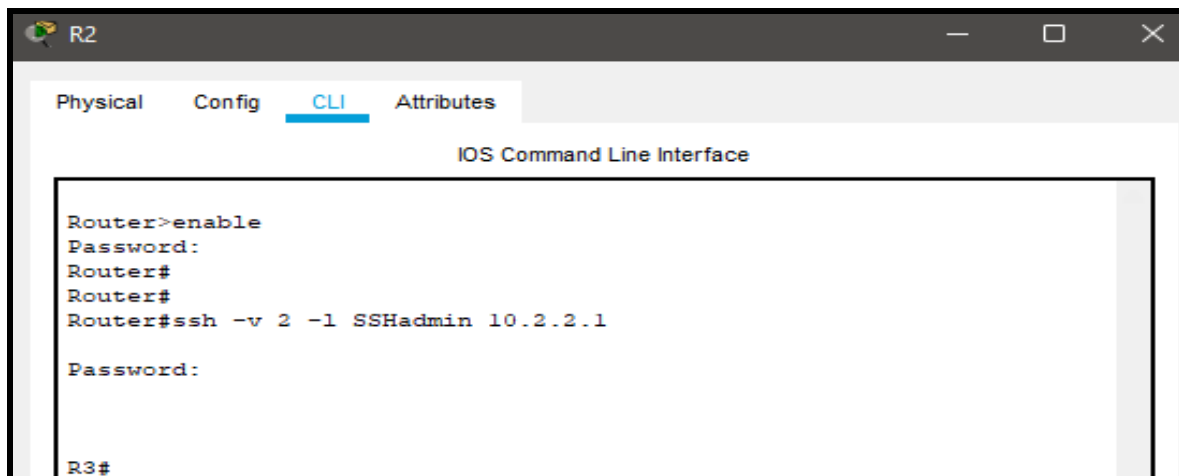
The screenshot shows a terminal window titled 'R3' with tabs for Physical, Config, CLI (selected), and Attributes. The CLI tab displays the 'IOS Command Line Interface' with the following text:

```
R3#
R3#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
R3#
R3#
```

Step 7: Configure SSH timeouts and authentication parameters.**R3(config)# ip ssh time-out 120****R3(config)# ip ssh authentication-retries 2****R3(config)# ip ssh version 2**Step 8: Attempt to connect to R3 via Telnet from PC-C.**PC> telnet 192.168.3.1**

This connection fail because R3 has been configured to accept only SSH connections on the virtual terminal lines.

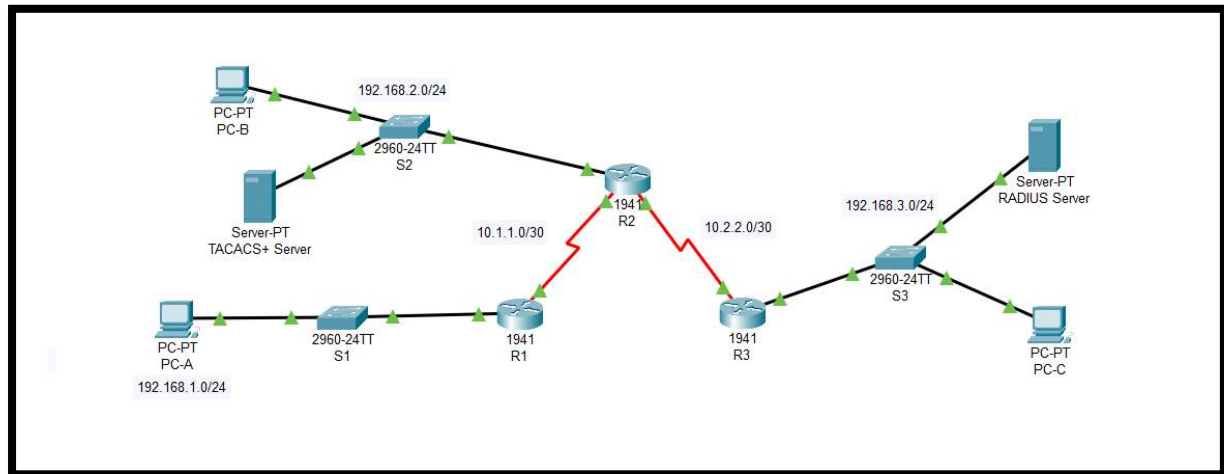
Step:9 Connect to R3 using SSH on PC-C.**PC> ssh -l SSHAdmin 192.168.3.1**

Step 10: Connect to R3 using SSH on R2.**R2# ssh -v 2 -l SSHadmin 10.2.2.1**

Practical 2

Q1} Packet Tracer - Configure AAA Authentication on cisco routers

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A	N/A
R2	G0/0	192.168.2.1	255.255.255.0	N/A	S2 F0/2
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.2	255.255.255.252	N/A	N/A
TACACS+ Server	NIC	192.168.2.2	255.255.255.0	192.168.2.1	S2 F0/6
RADIUS Server	NIC	192.168.3.2	255.255.255.0	192.168.3.1	S3 F0/1
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/2
PC-B	NIC	192.168.2.3	255.255.255.0	192.168.2.1	S2 F0/1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

Objectives

- Configure a local user account on R1 and configure authentication on the console and vty lines using local AAA.
- Verify local AAA authentication from the R1 console and the PC-A client.
- Configure server-based AAA authentication using TACACS+.
- Verify server-based AAA authentication from the PC-B client.
- Configure server-based AAA authentication using RADIUS.
- Verify server-based AAA authentication from the PC-C client.

Part 1: Configure Local AAA Authentication for Console Access on R1

Step 1: Test connectivity.

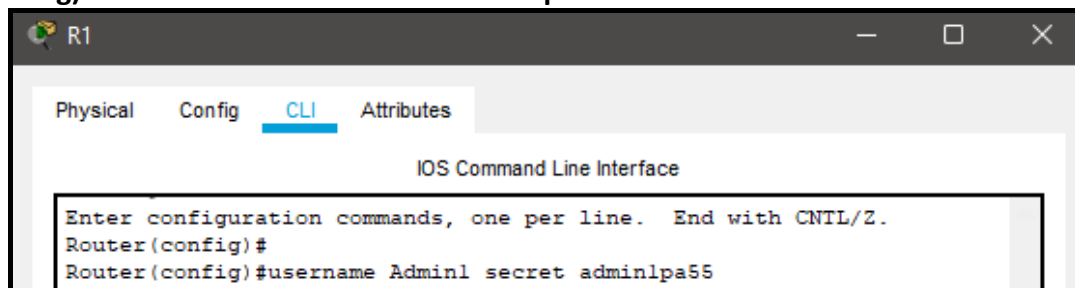
Ping from PC-A to PC-B.

Ping from PC-A to PC-C.

Ping from PC-B to PC-C.

Step 2: Configure a local username on R1. Configure a username of Admin1 with a secret password of admin1pa55.

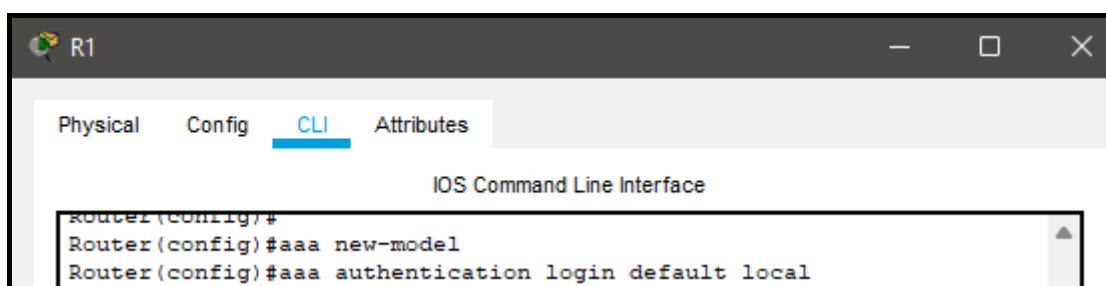
R1(config)# username Admin1 secret admin1pa55



Step 3: Configure local AAA authentication for console access on R1. Enable AAA on R1 and configure AAA authentication for the console login to use the local database.

R1(config)# aaa new-model

R1(config)# aaa authentication login default local

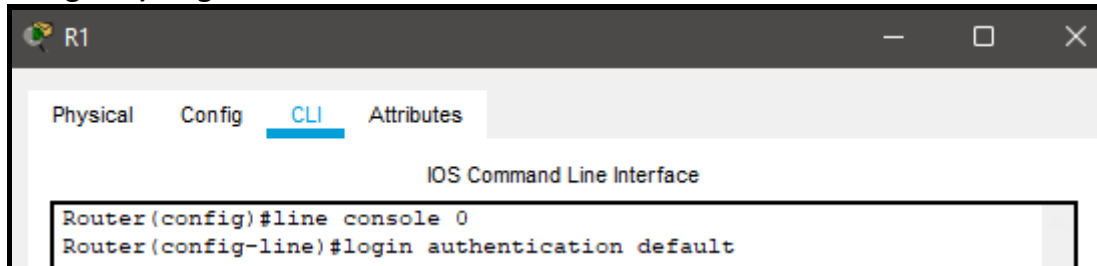


Step 4: Configure the line console to use the defined AAA authentication method.

Enable AAA on R1 and configure AAA authentication for the console login to use the default method list.

R1(config)# line console 0

R1(config-line)# login authentication default



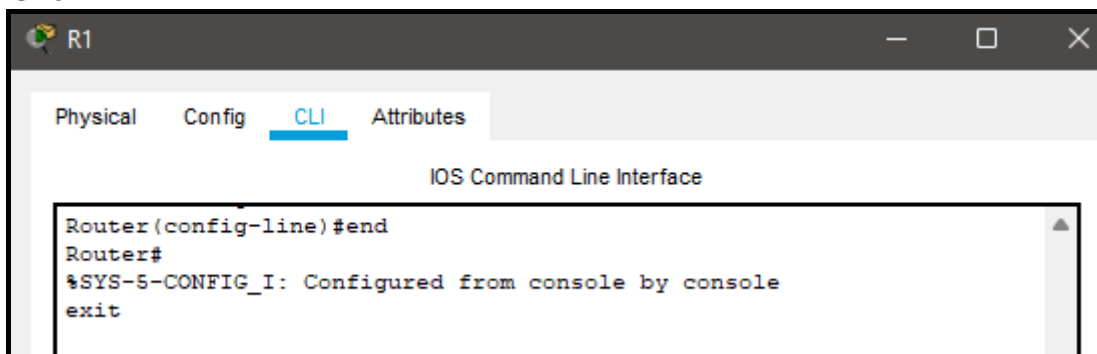
Step 5: Verify the AAA authentication method.

Verify the user EXEC login using the local database.

R1(config-line)# end

%SYS-5-CONFIG_I: Configured from console by console

R1# exit

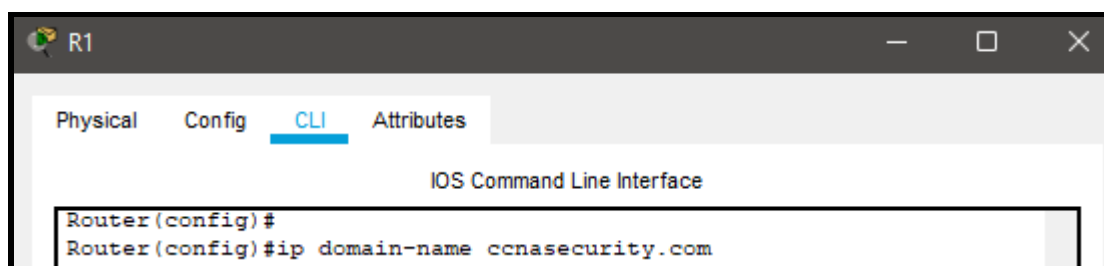


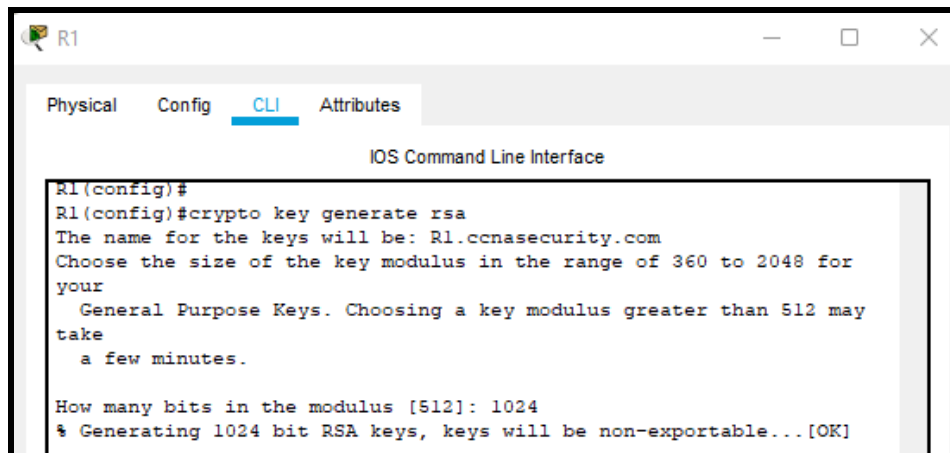
Part 2: Configure Local AAA Authentication for vty Lines on R1

Step 1: Configure domain name and crypto key for use with SSH.

a. Use ccnasecurity.com as the domain name on R1.

R1(config)# ip domain-name ccnasecurity.com

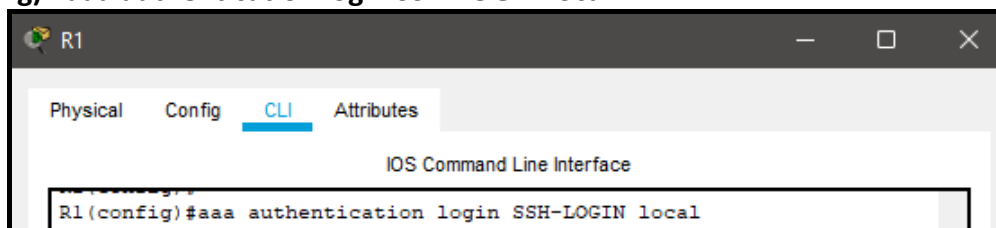


b. Create an RSA crypto key using 1024 bits.**R1(config)# crypto key generate rsa**

```
R1
Physical Config CLI Attributes
IOS Command Line Interface
R1(config)#
R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 2048 for
your
  General Purpose Keys. Choosing a key modulus greater than 512 may
take
  a few minutes.
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```

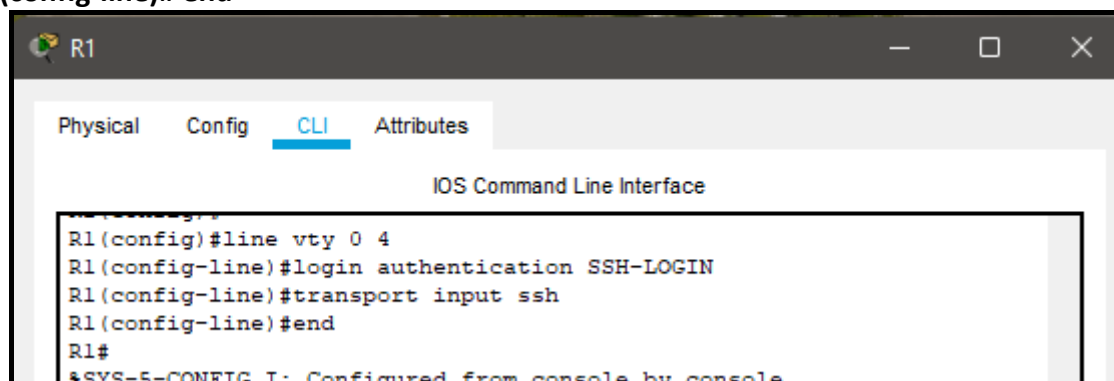
Step 2: Configure a named list AAA authentication method for the vty lines on R1.

Configure a named list called SSH-LOGIN to authenticate logins using local AAA.

R1(config)# aaa authentication login SSH-LOGIN local

```
R1
Physical Config CLI Attributes
IOS Command Line Interface
R1(config)#aaa authentication login SSH-LOGIN local
```

Step 3: Configure the vty lines to use the defined AAA authentication method.
Configure the vty lines to use the named AAA method and only allow SSH for remote access.

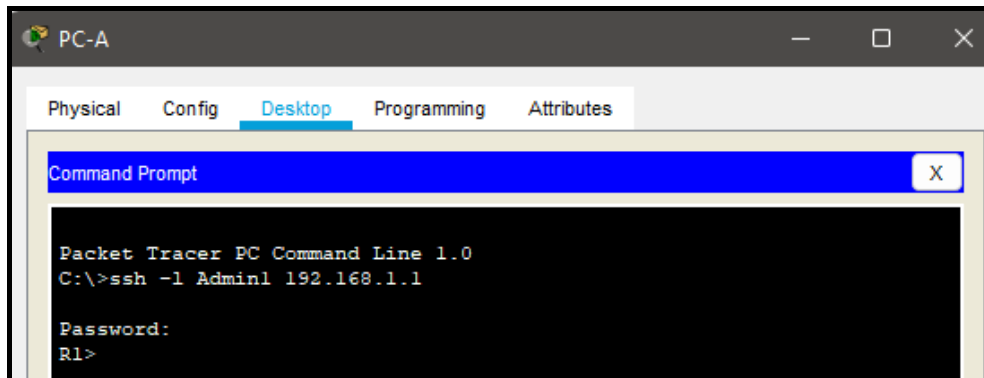
R1(config)# line vty 0 4**R1(config-line)# login authentication SSH-LOGIN****R1(config-line)# transport input ssh****R1(config-line)# end**

```
R1
Physical Config CLI Attributes
IOS Command Line Interface
R1(config)#line vty 0 4
R1(config-line)#login authentication SSH-LOGIN
R1(config-line)#transport input ssh
R1(config-line)#end
R1#
&SYS-5-CONFIG I: Configured from console by console
```

Step 4: Verify the AAA authentication method.

Verify the SSH configuration SSH to R1 from the command prompt of PC-A..

PC> ssh -l Admin1 192.168.1.1

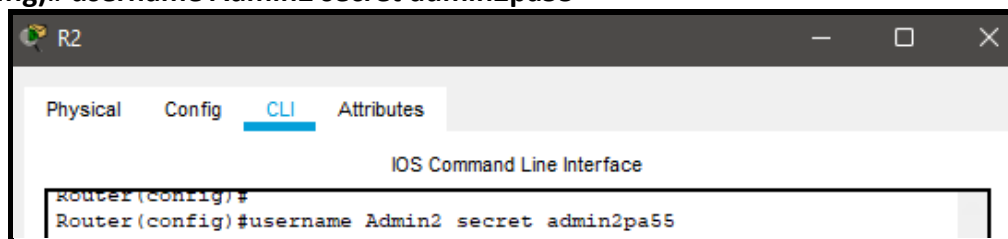


Part 3: Configure Server-Based AAA Authentication Using TACACS+ on R2

Step 1: Configure a backup local database entry called Admin.

For backup purposes, configure a local username of Admin2 and a secret password of admin2pa55.

R2(config)# username Admin2 secret admin2pa55



Step 2: Verify the TACACS+ Server configuration.

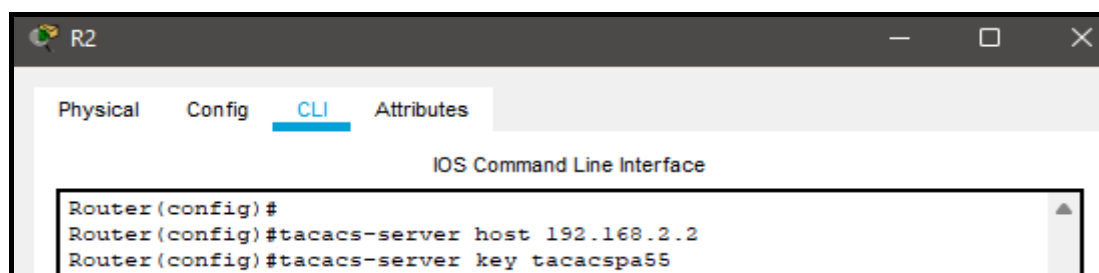
Click the TACACS+ Server. On the Services tab, click AAA. Notice that there is a Network configuration entry for R2 and a User Setup entry for Admin2.

Step 3: Configure the TACACS+ server specifics on R2.

Configure the AAA TACACS server IP address and secret key on R2.

R2(config)# tacacs-server host 192.168.2.2

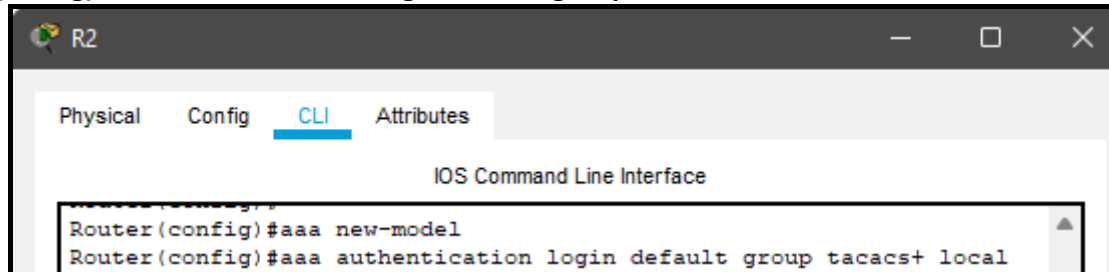
R2(config)# tacacs-server key tacacspa55



Step 4: Configure AAA login authentication for console access on R2.
Enable AAA on R2 and configure all logins to authenticate using the AAA TACACS+ server. If it is not available, then use the local database.

R2(config)# aaa new-model

R2(config)# aaa authentication login default group tacacs+ local

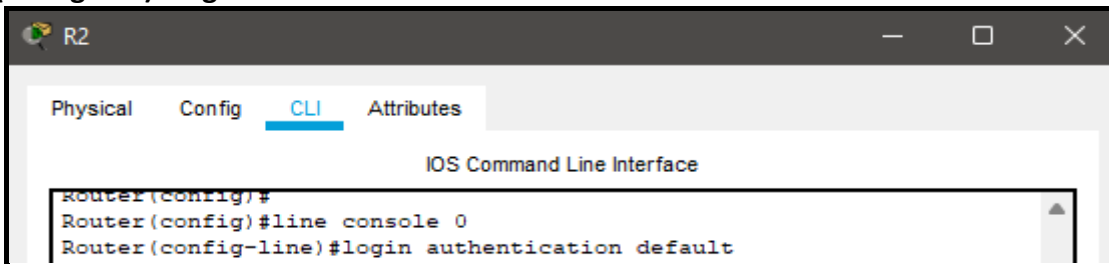


Step 5: Configure the line console to use the defined AAA authentication method.

Configure AAA authentication for console login to use the default AAA authentication method.

R2(config)# line console 0

R2(config-line)# login authentication default



Step 6: Verify the AAA authentication method.

Verify the user EXEC login using the AAA TACACS+ server.

R2(config-line)# end

%SYS-5-CONFIG_I: Configured from console by console

R2# exit

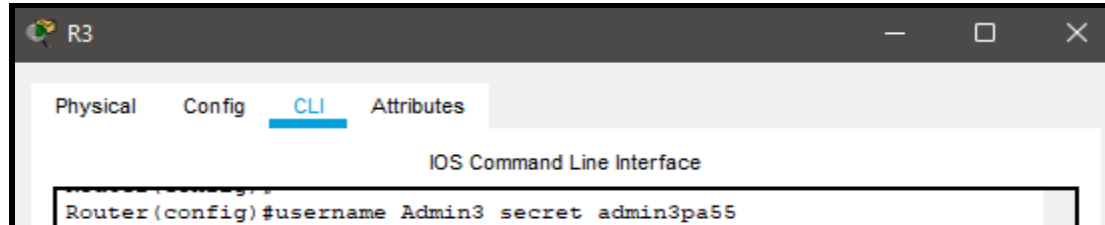


Part 4: Configure Server-Based AAA Authentication Using RADIUS on R3

Step 1: Configure a backup local database entry called Admin.

For backup purposes, configure a local username of Admin3 and a secret password of admin3pa55.

R3(config)# username Admin3 secret admin3pa55



Step 2: Verify the RADIUS Server configuration.

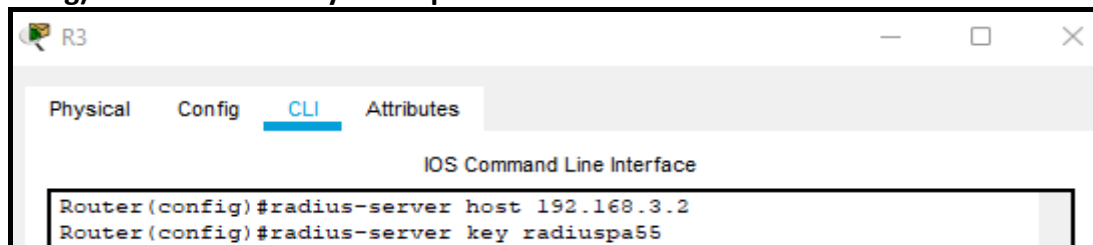
Click the RADIUS Server. On the Services tab, click AAA. Notice that there is a Network configuration entry for R3 and a User Setup entry for Admin3.

Step 3: Configure the RADIUS server specifics on R3.

Configure the AAA RADIUS server IP address and secret key on R3.

R3(config)# radius-server host 192.168.3.2

R3(config)# radius-server key radiuspa55

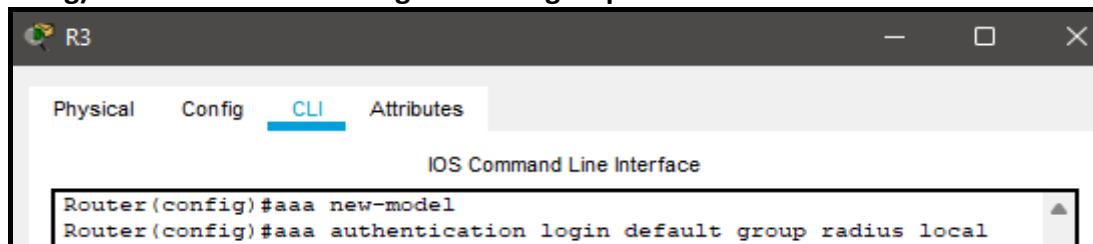


Step 4: Configure AAA login authentication for console access on R3.

Enable AAA on R3 and configure all logins to authenticate using the AAA RADIUS server. If it is not available, then use the local database.

R3(config)# aaa new-model

R3(config)# aaa authentication login default group radius local

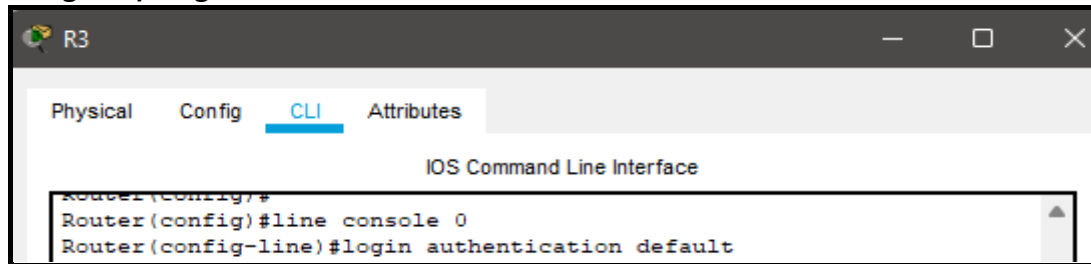


Step 5: Configure the line console to use the defined AAA authentication method.

Configure AAA authentication for console login to use the default AAA authentication method.

R3(config)# line console 0

R3(config-line)# login authentication default

A screenshot of the R3 router's CLI interface. The window title is 'R3'. At the top, there are tabs for 'Physical', 'Config', 'CLI' (which is selected and highlighted in blue), and 'Attributes'. Below the tabs, the text 'IOS Command Line Interface' is displayed. The command prompt shows the following sequence of commands: 'Router(config)#', 'Router(config)#line console 0', and 'Router(config-line)#login authentication default'.

```
R3
Physical Config CLI Attributes
IOS Command Line Interface
Router(config)#
Router(config)#line console 0
Router(config-line)#login authentication default
```

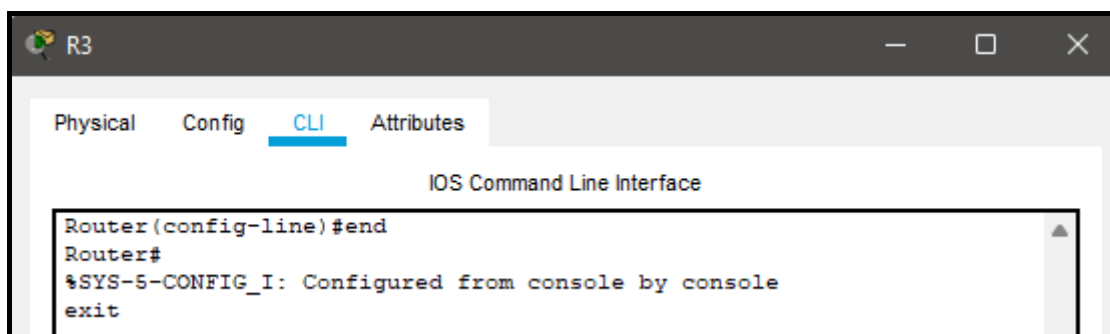
Step 6: Verify the AAA authentication method.

Verify the user EXEC login using the AAA RADIUS server.

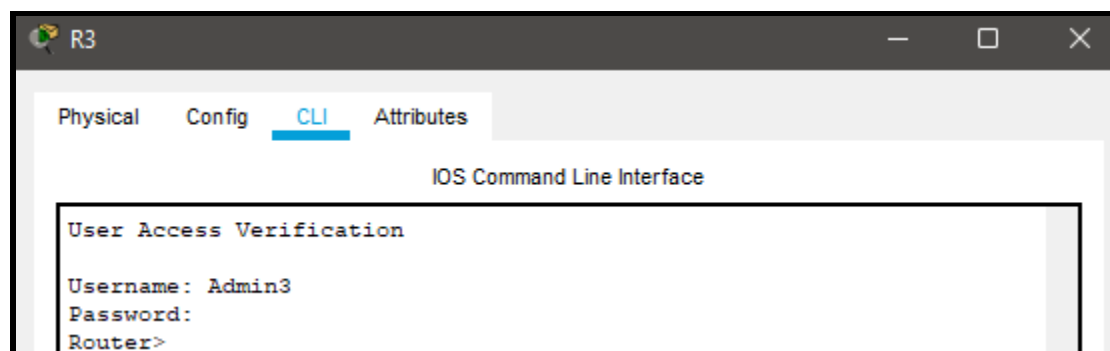
R3(config-line)# end

%SYS-5-CONFIG_I: Configured from console by console

R3# exit

A screenshot of the R3 router's CLI interface. The window title is 'R3'. At the top, there are tabs for 'Physical', 'Config', 'CLI' (which is selected and highlighted in blue), and 'Attributes'. Below the tabs, the text 'IOS Command Line Interface' is displayed. The command prompt shows the following sequence of commands: 'Router(config-line)#end', 'Router#', '%SYS-5-CONFIG_I: Configured from console by console', and 'exit'.

```
R3
Physical Config CLI Attributes
IOS Command Line Interface
Router(config-line)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
exit
```

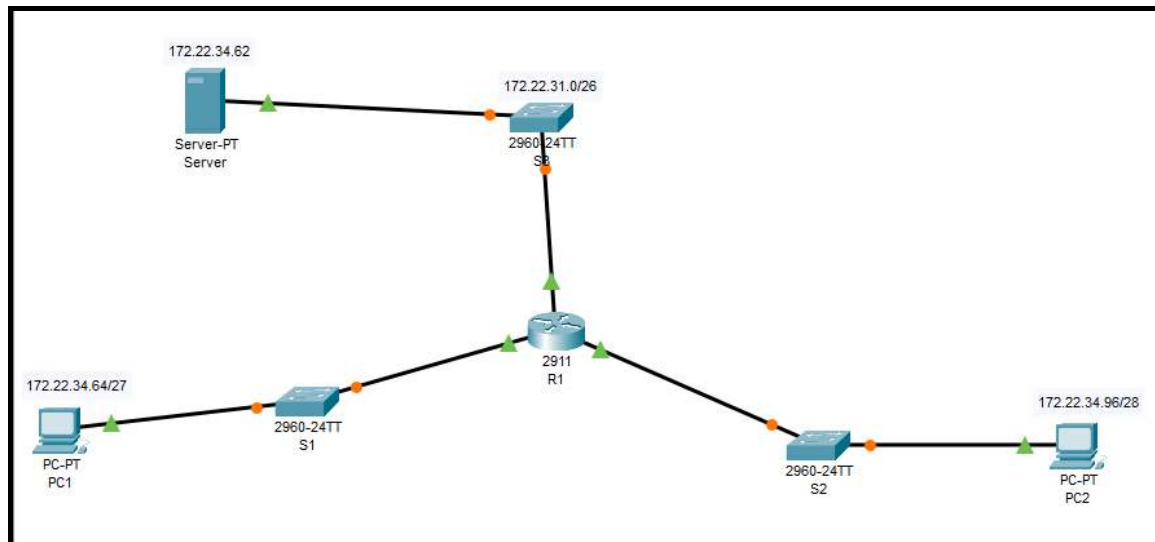
A screenshot of the R3 router's CLI interface. The window title is 'R3'. At the top, there are tabs for 'Physical', 'Config', 'CLI' (which is selected and highlighted in blue), and 'Attributes'. Below the tabs, the text 'IOS Command Line Interface' is displayed. The command prompt shows the following sequence of commands: 'User Access Verification', 'Username: Admin3', 'Password:', and 'Router>'.

```
R3
Physical Config CLI Attributes
IOS Command Line Interface
User Access Verification
Username: Admin3
Password:
Router>
```

Practical 3

Q1} Configure Extended ACL's-Scenario 1

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	172.22.34.65	255.255.255.224	N/A
	G0/1	172.22.34.97	255.255.255.240	N/A
	G0/2	172.22.34.1	255.255.255.192	N/A
Server	NIC	172.22.34.62	255.255.255.192	172.22.34.1
PC1	NIC	172.22.34.66	255.255.255.224	172.22.34.65
PC2	NIC	172.22.34.98	255.255.255.240	172.22.34.97

Objectives

Part 1: Configure, Apply and Verify an Extended Numbered ACL

Part 2: Configure, Apply and Verify an Extended Named ACL

Background / Scenario

Two employees need access to services provided by the server. PC1 needs only FTP access while PC2 needs only web access. Both computers are able to ping the server, but not each other.

Part 1: Configure, Apply and Verify an Extended Numbered ACL

Step 1: Configure an ACL to permit FTP and ICMP.

a. From global configuration mode on R1, enter the following command to determine the first valid number for an extended access list.

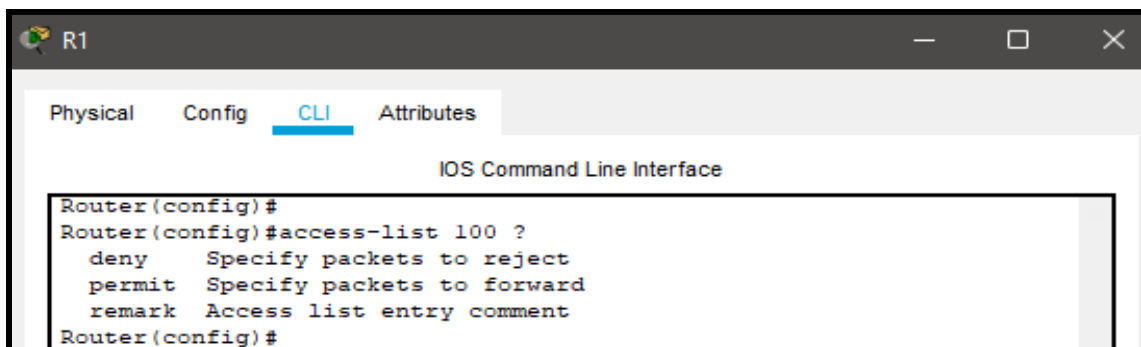
R1(config)# access-list ?



```
R1
Physical Config CLI Attributes
IOS Command Line Interface
Router(config)#access-list ?
<1-99>      IP standard access list
<100-199>   IP extended access list
Router(config)#
```

b. Add 100 to the command, followed by a question mark.

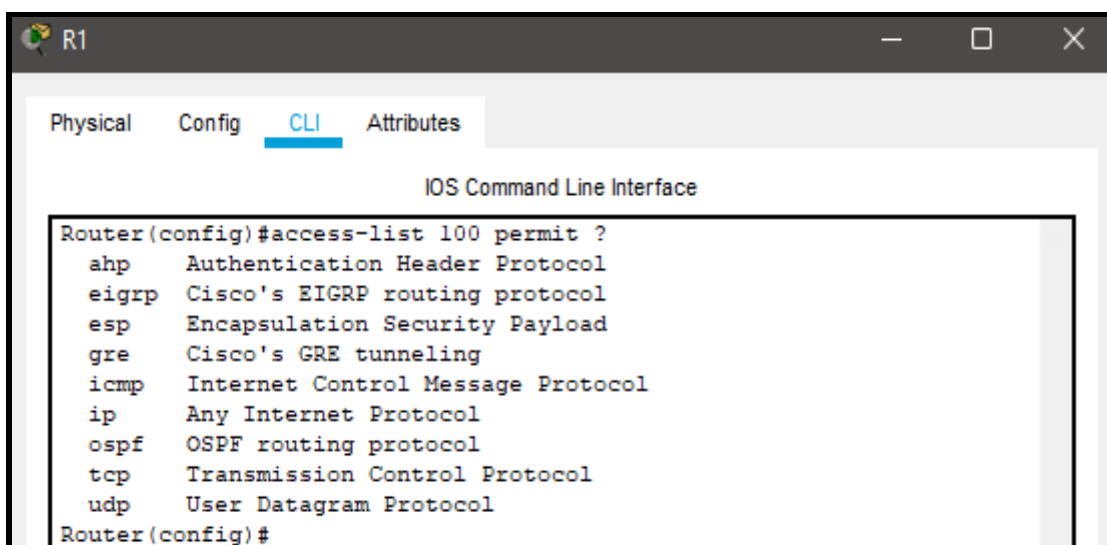
R1(config)# access-list 100 ?



```
R1
Physical Config CLI Attributes
IOS Command Line Interface
Router(config)#
Router(config)#access-list 100 ?
deny       Specify packets to reject
permit     Specify packets to forward
remark     Access list entry comment
Router(config)#
```

c. To permit FTP traffic, enter a permit, followed by a question mark.

R1(config)# access-list 100 permit ?



```
R1
Physical Config CLI Attributes
IOS Command Line Interface
Router(config)#access-list 100 permit ?
ahp        Authentication Header Protocol
eigrp      Cisco's EIGRP routing protocol
esp        Encapsulation Security Payload
gre        Cisco's GRE tunneling
icmp       Internet Control Message Protocol
ip         Any Internet Protocol
ospf       OSPF routing protocol
tcp        Transmission Control Protocol
udp        User Datagram Protocol
Router(config)#
```

d. This ACL permits FTP and ICMP. ICMP is listed above, but FTP is not, because FTP uses TCP.

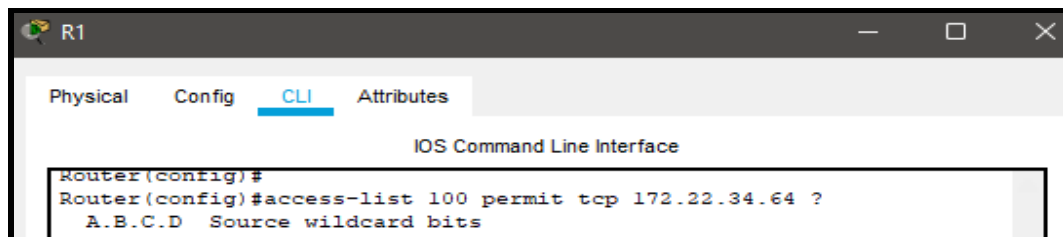
Therefore, enter tcp to further refine the ACL help.

R1(config)# access-list 100 permit tcp ?



e. Notice that we could filter just for PC1 by using the host keyword or we could allow any host. In this case, any device is allowed that has an address belonging to the 172.22.34.64/27 network. Enter the network address, followed by a question mark.

R1(config)# access-list 100 permit tcp 172.22.34.64 ?



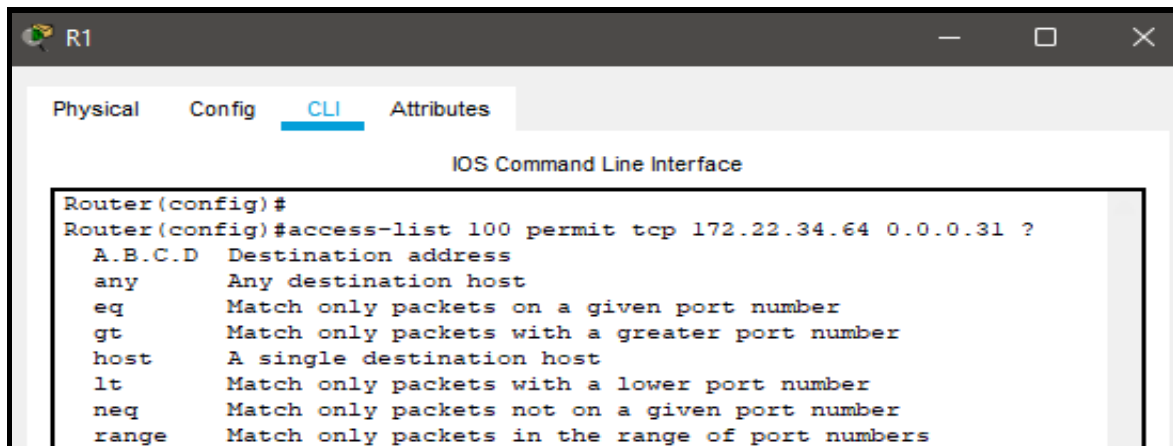
f. Calculate the wildcard mask determining the binary opposite of a subnet mask.

11111111.11111111.11111111.11100000 = 255.255.255.224

00000000.00000000.00000000.00011111 = 0.0.0.31

g. Enter the wildcard mask, followed by a question mark.

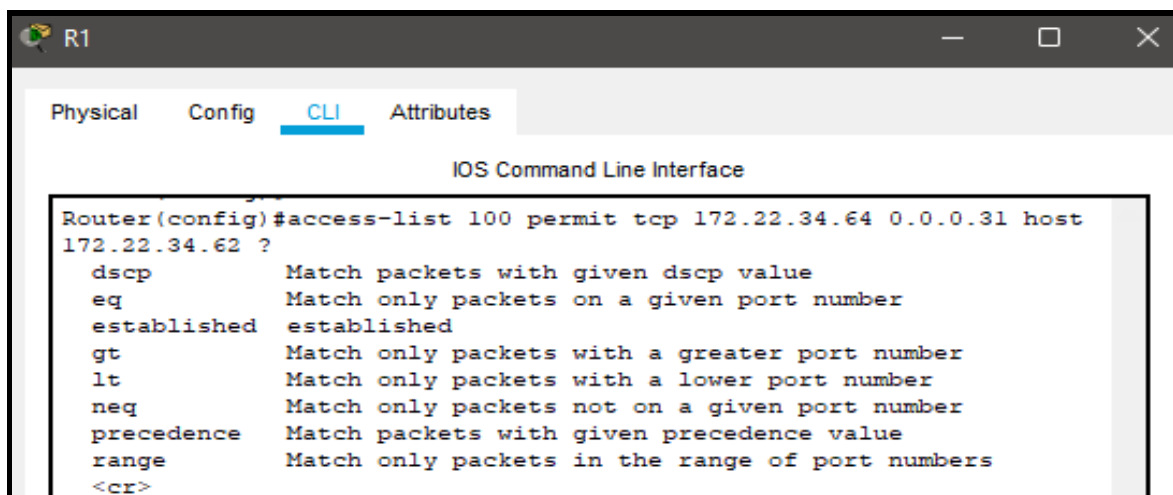
R1(config)# access-list 100 permit tcp 172.22.34.64 0.0.0.31 ?



```
Router(config)#
Router(config)#access-list 100 permit tcp 172.22.34.64 0.0.0.31 ?
A.B.C.D Destination address
any Any destination host
eq Match only packets on a given port number
gt Match only packets with a greater port number
host A single destination host
lt Match only packets with a lower port number
neq Match only packets not on a given port number
range Match only packets in the range of port numbers
```

h. Configure the destination address. In this scenario, we are filtering traffic for a single destination, which is the server. Enter the host keyword followed by the server's IP address.

R1(config)# access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 ?



```
Router(config)#access-list 100 permit tcp 172.22.34.64 0.0.0.31 host
172.22.34.62 ?
dscp Match packets with given dscp value
eq Match only packets on a given port number
established established
gt Match only packets with a greater port number
lt Match only packets with a lower port number
neq Match only packets not on a given port number
precedence Match packets with given precedence value
range Match only packets in the range of port numbers
<cr>
```

I. Notice that one of the options is <cr> (carriage return). In other words, you can press Enter and the

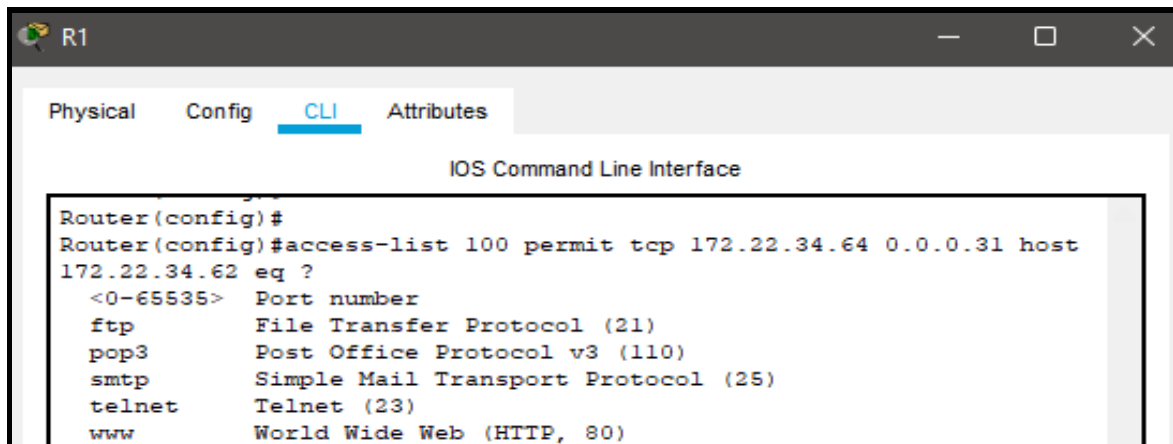
statement would permit all TCP traffic. However, we are only permitting FTP traffic; therefore, enter the

eq keyword, followed by a question mark to display the available options.

Then, enter ftp and press

Enter.

R1(config)# access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq ?



```
R1
Physical Config CLI Attributes
IOS Command Line Interface
Router(config)#
Router(config)#access-list 100 permit tcp 172.22.34.64 0.0.0.31 host
172.22.34.62 eq ?
<0-65535> Port number
ftp File Transfer Protocol (21)
pop3 Post Office Protocol v3 (110)
smtp Simple Mail Transport Protocol (25)
telnet Telnet (23)
www World Wide Web (HTTP, 80)
```

R1(config)# access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq ftp

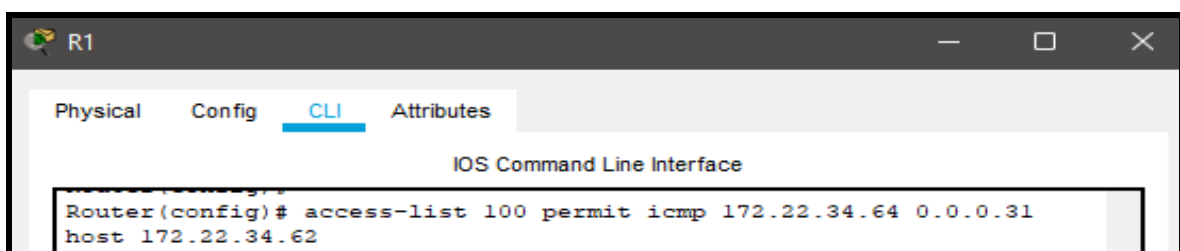


```
R1
Physical Config CLI Attributes
IOS Command Line Interface
Router(config)#access-list 100 permit tcp 172.22.34.64 0.0.0.31 host
172.22.34.62 eq ftp
```

j. Create a second access list statement to permit ICMP (ping, etc.) traffic from PC1 to Server. Note that

the access list number remains the same and no particular type of ICMP traffic needs to be specified.

R1(config)# access-list 100 permit icmp 172.22.34.64 0.0.0.31 host 172.22.34.62



```
R1
Physical Config CLI Attributes
IOS Command Line Interface
Router(config)# access-list 100 permit icmp 172.22.34.64 0.0.0.31
host 172.22.34.62
```

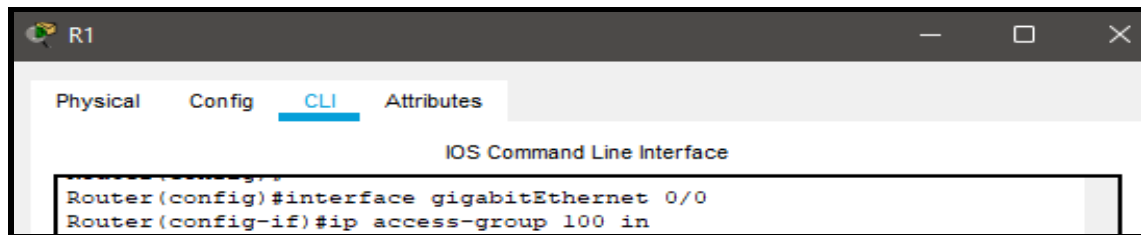
k. All other traffic is denied, by default.

Step 2: Apply the ACL on the correct interface to filter traffic.

From R1's perspective, the traffic that ACL 100 applies to is inbound from the network connected to Gigabit Ethernet 0/0 interface. Enter interface configuration mode and apply the ACL.

R1(config)# interface gigabitEthernet 0/0

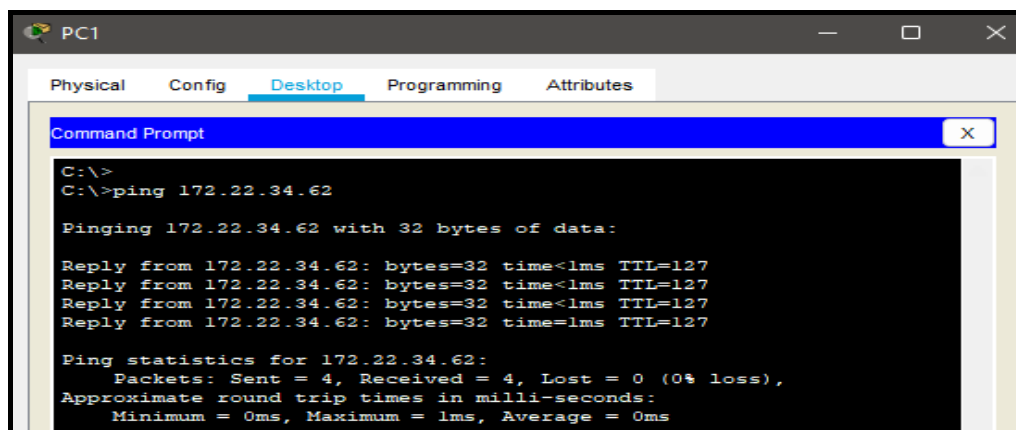
R1(config-if)# ip access-group 100 in



Step 3: Verify the ACL implementation.

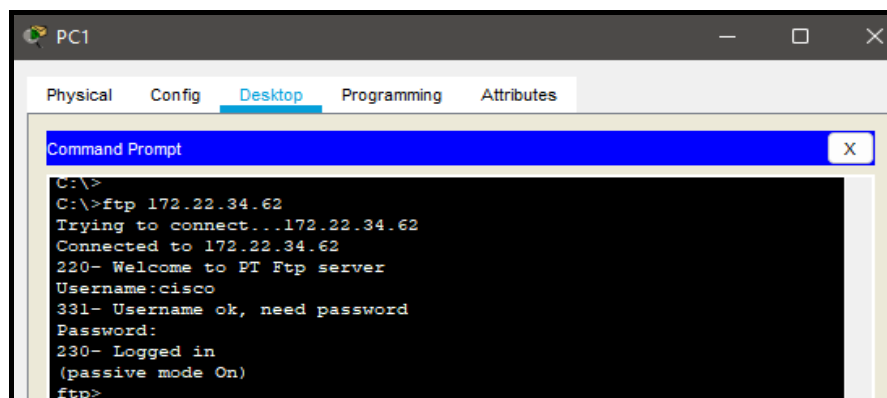
a. Ping from PC1 to Server. If the pings are unsuccessful, verify the IP addresses before continuing.

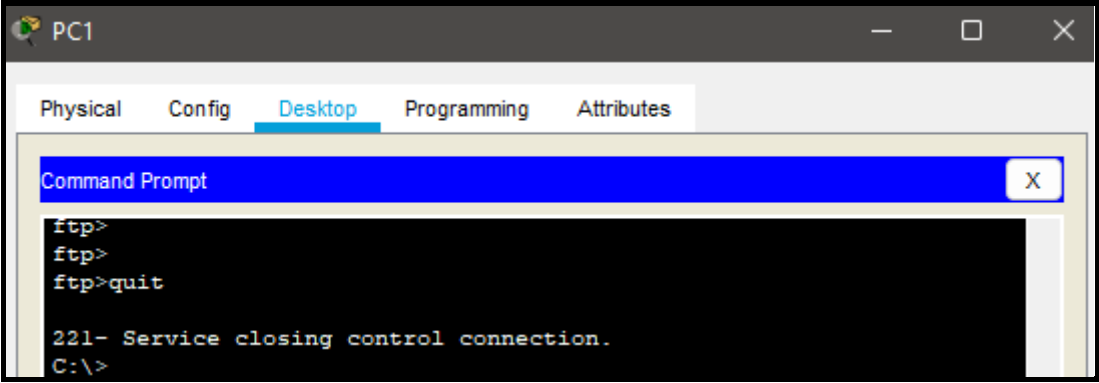
PC>ping 172.22.34.62



b. FTP from PC1 to Server. The username and password are both cisco.

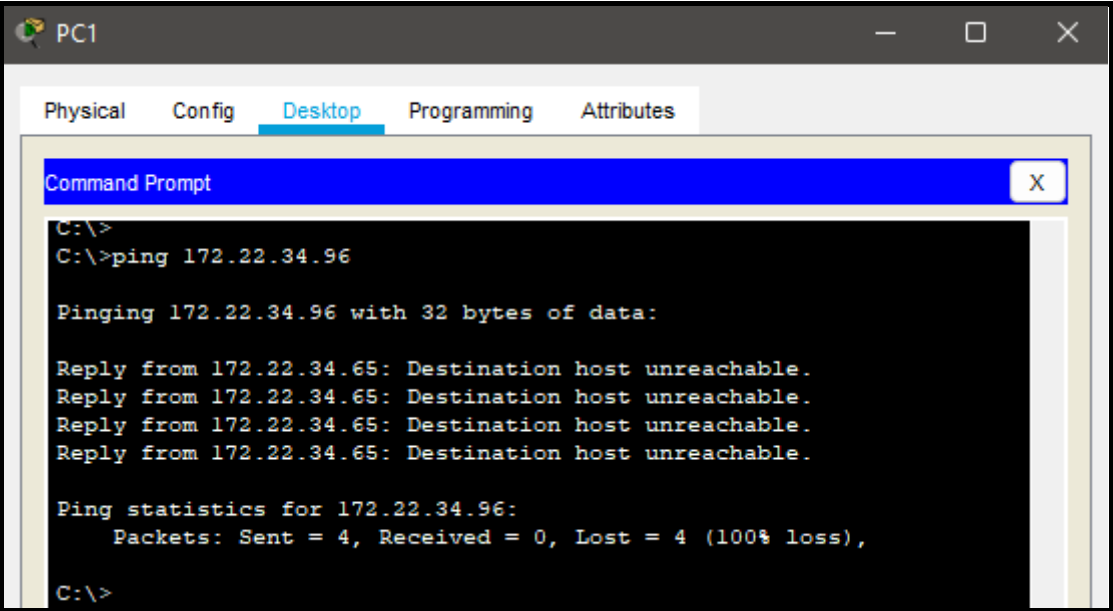
PC> ftp 172.22.34.62



c. Exit the FTP service of the Server.**ftp> quit**

The screenshot shows a window titled 'PC1' with tabs for Physical, Config, Desktop, Programming, and Attributes. The 'Desktop' tab is active, displaying a Command Prompt window. The Command Prompt shows the following text:

```
ftp>  
ftp>  
ftp>quit  
  
221- Service closing control connection.  
C:\>
```

d. Ping from PC1 to PC2. The destination host should be unreachable, because the traffic was not explicitly permitted.

The screenshot shows a window titled 'PC1' with tabs for Physical, Config, Desktop, Programming, and Attributes. The 'Desktop' tab is active, displaying a Command Prompt window. The Command Prompt shows the following text:

```
C:\>  
C:\>ping 172.22.34.96  
  
Pinging 172.22.34.96 with 32 bytes of data:  
  
Reply from 172.22.34.65: Destination host unreachable.  
Reply from 172.22.34.65: Destination host unreachable.  
Reply from 172.22.34.65: Destination host unreachable.  
Reply from 172.22.34.65: Destination host unreachable.  
  
Ping statistics for 172.22.34.96:  
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),  
C:\>
```

Part 2: Configure, Apply and Verify an Extended Named ACL

Step 1: Configure an ACL to permit HTTP access and ICMP.

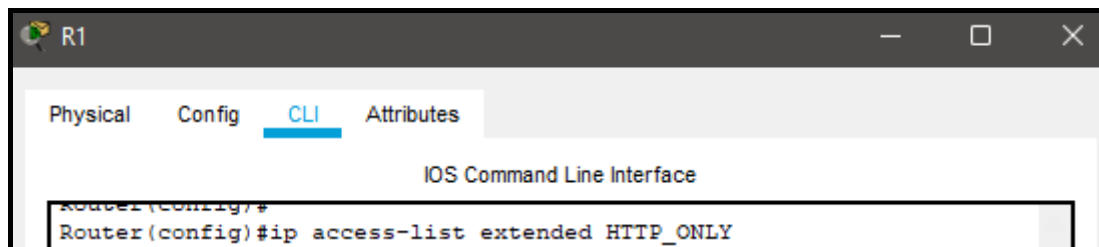
a. Named ACLs start with the ip keyword. From global configuration mode of R1, enter the following command, followed by a question mark.

R1(config)# ip access-list ?



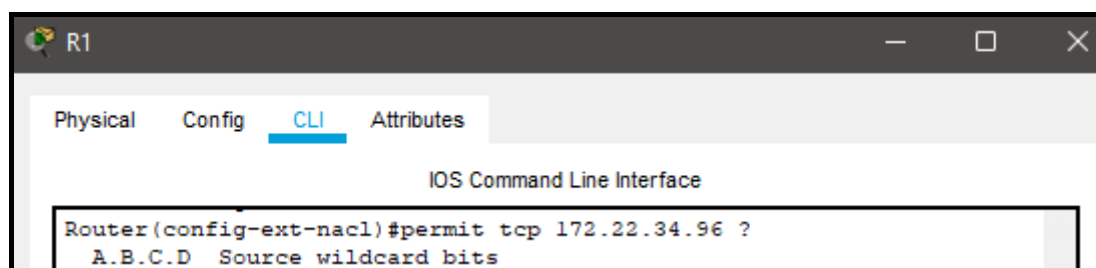
b. You can configure named standard and extended ACLs. This access list filters both source and destination IP addresses; therefore, it must be extended. Enter HTTP_ONLY as the name. (For Packet Tracer scoring, the name is case-sensitive.)

R1(config)# ip access-list extended HTTP_ONLY



c. The prompt changes. You are now in extended named ACL configuration mode. All devices on the PC 2 LAN need TCP access. Enter the network address, followed by a question mark.

R1(config-ext-nacl)# permit tcp 172.22.34.96 ?



d. An alternative way to calculate a wildcard is to subtract the subnet mask from

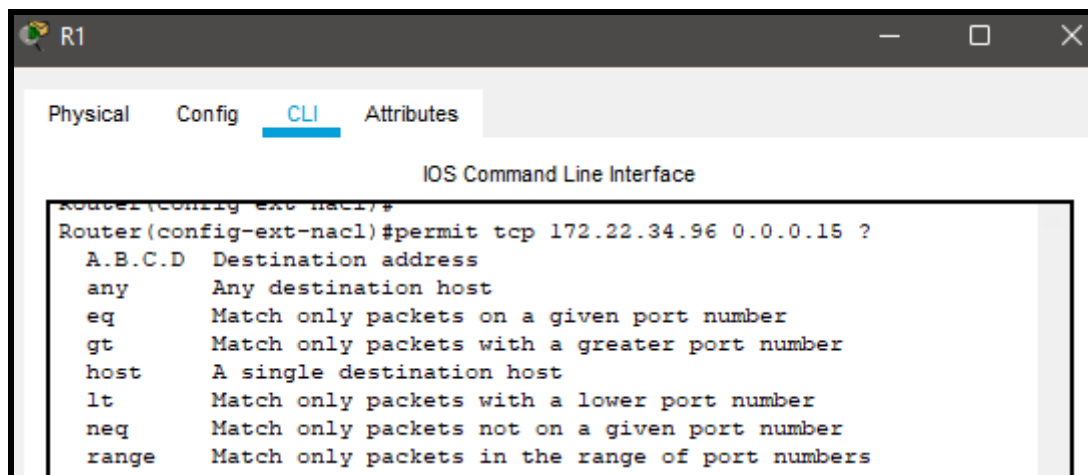
255.255.255.255.

255.255.255.255

- 255.255.255.240

= 0. 0. 0. 15

R1(config-ext-nacl)# permit tcp 172.22.34.96 0.0.0.15 ?



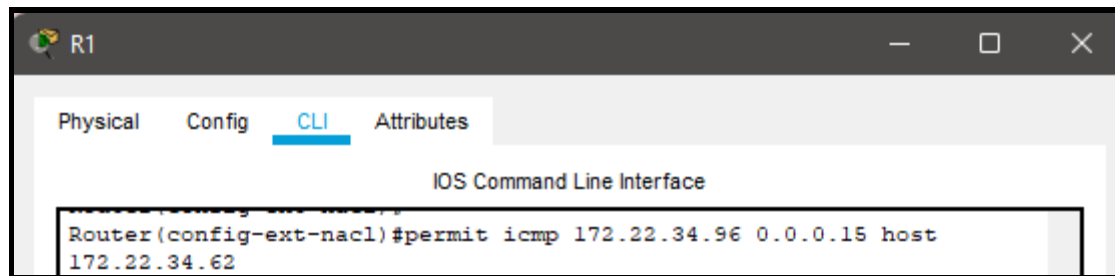
e. Finish the statement by specifying the server address as you did in Part 1 and filtering www traffic.

R1(config-ext-nacl)# permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.62 eq www

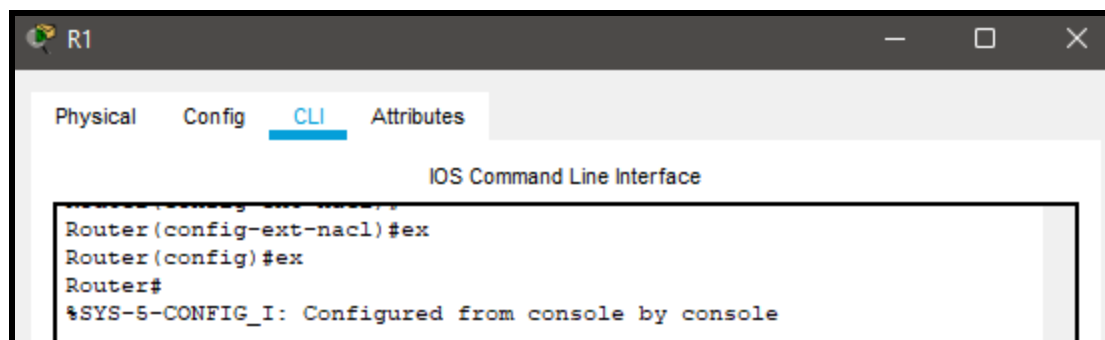


f. Create a second access list statement to permit ICMP (ping, etc.) traffic from PC2 to Server. Note: The prompt remains the same and a specific type of ICMP traffic does not need to be specified.

R1(config-ext-nacl)# permit icmp 172.22.34.96 0.0.0.15 host 172.22.34.62



g. All other traffic is denied, by default. Exit out of extended named ACL configuration mode.

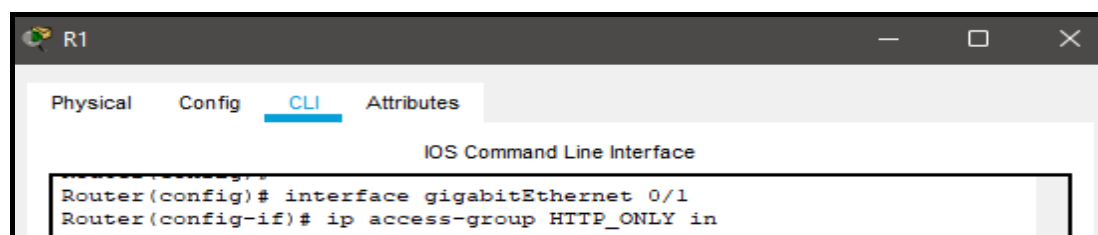


Step 2: Apply the ACL on the correct interface to filter traffic.

From R1's perspective, the traffic that access list HTTP_ONLY applies to is inbound from the network connected to Gigabit Ethernet 0/1 interface. Enter the interface configuration mode and apply the ACL.

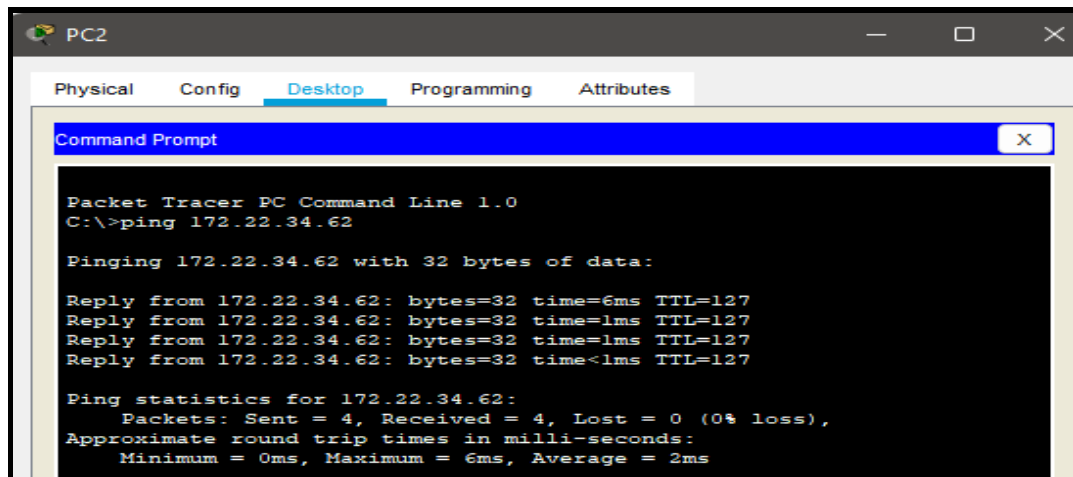
R1(config)# interface gigabitEthernet 0/1

R1(config-if)# ip access-group HTTP_ONLY in

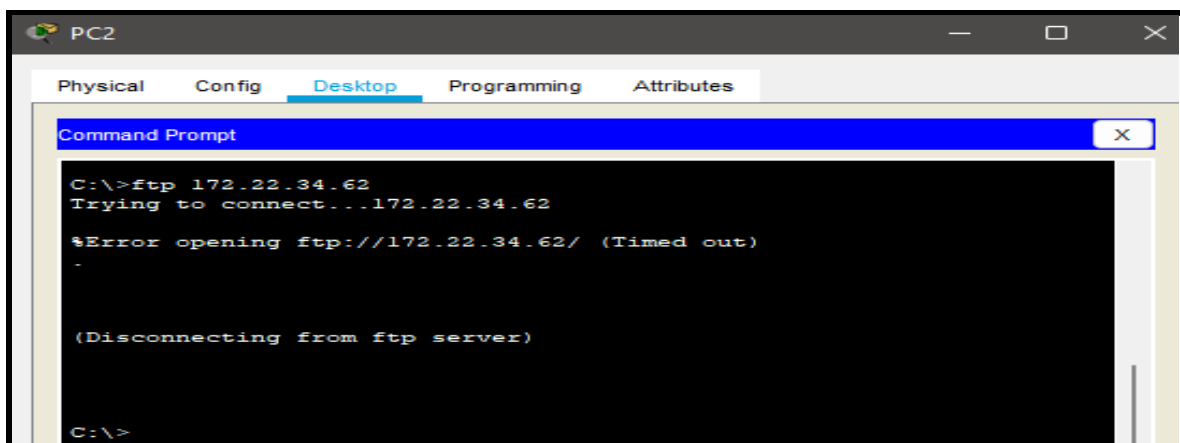


Step 3: Verify the ACL implementation.

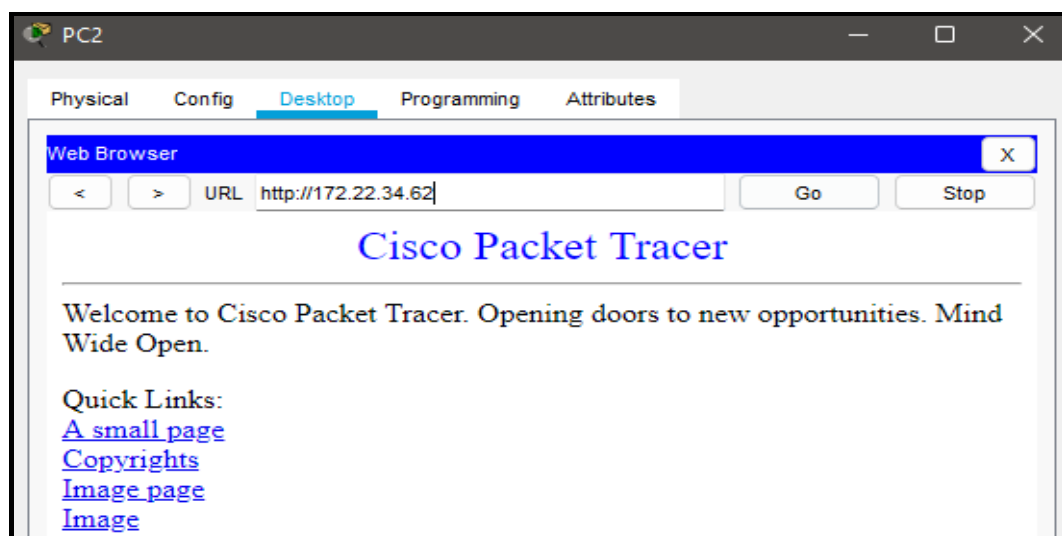
a. Ping from PC2 to Server. The ping should be successful, if the ping is unsuccessful, verify the IP addresses before continuing.



b. FTP from PC2 to Server. The connection should fail.



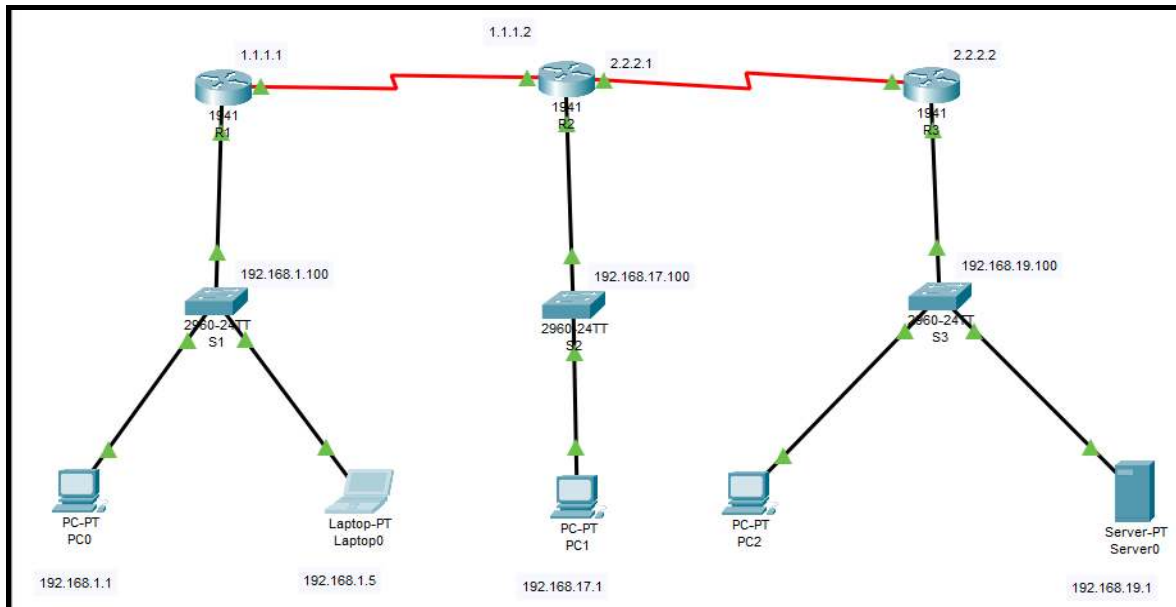
c. Open the web browser on PC2 and enter the IP address of Server as the URL. The connection should be successful.



Practical 4

Q1} Configure Extended ACL's-Scenario 2

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	Fa0/0	192.168.1.100	255.255.255.0	NIA
	Se2/0	1.1.1.1	255.255.255.252	NIA
R2	Fa0/0	192.168.17.100	255.255.255.0	NIA
	Se2/0	1.1.1.2	255.255.255.252	NIA
	Se3/0	2.2.2.1	255.255.255.252	NIA
R3	Fa0/0	192.168.19.100	255.255.255.0	NIA
	Se2/0	2.2.2.2	255.255.255.252	NIA
PC0	NIC	192.168.1.1	255.255.255.0	192.168.1.100
Laptop0	NIC	192.168.1.5	255.255.255.0	192.168.1.100
PC1	NIC	192.168.17.1	255.255.255.0	192.168.17.100
PC2	NIC			
Server0	NIC	192.168.19.1	255.255.255.0	192.168.19.100

The screenshot shows the configuration window for router R1. The 'Config' tab is active, and the 'ROUTING' section is expanded, with 'RIP' selected. The 'RIP Routing' section shows a table with two entries: '1.0.0.0' and '192.168.1.0'. A 'Remove' button is visible at the bottom right of the table. Below the table, the 'Equivalent IOS Commands' section displays the following commands:

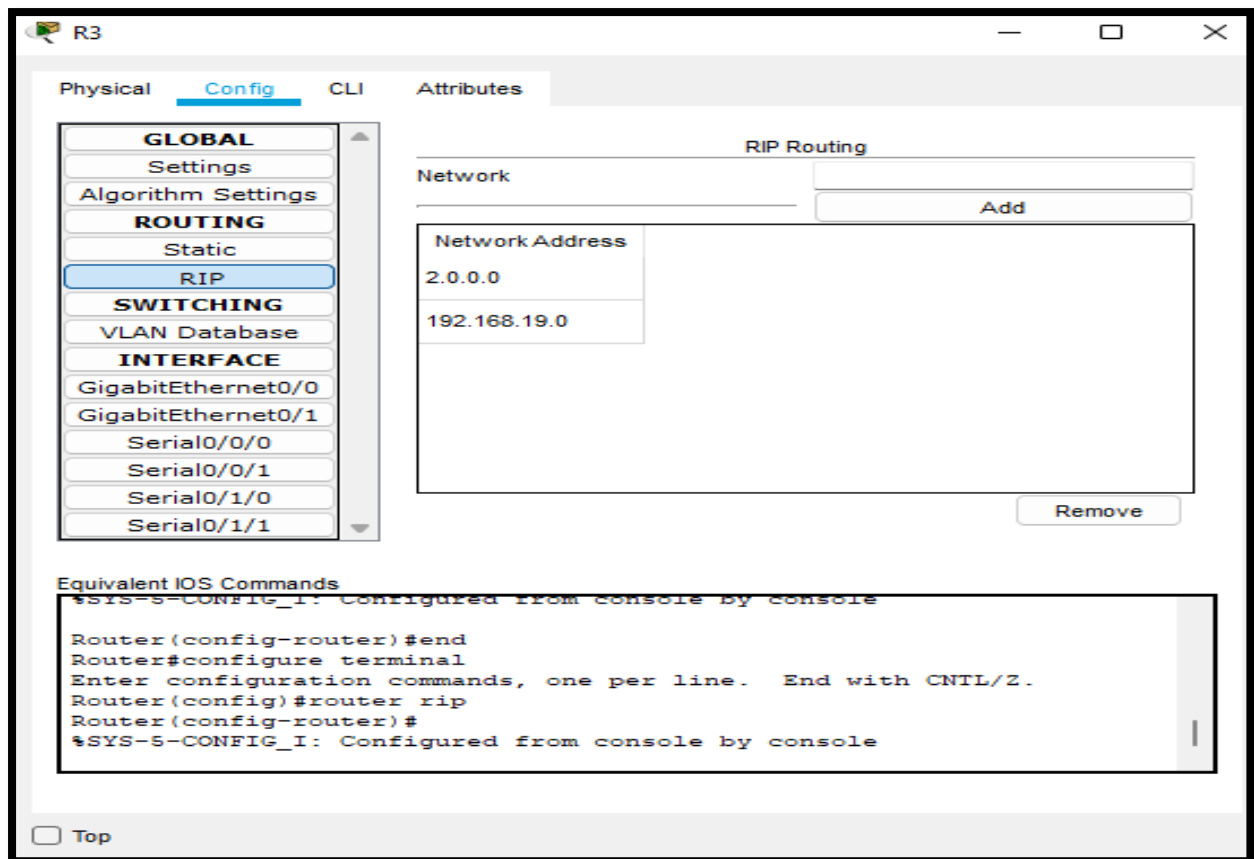
```
Router(config-router)#network 192.168.1.0
Router(config-router)#
Router(config-router)#end
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#
%SYS-5-CONFIG_I: Configured from console by console
```

A 'Top' button is located at the bottom left of the window.

The screenshot shows the configuration window for router R2. The 'Config' tab is active, and the 'ROUTING' section is expanded, with 'RIP' selected. The 'RIP Routing' section shows a table with three entries: '1.0.0.0', '2.0.0.0', and '192.168.17.0'. A 'Remove' button is visible at the bottom right of the table. Below the table, the 'Equivalent IOS Commands' section displays the following commands:

```
Router(config-router)#network 192.168.17.0
Router(config-router)#
Router(config-router)#end
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#
%SYS-5-CONFIG_I: Configured from console by console
```

A 'Top' button is located at the bottom left of the window.



Steps for the following Conditions:

1. Blocking Web browser communication from PC0 to Server0.
Router(Config)#access-list 100 deny tcp host ip-address host ip-address eq www
2. Blocking FTP connection from PC1 to Server0.
Router(Config)#access-list 100 deny tcp host ip-address host ip-address eq ftp
3. Blocking ICMP connection from laptop to Server0.
Router(Config)#access-list 100 deny icmp host ip-address host ip-address
4. All other IP traffic is denied, by default. So use the command
Router(Config)#access-list 100 permit ip any any

5. From R1's perspective, the traffic that access list HTTP_ONLY applies to is inbound from the network connected to Serial0/0/1 interface. Enter the interface configuration mode and apply the ACL.

Router(Config)#interface Serial0/0/1

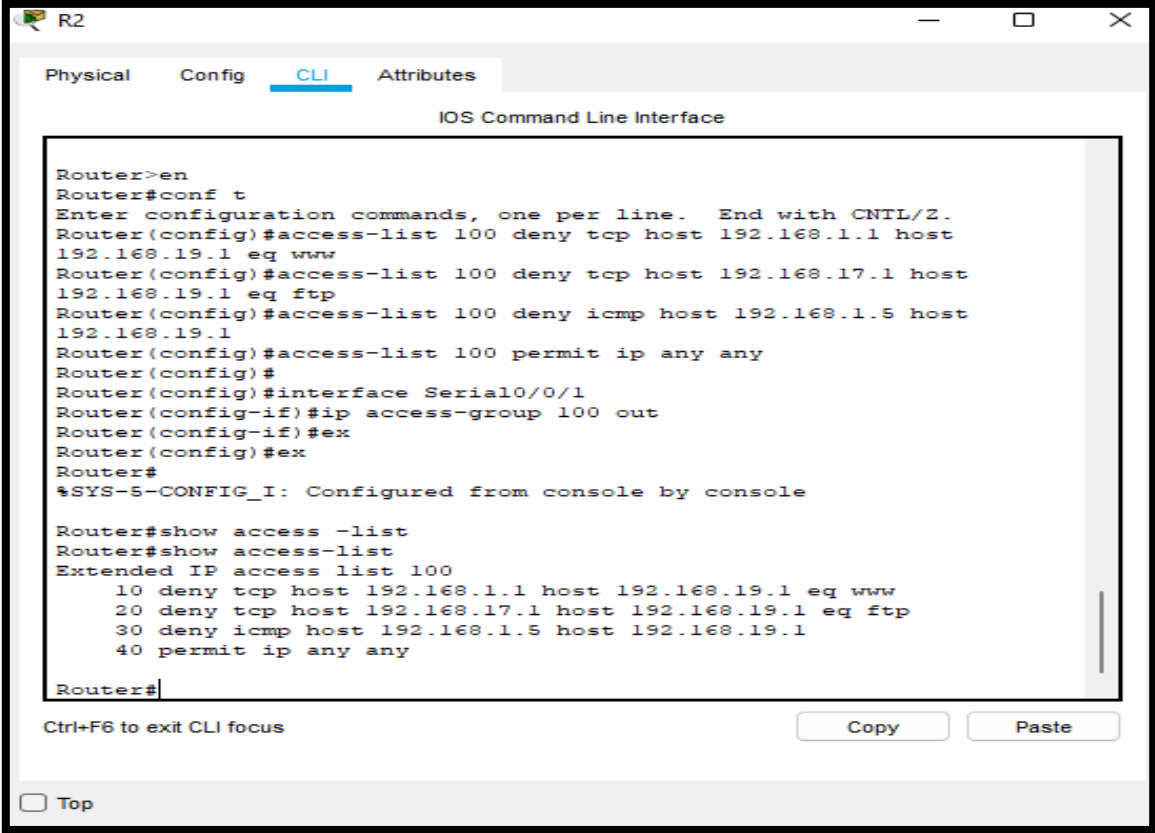
Router(Config-if)#ip access-group 100 out

Router(Config-if)#ex

Router(Config)#ex

6. To Check if the following conditions have been implemented or not use the command.

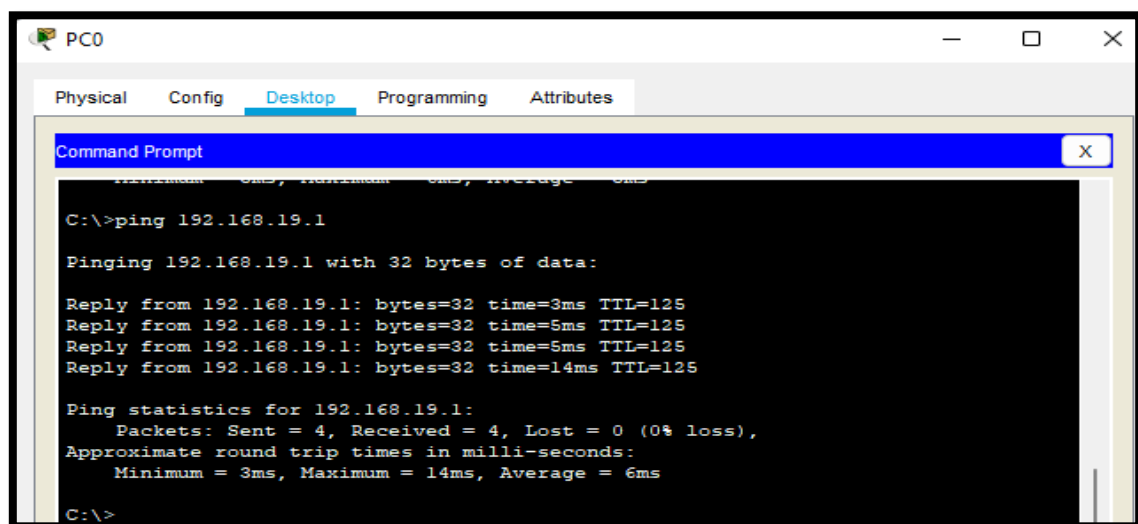
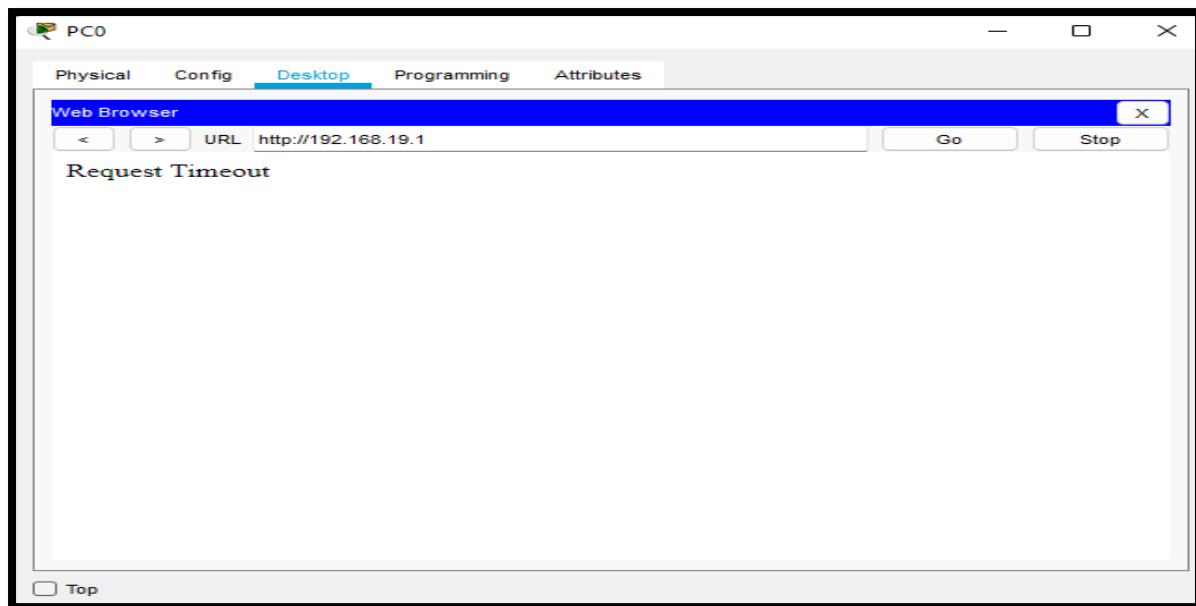
Router#show access-list



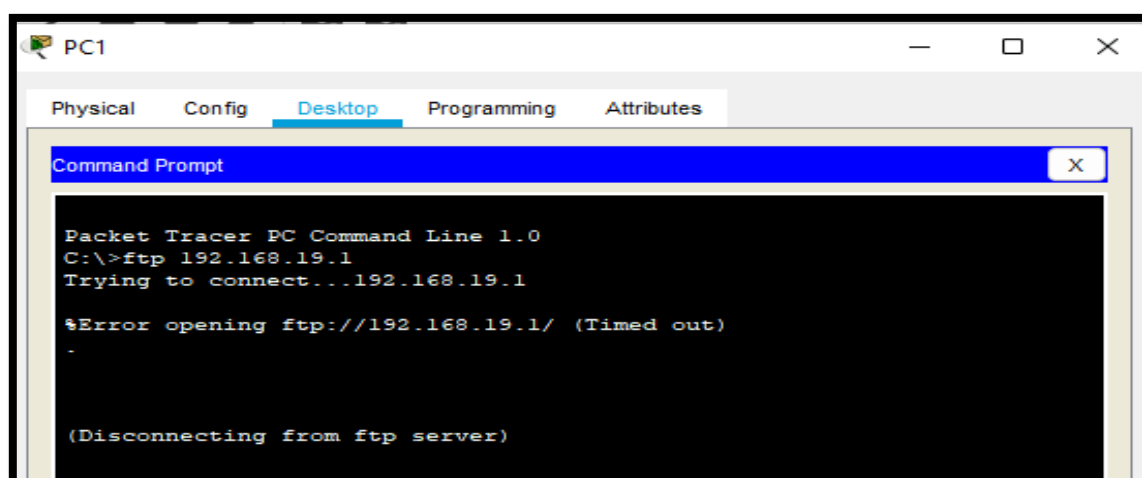
```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 deny tcp host 192.168.1.1 host 192.168.19.1 eq www
Router(config)#access-list 100 deny tcp host 192.168.17.1 host 192.168.19.1 eq ftp
Router(config)#access-list 100 deny icmp host 192.168.1.5 host 192.168.19.1
Router(config)#access-list 100 permit ip any any
Router(config)#
Router(config)#interface Serial0/0/1
Router(config-if)#ip access-group 100 out
Router(config-if)#ex
Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

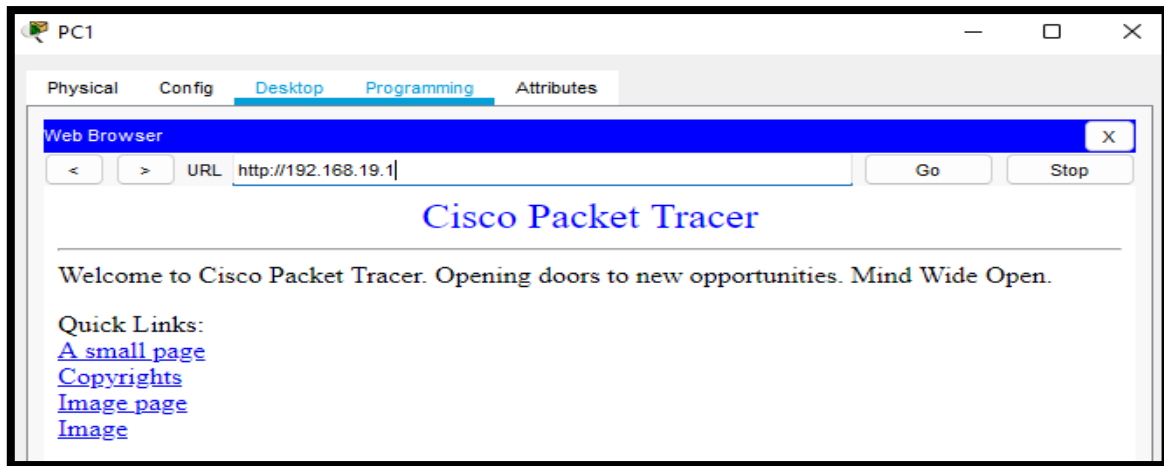
Router#show access -list
Router#show access-list
Extended IP access list 100
 10 deny tcp host 192.168.1.1 host 192.168.19.1 eq www
 20 deny tcp host 192.168.17.1 host 192.168.19.1 eq ftp
 30 deny icmp host 192.168.1.5 host 192.168.19.1
 40 permit ip any any
Router#
```

Condition 01: Blocking Web browser communication from PC0 to Server0.

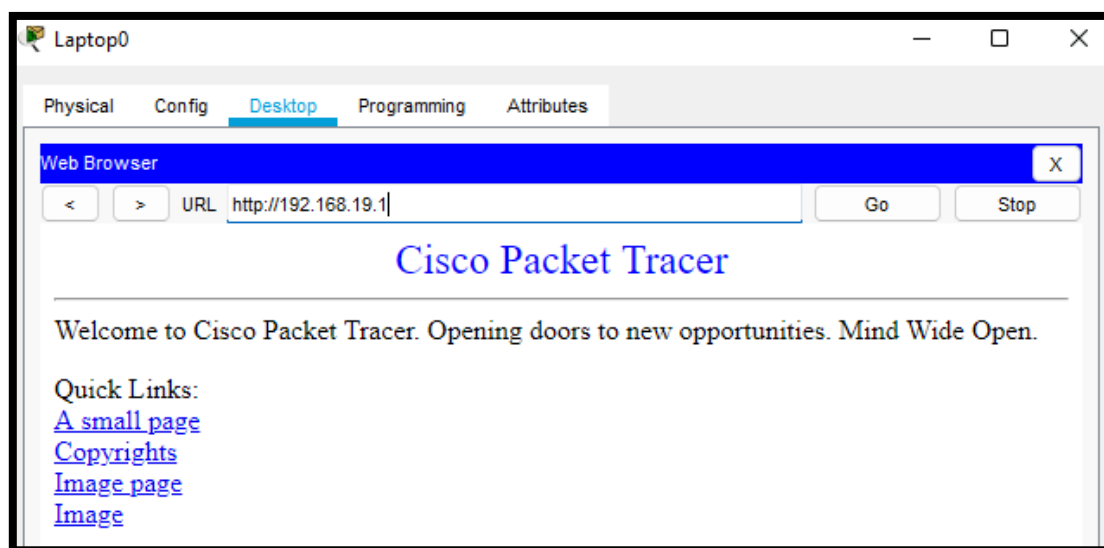
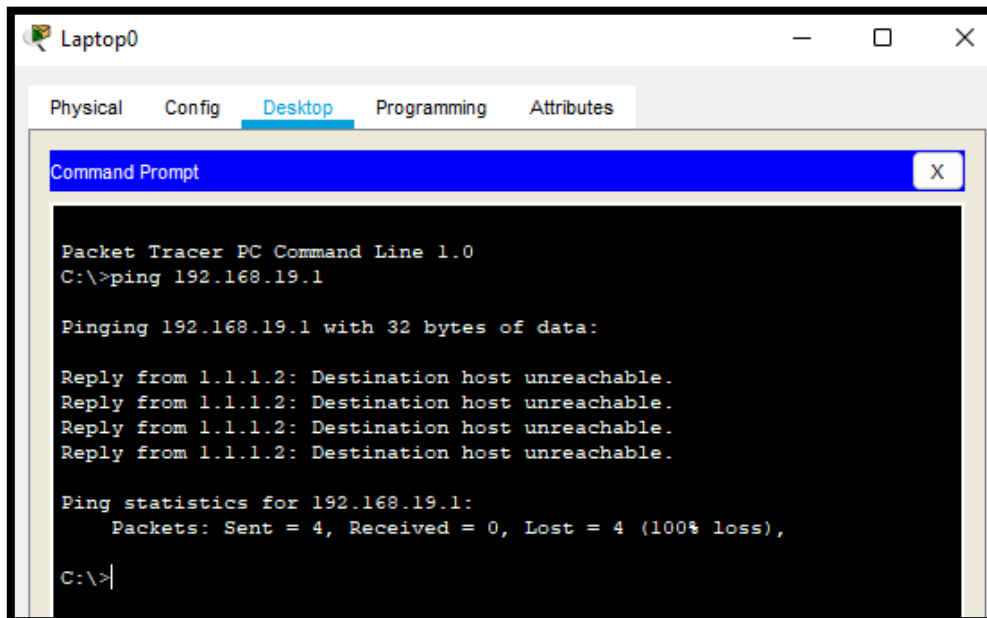


Condition 02: Blocking FTP connection from PC1 to Server0.





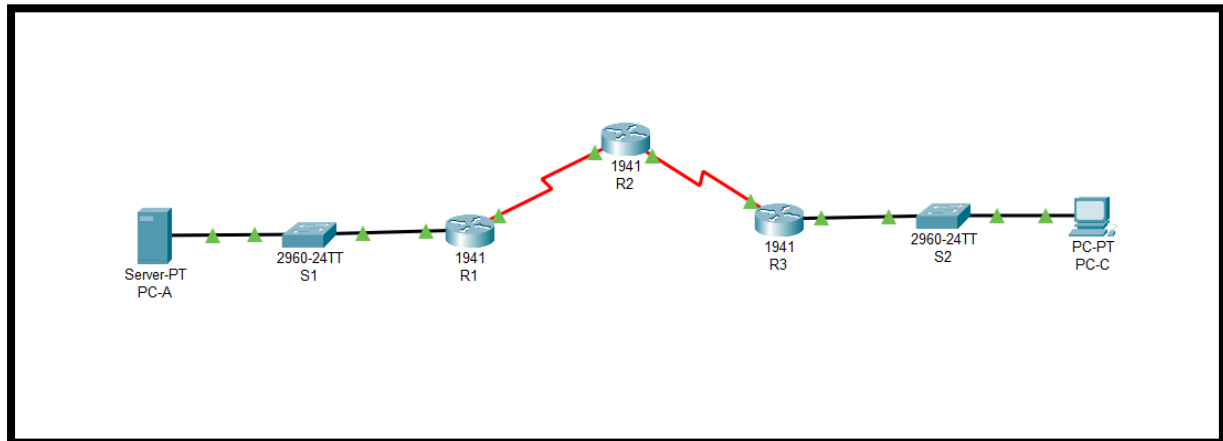
Condition 03: Blocking ICMP connection from laptop to Server0.



Practical 5

Q1} Configure IP ACL's to Mitigate Attacks

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
	Lo0	192.168.2.1	255.255.255.0	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

Objectives

- Verify connectivity among devices before firewall configuration.
- Use ACLs to ensure remote access to the routers is available only from management station PC-C.
- Configure ACLs on R1 and R3 to mitigate attacks.
- Verify ACL functionality.

All routers R1,R2,R3

User Access Verification

Password: ciscoconpa55

Router>enable

Router#conf t

Router(config)#enable password ciscoenpa55

Router(config)#exit

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#enable password ciscoenpa55
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#ip domain-name ccnasecurity.com

Router(config)#username SSHadmin privilege 15 secret ciscosshpa55

Router(config)#line vty 0 4

Router(config-line)#login local

Router(config-line)#transport input ssh

Router(config-line)#hostname R1

R1(config)#crypto key zeroize rsa

R1(config)#crypto key generate rsa

The name for the keys will be: R1.ccnasecurity.com

Choose the size of the key modulus in the range of 360 to 2048 for your

General Purpose Keys. Choosing a key modulus greater than 512 may take a few minutes.

How many bits in the modulus [512]: 1024

% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#exit

***Mar 1 0:16:40.574: %SSH-5-ENABLED: SSH 1.99 has been enabled**

R1#

```
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip domain-name ccnasecurity.com
Router(config)#username SSHadmin privilege 15 secret ciscosshpa55
Router(config)#line vty 0 4
Router(config-line)#transport input ssh
Router(config-line)#hostname R1
R1(config)#crypto key zeroize rsa
% No Signature RSA Keys found in configuration.

R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 2048 for
your
  General Purpose Keys. Choosing a key modulus greater than 512 may
take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#exit
*Mar 1 0:24:2.605: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

Repeat the above step in all the routers

R1#show ip ssh

SSH Enabled - version 1.99

Authentication timeout: 120 secs; Authentication retries: 3

```
R1#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
R1#
```

Ctrl+F6 to exit CLI focus

Copy Paste

Only on R2 to set loopback address

R2(config)# interface loopback 0

R2(config-if)#ip address 192.168.2.1 255.255.255.0

R2(config-if)#exit

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# interface loopback 0

Router(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed
state to up

Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router(config)#ip route 192.168.1.0 255.255.255.0 10.1.1.1
Router(config)#ip route 192.168.3.0 255.255.255.0 10.2.2.1
Router(config)#
```

Update routing Table

R1(config)#ip route 192.168.3.0 255.255.255.0 10.1.1.2

```
R1(config)#ip route 192.168.3.0 255.255.255.0 10.1.1.2
R1(config)#ip route 10.2.2.0 255.255.255.252 10.1.1.2
R1(config)#ip route 192.168.2.0 255.255.255.0 10.1.1.2
R1(config)#
```

R3(config)#ip route 192.168.2.0 255.255.255.0 10.2.2.2

```
R3(config)#  
R3(config)#ip route 192.168.1.0 255.255.255.0 10.2.2.2  
R3(config)#ip route 192.168.2.0 255.255.255.0 10.2.2.2  
R3(config)#ip route 10.1.1.0 255.255.255.252 10.2.2.2  
R3(config)#
```

Part 1: Verify Basic Network Connectivity

Verify network connectivity prior to configuring the IP ACLs.

Step 1: From PC-A, verify connectivity to PC-C and R2.

a. From the command prompt, ping PC-C (192.168.3.3).

ping 192.168.3.3

```
C:\>ping 192.168.3.3  
  
Pinging 192.168.3.3 with 32 bytes of data:  
  
Reply from 192.168.3.3: bytes=32 time=2ms TTL=125  
Reply from 192.168.3.3: bytes=32 time=3ms TTL=125  
Reply from 192.168.3.3: bytes=32 time=2ms TTL=125  
Reply from 192.168.3.3: bytes=32 time=2ms TTL=125  
  
Ping statistics for 192.168.3.3:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 2ms, Maximum = 3ms, Average = 2ms  
  
C:\>
```

b. From the command prompt, establish an SSH session to R2 Lo0 interface (192.168.2.1) using username SSHAdmin and password ciscosshpa55. When finished, exit the SSH session.

SERVER> ssh -l SSHAdmin 192.168.2.1

```
C:\>ssh -l SSHAdmin 192.168.2.1  
  
Password:  
  
R2#  
R2#  
R2#  
R2#  
R2#exit  
  
[Connection to 192.168.2.1 closed by foreign host]  
C:\>
```

Step 2: From PC-C, verify connectivity to PC-A and R2.

a. From the command prompt, ping PC-A (192.168.1.3).

Ping 192.168.1.3

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=3ms TTL=125
Reply from 192.168.1.3: bytes=32 time=2ms TTL=125
Reply from 192.168.1.3: bytes=32 time=2ms TTL=125
Reply from 192.168.1.3: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

b. From the command prompt, establish an SSH session to R2 Lo0 interface (192.168.2.1) using username SSHadmin and password ciscosshpa55. Close the SSH session when finished.

PC> ssh -l SSHadmin 192.168.2.1

```
C:\>ssh -l SSHadmin 192.168.2.1

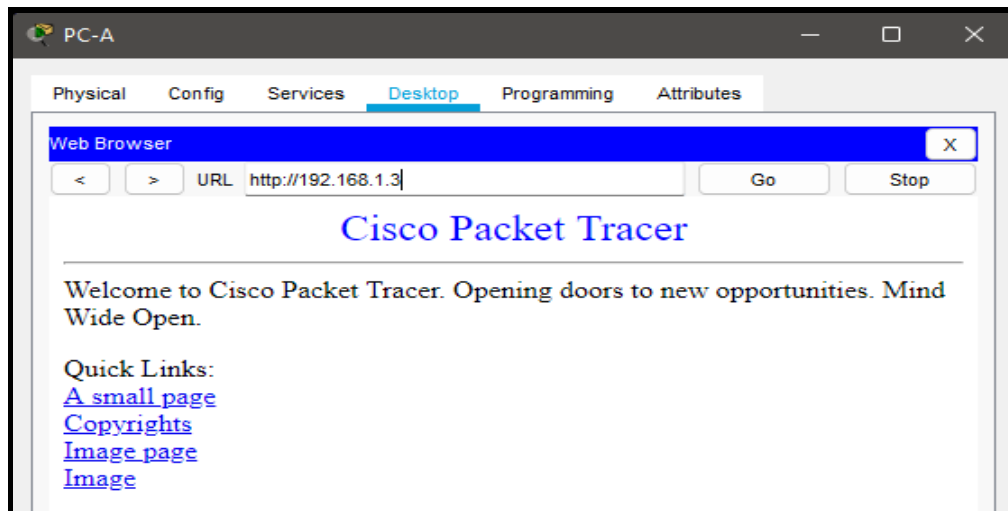
Password:

R2#
R2#exit

[Connection to 192.168.2.1 closed by foreign host]
C:\>
```

c. Open a web browser to the PC-A server (192.168.1.3) to display the web page. Close the browser when done.

http:// 192.168.1.3



Part 2: Secure Access to Routers

Step 1: Configure ACL 10 to block all remote access to the routers except from PC-C. Use the access-list command to create a numbered IP ACL on R1, R2, and R3.

- R1(config)# access-list 10 permit host 192.168.3.3
- R2(config)# access-list 10 permit host 192.168.3.3
- R3(config)# access-list 10 permit host 192.168.3.3

```
R1(config)#access-list 10 permit host 192.168.3.3
R1(config)#
```

```
R2(config)#access-list 10 permit host 192.168.3.3
R2(config)#
```

```
R3(config)#access-list 10 permit host 192.168.3.3
R3(config)#
```

Step 2: Apply ACL 10 to ingress traffic on the VTY lines. Use the access-class command to apply the access list to incoming traffic on the VTY lines.

- R1(config-line)# access-class 10 in
- R2(config-line)# access-class 10 in
- R3(config-line)# access-class 10 in

```
R1(config-line)#access-class 10 in
```

```
R2(config-line)#access-class 10 in
R2(config-line)#
```

```
R3(config-line)#access-class 10 in  
R3(config-line)#
```

Step 3: Verify exclusive access from management station PC-C.

a. Establish an SSH session to 192.168.2.1 from PC-C (should be successful).

- PC> ssh -l SSHAdmin 192.168.2.1 (PC-C...successful)

```
C:\>ssh -l SSHAdmin 192.168.2.1  
  
Password:  
  
R2#  
R2#exit  
  
[Connection to 192.168.2.1 closed by foreign host]  
C:\>
```

- PC> ssh -l SSHAdmin 192.168.2.1 (PC-A....failed)

```
C:\>ssh -l SSHAdmin 192.168.2.1  
  
% Connection refused by remote host  
C:\>
```

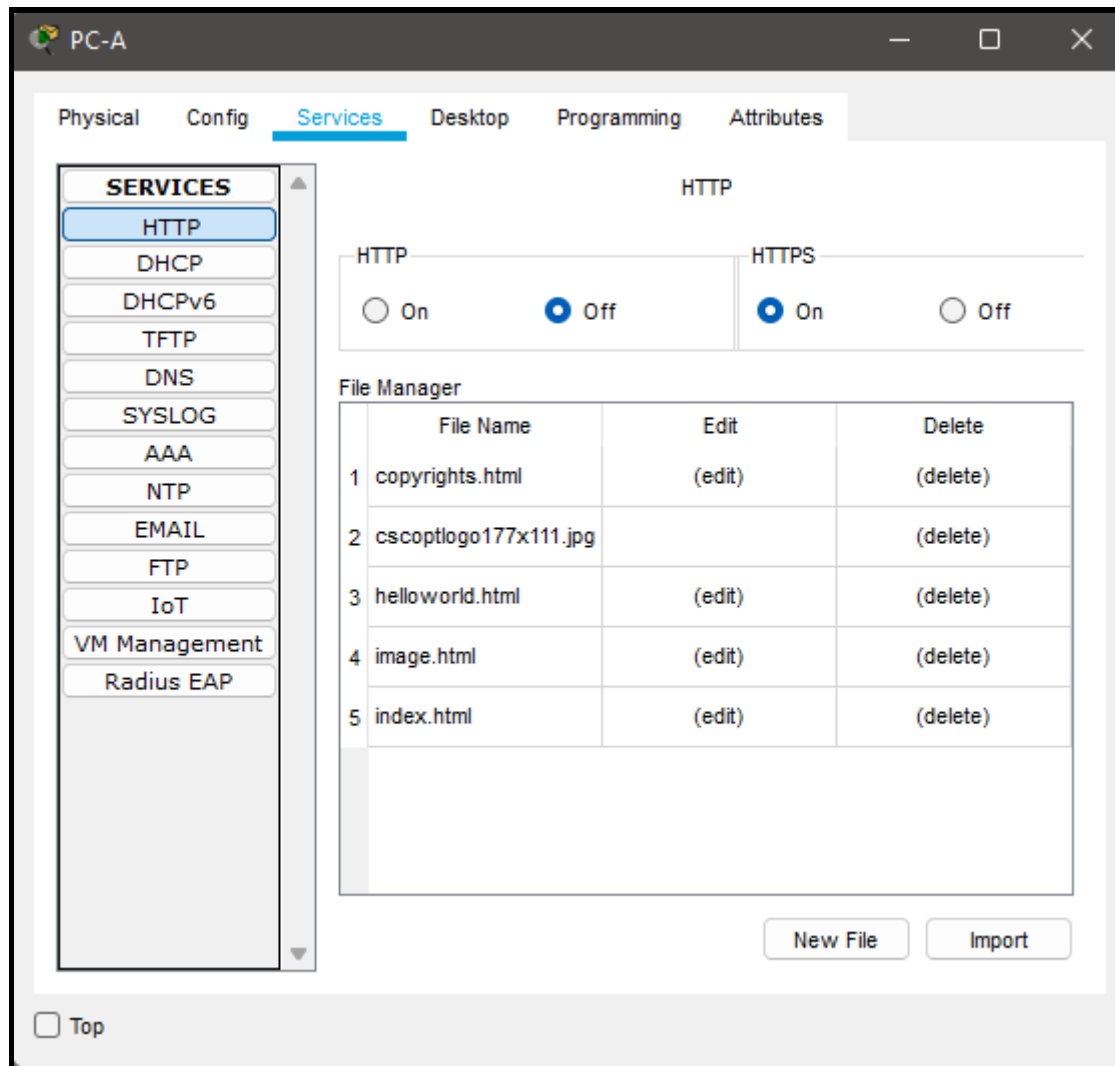
Part 3: Create a Numbered IP ACL 120 on R1

Create an IP ACL numbered 120 with the following rules:

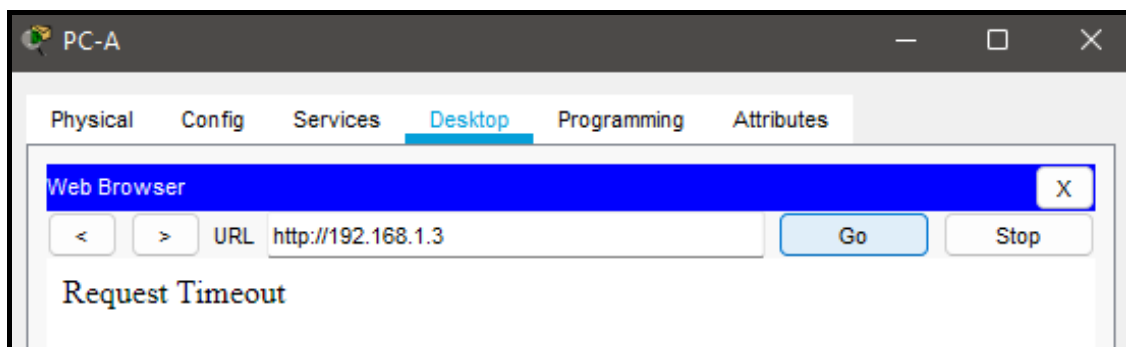
- Permit any outside host to access DNS, SMTP, and FTP services on server PC-A.
- Deny any outside host access to HTTPS services on PC-A.
- Permit PC-C to access R1 via SSH.

Step 1: Verify that PC-C can access the PC-A via HTTPS using the web browser.

Be sure to disable HTTP and enable HTTPS on server PC-A.



- <http://192.168.1.3>



Step 2: Configure ACL 120 to specifically permit and deny the specified traffic.
Use the access-list command to create a numbered IP ACL.

- R1(config)# access-list 120 permit udp any host 192.168.1.3 eq domain
- R1(config)# access-list 120 permit tcp any host 192.168.1.3 eq smtp
- R1(config)# access-list 120 permit tcp any host 192.168.1.3 eq ftp
- R1(config)# access-list 120 deny tcp any host 192.168.1.3 eq 443
- R1(config)# access-list 120 permit tcp host 192.168.3.3 host 10.1.1.1 eq 22

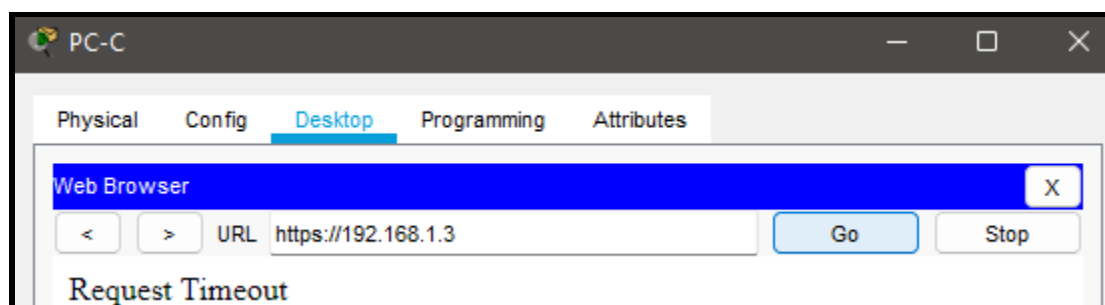
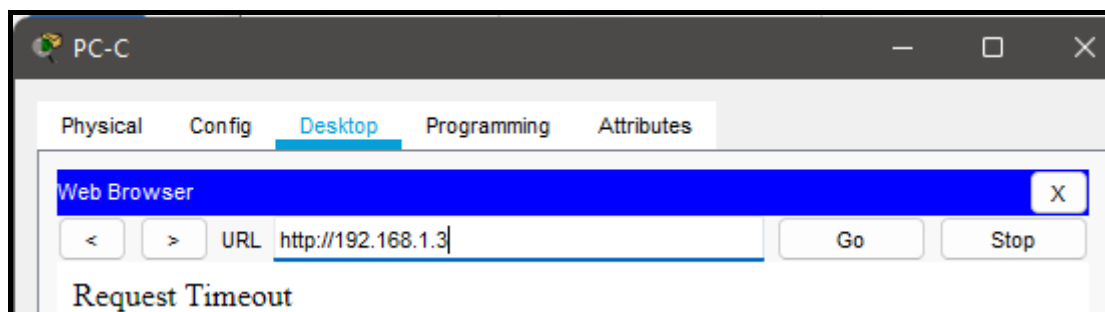
```
R1>en
Password:
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 120 permit udp any host 192.168.1.3 eq domain
R1(config)#access-list 120 permit tcp any host 192.168.1.3 eq smtp
R1(config)#access-list 120 permit tcp any host 192.168.1.3 eq ftp
R1(config)#access-list 120 deny tcp any host 192.168.1.3 eq 443
R1(config)#access-list 120 permit tcp host 192.168.3.3 host 10.1.1.1
eq 22
R1(config)#
```

Step 3: Apply the ACL to interface S0/0/0. Use the ip access-group command to apply the access list to incoming traffic on interface S0/0/0.

- R1(config)# interface s0/0/0
- R1(config-if)# ip access-group 120 in

```
R1(config)#int se0/0/0
R1(config-if)#ip access-group 120 in
R1(config-if)#exit
R1(config)#
```

- <http://192.168.1.3>



Part 4: Modify an Existing ACL on R1

Step 1: Verify that PC-A cannot successfully ping the loopback interface on R2.

- ping 192.168.2.1


```
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Step 2: Make any necessary changes to ACL 120 to permit and deny the specified traffic. Use the access-list command to create a numbered IP ACL.

- R1(config)# access-list 120 permit icmp any any echo-reply
- R1(config)# access-list 120 permit icmp any any unreachable
- R1(config)# access-list 120 deny icmp any any
- R1(config)# access-list 120 permit ip any any

```
R1(config)#access-list 120 permit icmp any any echo-reply
R1(config)#access-list 120 permit icmp any any unreachable
R1(config)#access-list 120 deny icmp any any
R1(config)#access-list 120 permit ip any any
R1(config)#
```

Step 3: Verify that PC-A can successfully ping the loopback interface on R2.

- ping 192.168.2.1

```
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=2ms TTL=254
Reply from 192.168.2.1: bytes=32 time=1ms TTL=254
Reply from 192.168.2.1: bytes=32 time=1ms TTL=254
Reply from 192.168.2.1: bytes=32 time=2ms TTL=254

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>
```

Part 5 : Create a Numbered IP ACL 110 on R3

Deny all outbound packets with source address outside the range of internal IP addresses on R3

Step 1: Configure ACL 110 to permit only traffic from the inside network. Use the access-list command to create a numbered IP ACL.

- R3(config)# access-list 110 permit ip 192.168.3.0 0.0.0.255 any

```
R3(config)#access-list 110 permit ip 192.168.3.0 0.0.0.255 any
R3(config)#
```

Step 2: Apply the ACL to interface G0/1.

Use the ip access-group command to apply the access list to incoming traffic on interface G0/1.

- R3(config)# interface f0/1
- R3(config-if)# ip access-group 110 in

```
R3(config)#int G0/0
R3(config-if)#ip access-group 110 in
R3(config-if)#
```

Part 6: Create a Numbered IP ACL 100 on R3

On R3, block all packets containing the source IP address from the following pool of addresses: any RFC 1918, private addresses, 127.0.0.0/8, and any IP multicast address.

Since PC-C is being used for remote administration, permit SSH traffic from the 10.0.0.0/8 network to return to the host PC-C.

Step 1: Configure ACL 100 to block all specified traffic from the outside network.

You should also block traffic sourced from your own internal address space if it is not an RFC 1918 address. In this activity, your internal address space is part of the private address space specified in RFC 1918. Use the access-list command to create a numbered IP ACL

- R3(config)# access-list 100 permit tcp 10.0.0.0 0.255.255.255 eq 22 host 192.168.3.3
- R3(config)# access-list 100 deny ip 10.0.0.0 0.255.255.255 any
- R3(config)# access-list 100 deny ip 172.16.0.0 0.15.255.255 any
- R3(config)# access-list 100 deny ip 192.168.0.0 0.0.255.255 any
- R3(config)# access-list 100 deny ip 127.0.0.0 0.255.255.255 any
- R3(config)# access-list 100 deny ip 224.0.0.0 15.255.255.255 any
- R3(config)# access-list 100 permit ip any any

```
R3(config)#access-list 100 permit tcp 10.0.0.0 0.255.255.255 eq 22
host 192.168.3.3
R3(config)#access-list 100 deny ip 10.0.0.0 0.255.255.255 any
R3(config)#access-list 100 deny ip 172.16.0.0 0.15.255.255 any
R3(config)#access-list 100 deny ip 192.168.0.0 0.0.255.255 any
R3(config)#access-list 100 deny ip 127.0.0.0 0.255.255.255 any
R3(config)#access-list 100 deny ip 224.0.0.0 15.255.255.255 any
R3(config)#access-list 100 permit ip any any
R3(config)#
```

Step 2: Apply the ACL to interface Serial 0/0/1. Use the ip access-group command to apply the access list to incoming traffic on interface Serial 0/0/1.

- R3(config)# interface s2/0
- R3(config-if)# ip access-group 100 in

```
R3(config)#int se0/0/1
R3(config-if)#ip access-group 100 in
R3(config-if)#
```

Step 3: Confirm that the specified traffic entering interface Serial 0/0/1 is handled correctly.

a. From the PC-C command prompt, ping the PC-A server. The ICMP echo replies are blocked by the ACL since they are sourced from the 192.168.0.0/16 address space.

b. Establish an SSH session to 192.168.2.1 from PC-C (should be successful).

- Ping 192.168.1.3(Fails)

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

- ssh -l SSHadmin 192.168.2.1

```
C:\>ssh -l SSHadmin 192.168.2.1

Password:

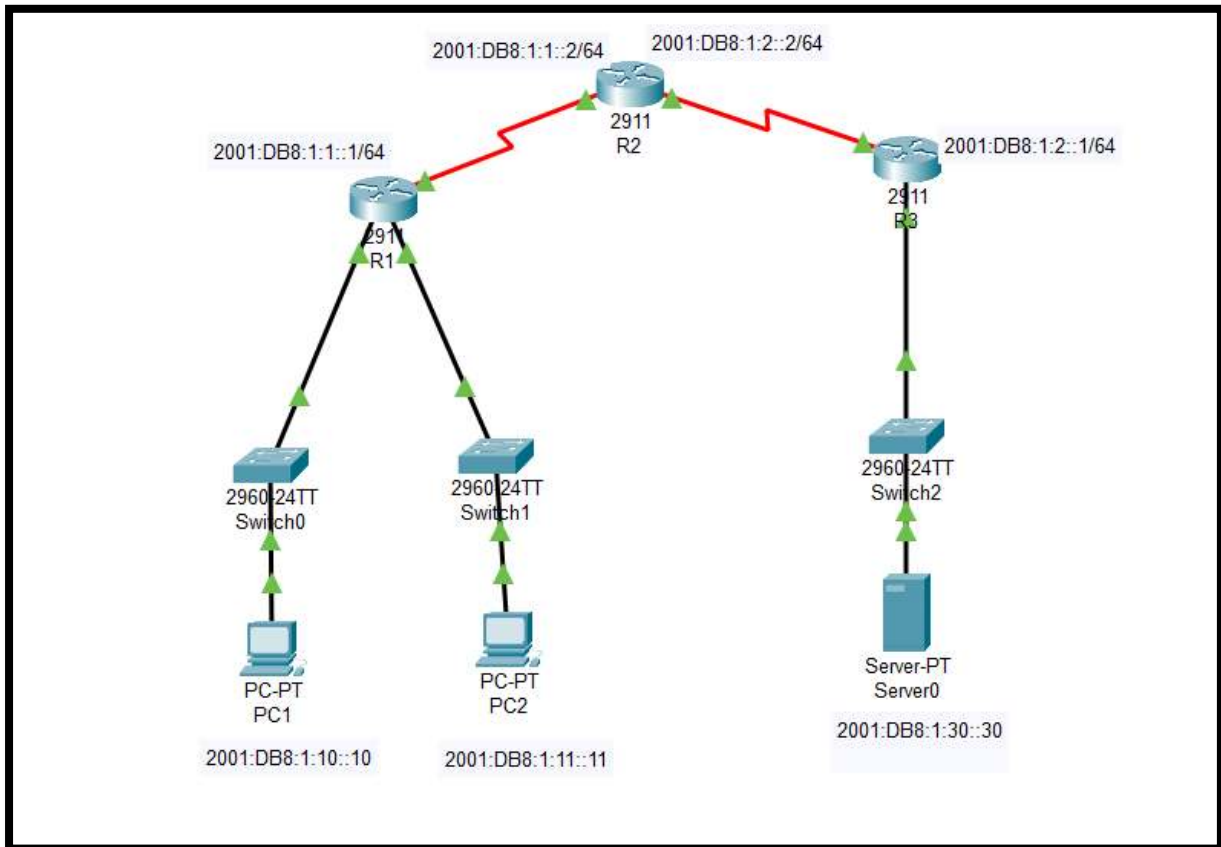
R2#
R2#exit

[Connection to 192.168.2.1 closed by foreign host]
C:\>
```

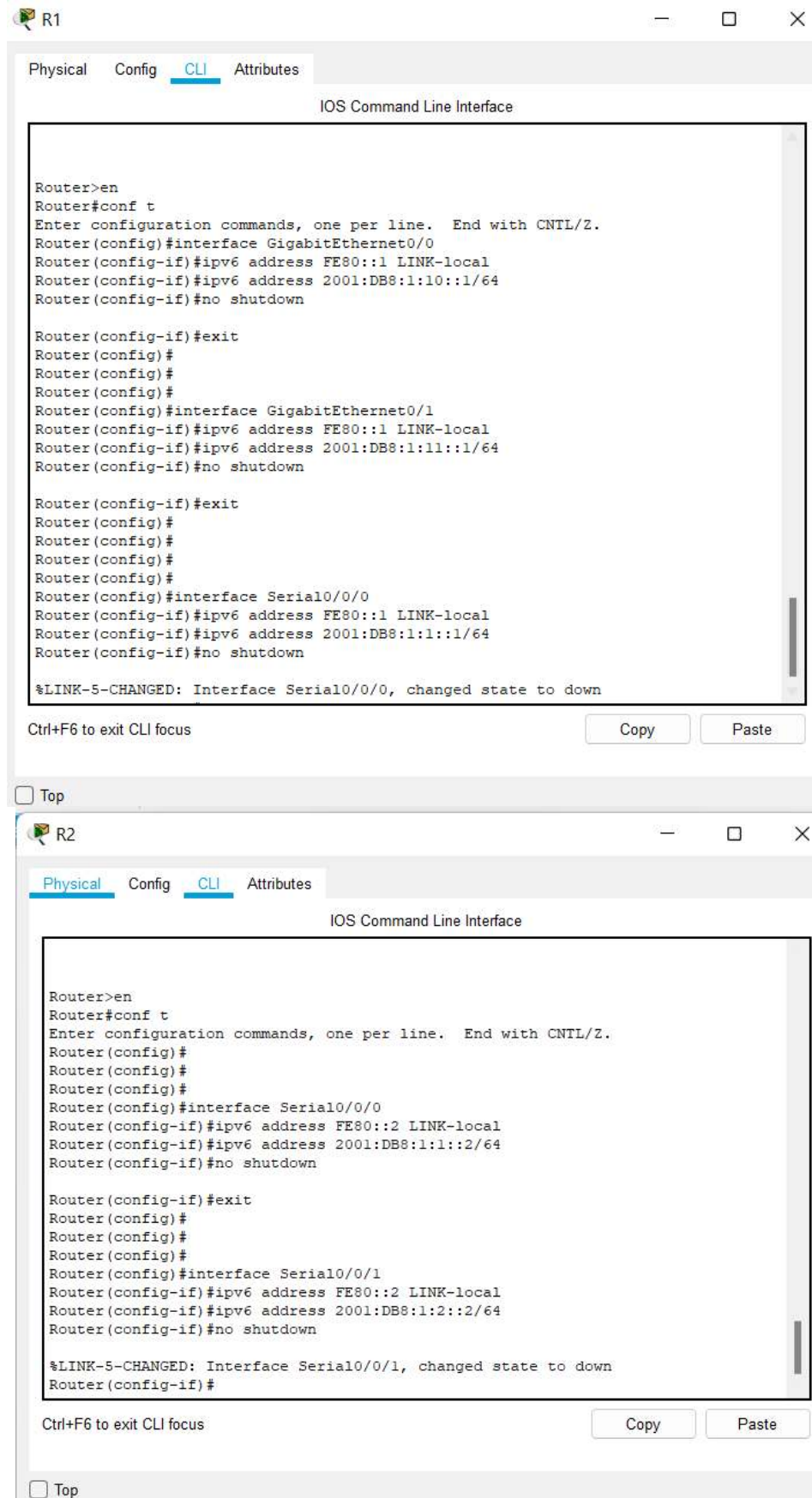
Practical 6

Q1} Configuring IPv6 ACLs

Topology



Addressing Table

Part1: Configure, Apply, and Verify an IPv6 ACL

R1

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ipv6 address FE80::1 LINK-local
Router(config-if)#ipv6 address 2001:DB8:1:10::1/64
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ipv6 address FE80::1 LINK-local
Router(config-if)#ipv6 address 2001:DB8:1:11::1/64
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#interface Serial0/0/0
Router(config-if)#ipv6 address FE80::1 LINK-local
Router(config-if)#ipv6 address 2001:DB8:1:1::1/64
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
```

Ctrl+F6 to exit CLI focus

Copy Paste

☐ Top

R2

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#interface Serial0/0/0
Router(config-if)#ipv6 address FE80::2 LINK-local
Router(config-if)#ipv6 address 2001:DB8:1:1::2/64
Router(config-if)#no shutdown

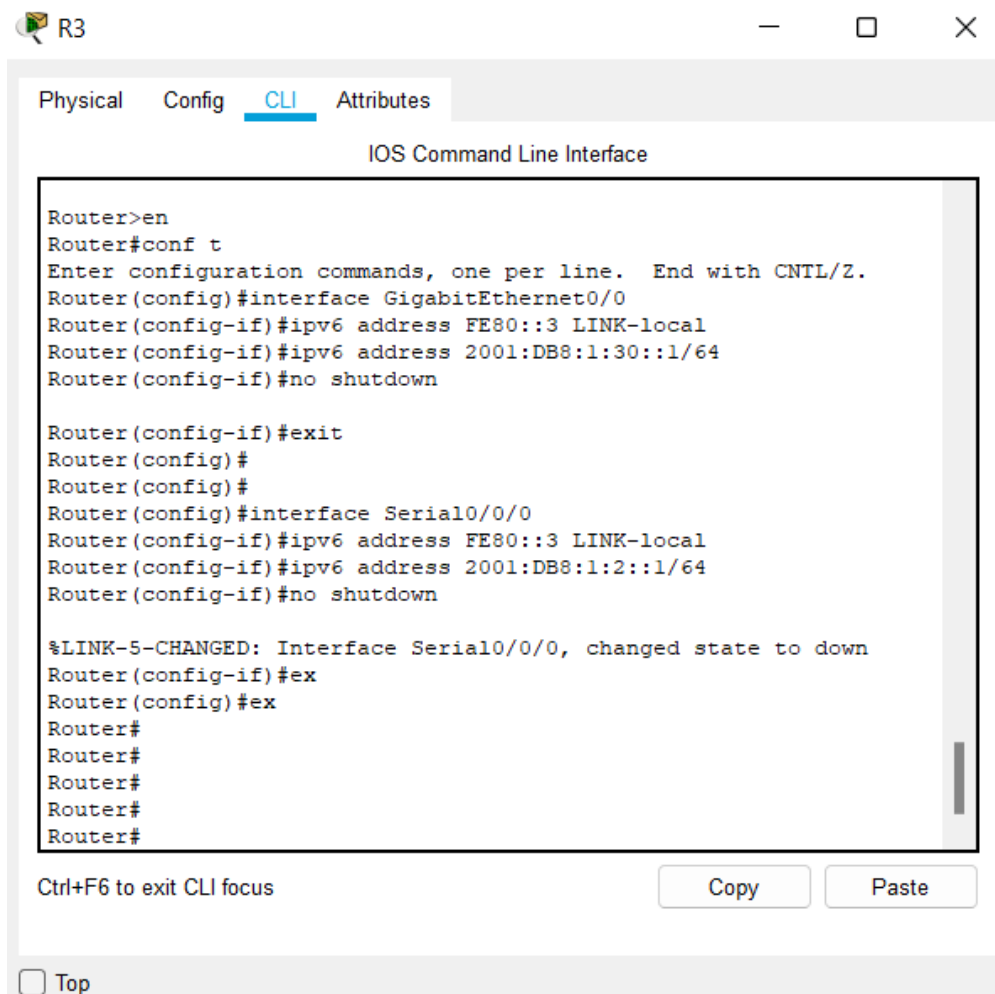
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#interface Serial0/0/1
Router(config-if)#ipv6 address FE80::2 LINK-local
Router(config-if)#ipv6 address 2001:DB8:1:2::2/64
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
Router(config-if)#
```

Ctrl+F6 to exit CLI focus

Copy Paste

☐ Top



The screenshot shows a window titled 'R3' with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the 'IOS Command Line Interface'. The terminal output shows the following commands and responses:

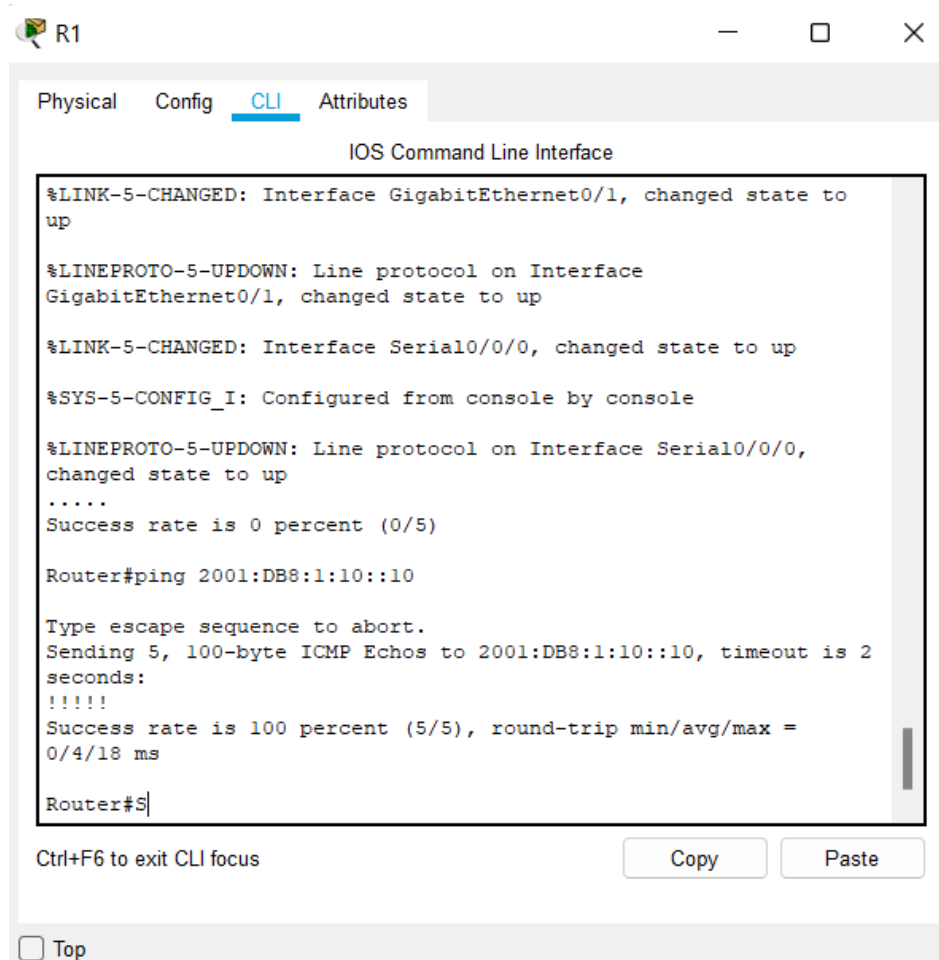
```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ipv6 address FE80::3 LINK-local
Router(config-if)#ipv6 address 2001:DB8:1:30::1/64
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#interface Serial0/0/0
Router(config-if)#ipv6 address FE80::3 LINK-local
Router(config-if)#ipv6 address 2001:DB8:1:2::1/64
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Router(config-if)#ex
Router(config)#ex
Router#
Router#
Router#
Router#
Router#
```

At the bottom of the CLI window, there is a text label 'Ctrl+F6 to exit CLI focus' and two buttons: 'Copy' and 'Paste'. Below the CLI window, there is a checkbox labeled 'Top'.

Ping output:**R1:****Router>en****Router#ping 2001:DB8:1:10::10**



The screenshot shows a Cisco IOS CLI window for router R1. The 'CLI' tab is selected. The terminal output displays the following messages:

```
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface Serial10/0/0, changed state to up
%SYS-5-CONFIG_I: Configured from console by console
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial10/0/0, changed state to up
.....
Success rate is 0 percent (0/5)

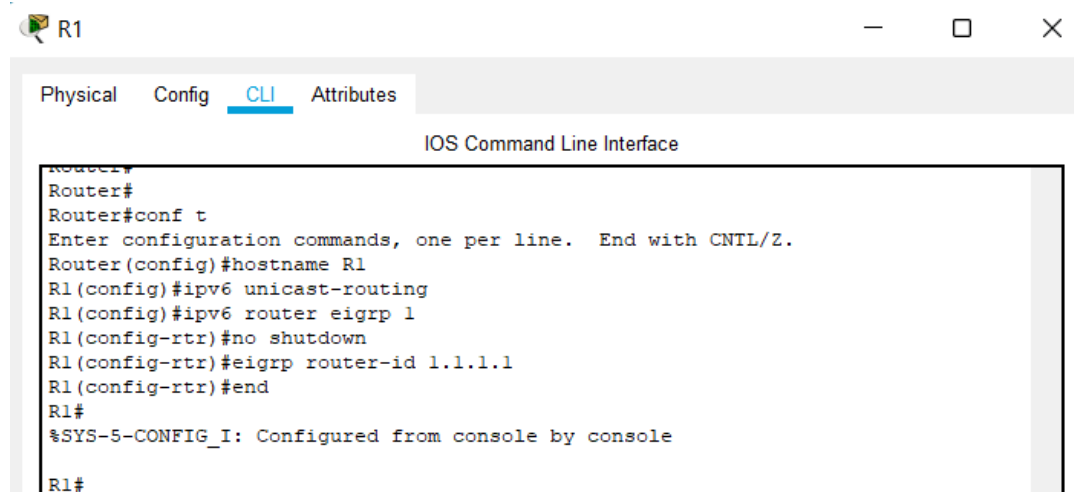
Router#ping 2001:DB8:1:10::10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:1:10::10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/4/18 ms

Router#S
```

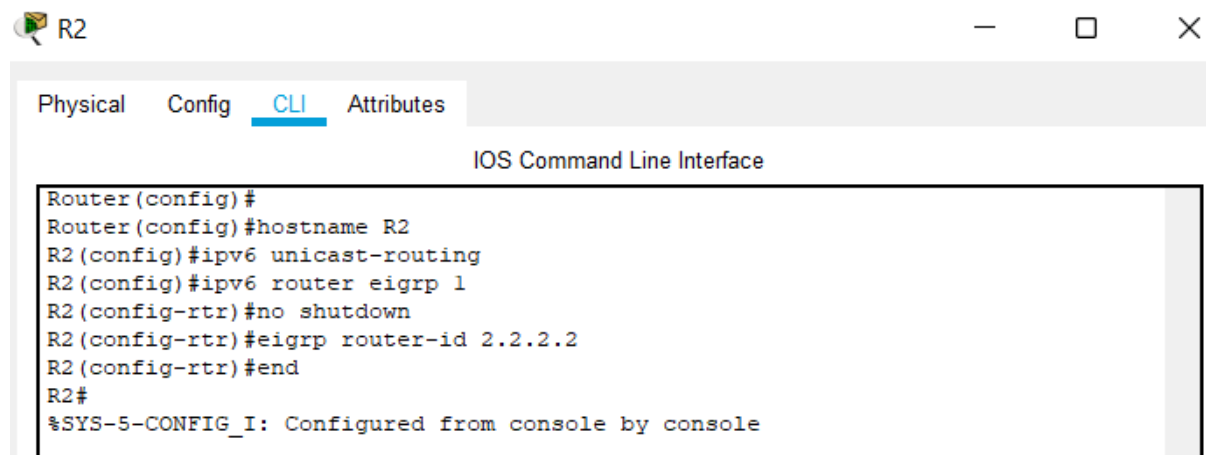
Below the terminal window, there are buttons for 'Copy' and 'Paste', and a 'Top' link.

Set router Id:**R1:****Router#****Router#configure terminal****Enter configuration commands, one per line. End with CNTL/Z.****Router(config)#hostname R1****R1(config)#ipv6 unicast-routing****R1(config)#ipv6 router eigrp 1****R1(config-rtr)#no shutdown****R1(config-rtr)#eigrp router-id 1.1.1.1****R1(config-rtr)#****R1(config-rtr)#end**

A screenshot of a network configuration window for router R1. The window has tabs for Physical, Config, CLI (selected), and Attributes. The CLI tab shows the IOS Command Line Interface with the following commands entered: Router#, Router#conf t, Enter configuration commands, one per line. End with CNTL/Z., Router(config)#hostname R1, R1(config)#ipv6 unicast-routing, R1(config)#ipv6 router eigrp 1, R1(config-rtr)#no shutdown, R1(config-rtr)#eigrp router-id 1.1.1.1, R1(config-rtr)#end, R1#, and %SYS-5-CONFIG_I: Configured from console by console. The prompt R1# is visible at the bottom.

```
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#ipv6 unicast-routing
R1(config)#ipv6 router eigrp 1
R1(config-rtr)#no shutdown
R1(config-rtr)#eigrp router-id 1.1.1.1
R1(config-rtr)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
R1#
```

R2(config)#ipv6 unicast-routing
R2(config)#ipv6 router eigrp 1
R2(config-rtr)#
R2(config-rtr)#no shutdown
R2(config-rtr)#eigrp router-id 2.2.2.2
R2(config-rtr)#

A screenshot of a network configuration window for router R2. The window has tabs for Physical, Config, CLI (selected), and Attributes. The CLI tab shows the IOS Command Line Interface with the following commands entered: Router(config)#, Router(config)#hostname R2, R2(config)#ipv6 unicast-routing, R2(config)#ipv6 router eigrp 1, R2(config-rtr)#no shutdown, R2(config-rtr)#eigrp router-id 2.2.2.2, R2(config-rtr)#end, R2#, and %SYS-5-CONFIG_I: Configured from console by console. The prompt R2# is visible at the bottom.

```
Router(config)#
Router(config)#hostname R2
R2(config)#ipv6 unicast-routing
R2(config)#ipv6 router eigrp 1
R2(config-rtr)#no shutdown
R2(config-rtr)#eigrp router-id 2.2.2.2
R2(config-rtr)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
R2#
```

R3:
R3(config)#ipv6 unicast-routing
R3(config)#ipv6 router eigrp 1
R3(config-rtr)#no shutdown
R3(config-rtr)#eigrp router-id 3.3.3.3
R3(config-rtr)#



```
Physical  Config  CLI  Attributes
IOS Command Line Interface
Router#
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#ipv6 unicast-routing
R3(config)#ipv6 router eigrp 1
R3(config-rtr)#no shutdown
R3(config-rtr)#eigrp router-id 3.3.3.3
R3(config-rtr)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console
```

Inform it to all the interface:

R1:

R1(config)#interface GigabitEthernet0/0

R1(config-if)#

%SYS-5-CONFIG_I: Configured from console by console

R1(config-if)#ipv6 eigrp 1

R1(config-if)#ex

R1(config)#interface GigabitEthernet0/1

R1(config-if)#ipv6 eigrp 1

R1(config-if)#ex

R1(config)#interface Serial0/0/0

R1(config-if)#ipv6 eigrp 1

R1(config-if)#ex



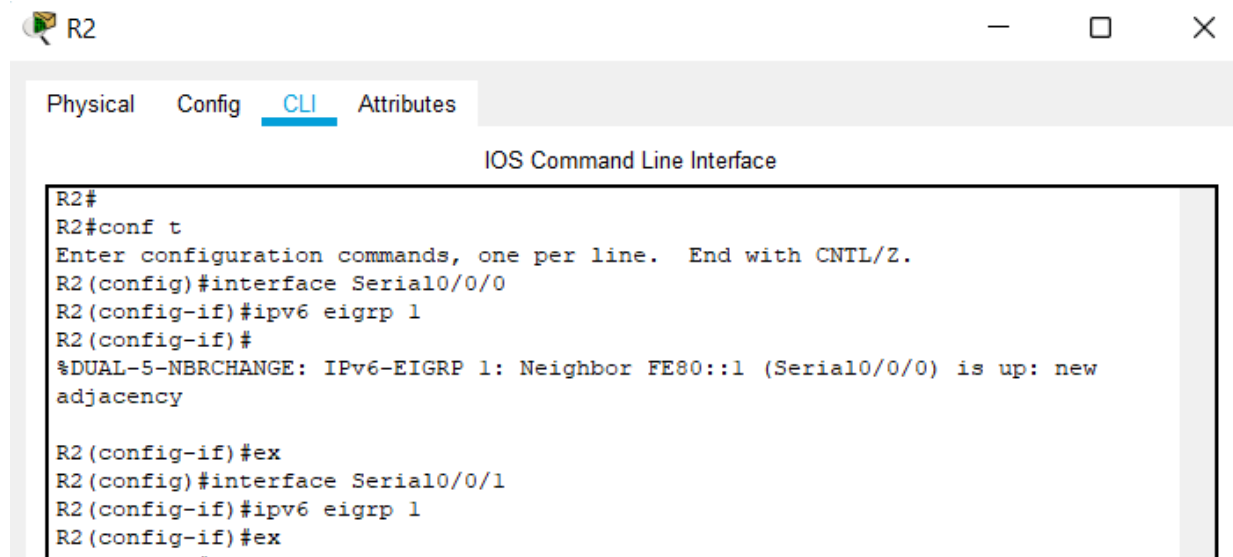
```
Physical  Config  CLI  Attributes
IOS Command Line Interface
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface GigabitEthernet0/0
R1(config-if)#ipv6 eigrp 1
R1(config-if)#ex
R1(config)#interface GigabitEthernet0/1
R1(config-if)#ipv6 eigrp 1
R1(config-if)#ex
R1(config)#interface Serial0/0/0
R1(config-if)#ipv6 eigrp 1
R1(config-if)#ex
R1(config-if)#
```

R2(config)#interface Serial0/0/0

R2(config-if)#ipv6 eigrp 1

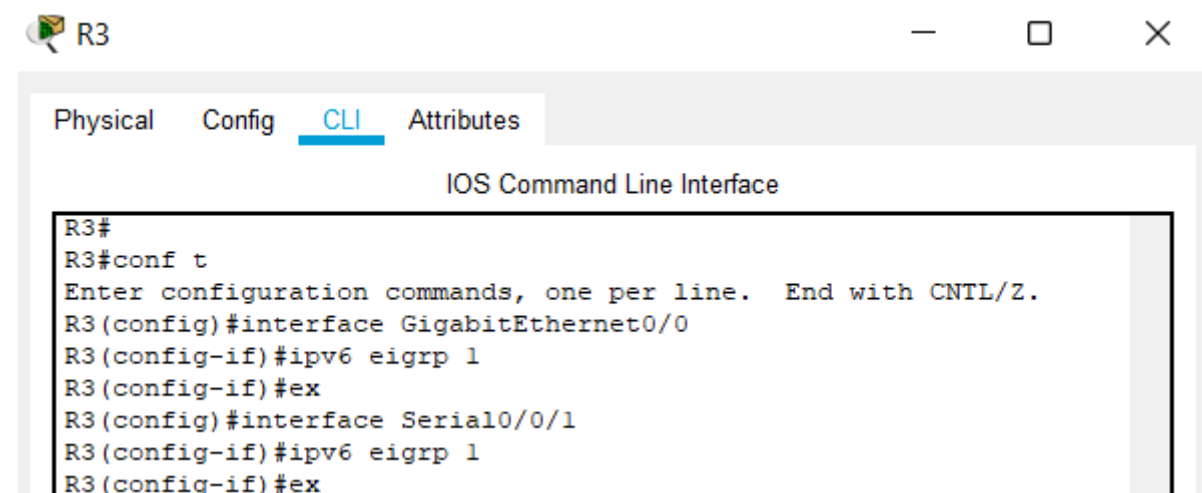
R2(config-if)#ex

R2(config)#interface Serial0/0/1

R2(config-if)#ipv6 eigrp 1**R2(config-if)#ex**

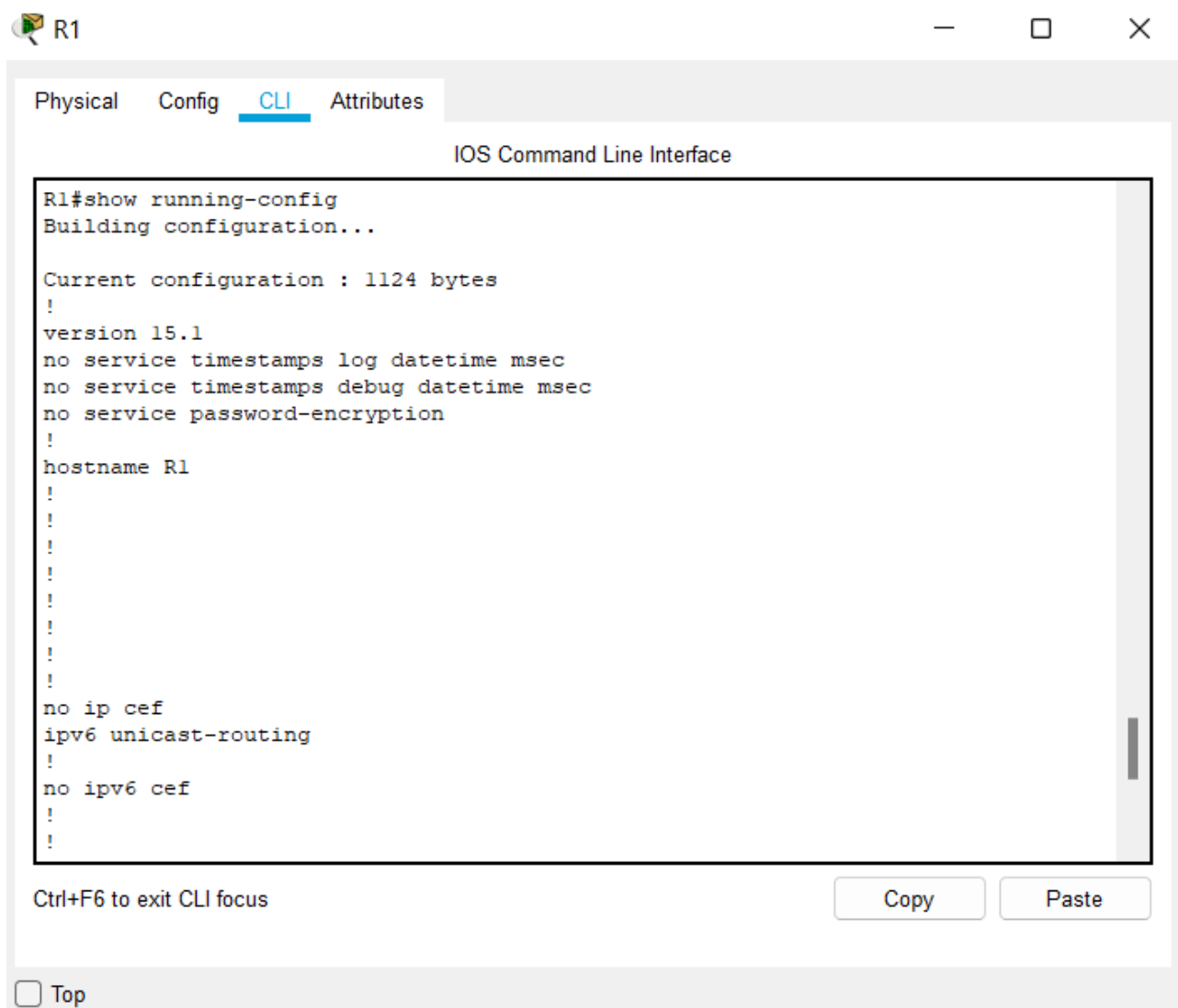
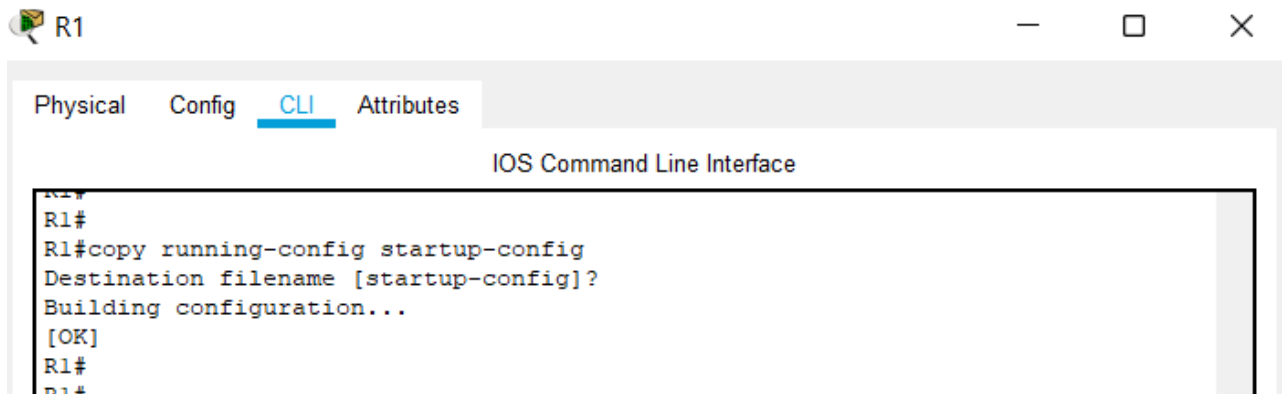
The screenshot shows a terminal window titled 'R2' with tabs for Physical, Config, CLI (selected), and Attributes. The CLI tab displays the following commands and output:

```
R2#
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#interface Serial0/0/0
R2(config-if)#ipv6 eigrp 1
R2(config-if)#
%DUAL-5-NBRCHANGE: IPv6-EIGRP 1: Neighbor FE80::1 (Serial0/0/0) is up: new adjacency
R2(config-if)#ex
R2(config)#interface Serial0/0/1
R2(config-if)#ipv6 eigrp 1
R2(config-if)#ex
```

R3(config)#interface GigabitEthernet0/0**R3(config-if)#ipv6 eigrp 1****R3(config-if)#ex****R3(config)#interface Serial0/0/1****R3(config-if)#ipv6 eigrp 1****R3(config-if)#ex**

The screenshot shows a terminal window titled 'R3' with tabs for Physical, Config, CLI (selected), and Attributes. The CLI tab displays the following commands:

```
R3#
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface GigabitEthernet0/0
R3(config-if)#ipv6 eigrp 1
R3(config-if)#ex
R3(config)#interface Serial0/0/1
R3(config-if)#ipv6 eigrp 1
R3(config-if)#ex
```



Part2: Configure, Apply, and Verify a Second IPv6 ACL

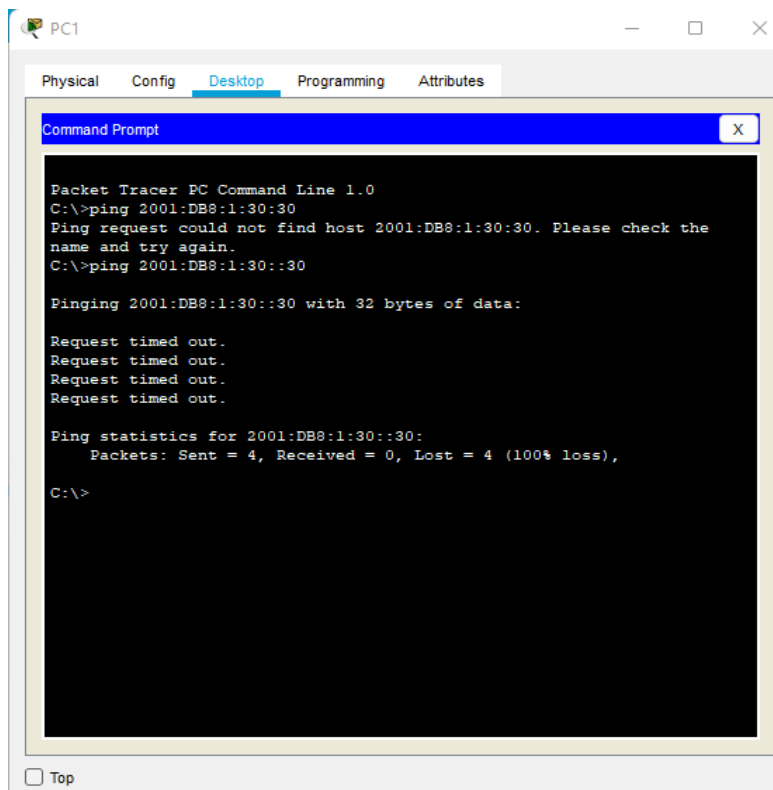
```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ipv6 access-list BLOCK_HTTP
R1(config-ipv6-acl)#deny tcp any host 2001:DB8:1:30::30 eq www
R1(config-ipv6-acl)#deny tcp any host 2001:DB8:1:30::30 eq 443
R1(config-ipv6-acl)#permit ipv6 any any
R1(config-ipv6-acl)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

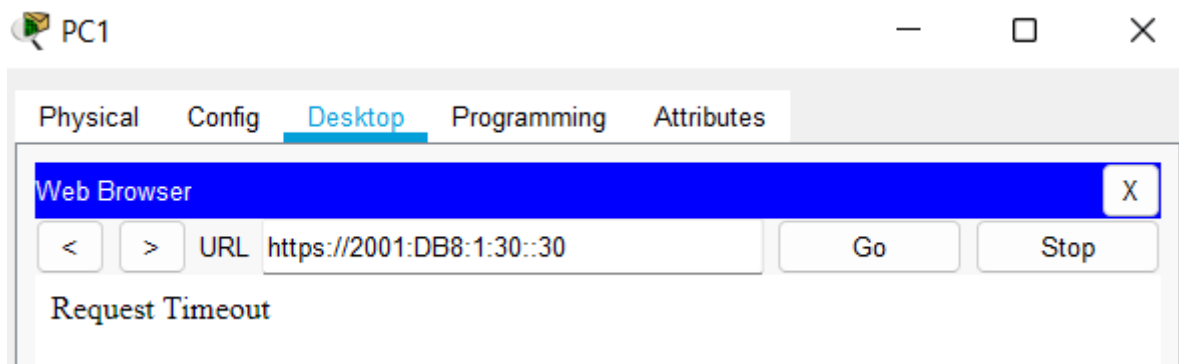
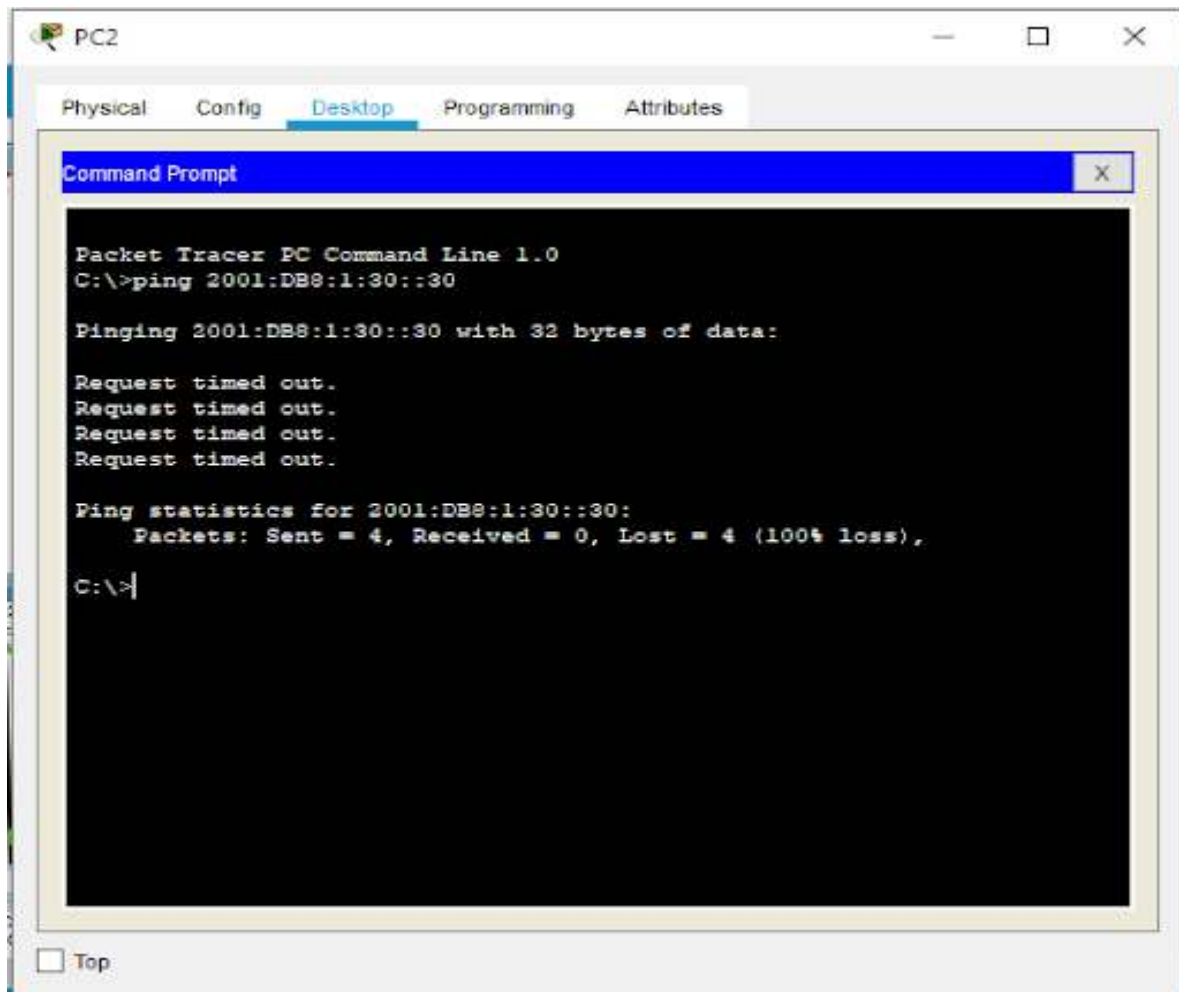
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface GigabitEthernet0/1
R1(config-if)#ipv6 traffic-filter BLOCK_HTTP in
R1(config-if)#
```

```
R3(config)#ipv6 access-list BLOCK_ICMP
R3(config-ipv6-acl)#deny icmp any any
R3(config-ipv6-acl)#permit ipv6 any any
R3(config-ipv6-acl)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ipv6 traffic-filter BLOCK_ICMP out
R3(config-if)#
^
% Invalid input detected at '^' marker.

R3(config)#interface GigabitEthernet0/0
R3(config-if)#ipv6 traffic-filter BLOCK_ICMP out
R3(config-if)#
```

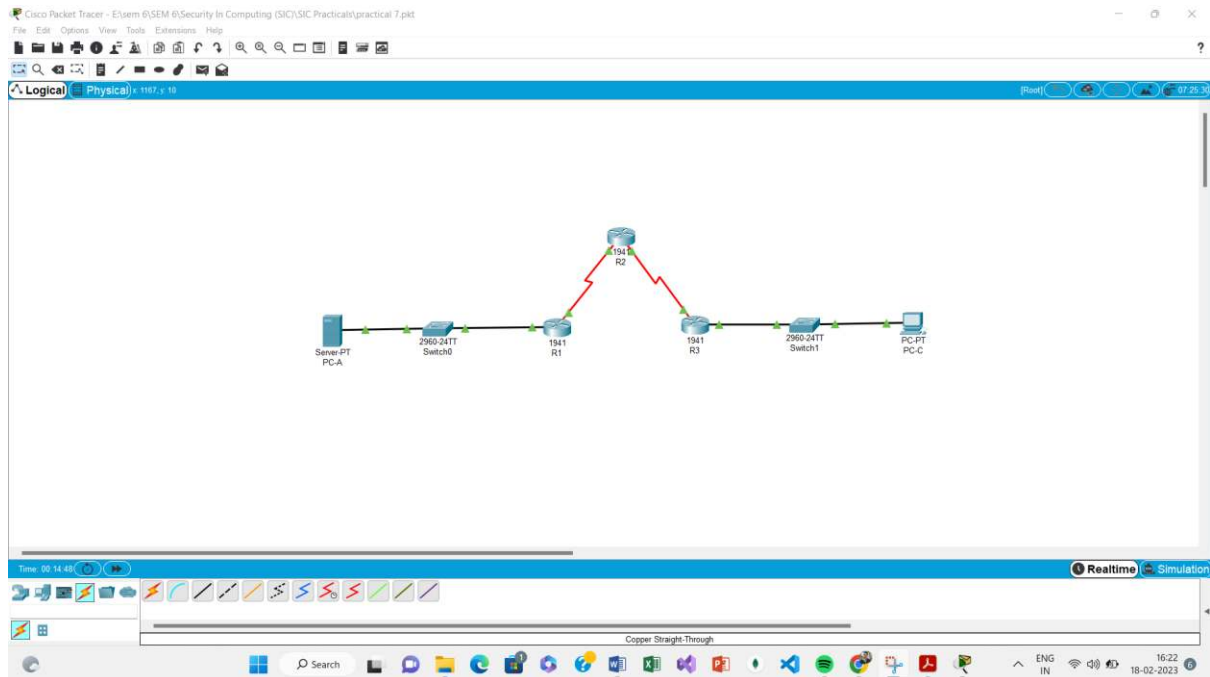




Practical 7

Q1} Configuring a Zone-Based Policy Firewall (ZPF)

Topology



Addressing table

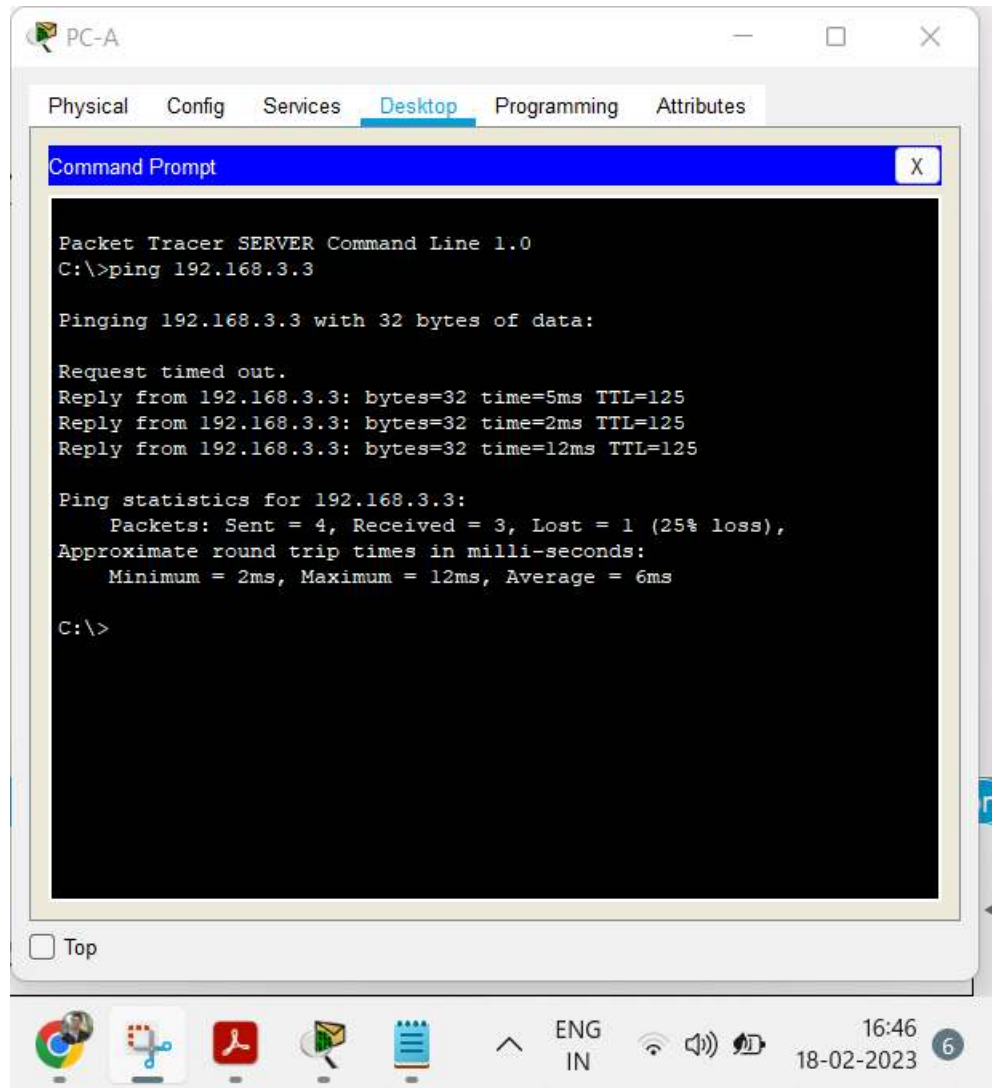
Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

Objectives

- Verify connectivity among devices before firewall configuration.
- Configure a zone-based policy (ZPF) firewall on R3.
- Verify ZPF firewall functionality using ping, SSH, and a web browser.

Part 1: Verify Basic Network Connectivity

Step 1: From the PC-A command prompt, ping PC-C at 192.168.3.3.



Step 2: Access R2 using SSH.

- a. From the PC-C command prompt, SSH to the S0/0/1 interface on R2 at 10.2.2.2. Use the username Admin and password Adminpa55 to log in.
PC> ssh -l Admin 10.2.2.2
- b. Exit the SSH session.

Step 3: From PC-C, open a web browser to the PC-A server.

- a. Click the Desktop tab and then click the Web Browser application. Enter the PC-A IP address 192.168.1.3 as the URL. The Packet Tracer welcome page from the web server should be displayed.
- b. Close the browser on PC-C.

Part 2: Create the Firewall Zones on R3**Step 1: Enable the Security Technology package.**

- a. On R3, issue the show version command to view the Technology Package license information.
- b. If the Security Technology package has not been enabled, use the following command to enable the package.

R3(config)# license boot module c1900 technology-package securityk9
- c. Accept the end-user license agreement.
- d. Save the running-config and reload the router to enable the security license.
- e. Verify that the Security Technology package has been enabled by using the show version command.

Step 2: Create an internal zone. Use the zone security command to create a zone named IN-ZONE.

- a. R3(config)# zone security IN-ZONE
- b. R3(config-sec-zone) exit
- c. Step 3: Create an external zone.
- d. Use the zone security command to create a zone named OUT-ZONE.
- e. R3(config-sec-zone)# zone security OUT-ZONE
- f. R3(config-seczone)# exit

Part 3: Identify Traffic Using a Class-Map**Step 1: Create an ACL that defines internal traffic.**

Use the access-list command to create extended ACL 101 to permit all IP protocols from the 192.168.3.0/24 source network to any destination.

R3(config)# access-list 101 permit ip 192.168.3.0 0.0.0.255 any

Step 2: Create a class map referencing the internal traffic ACL.

Use the class-map type inspect command with the match-all option to create a class map named INNETCLASS-MAP. Use the match access-group command to match ACL 101.

- a. R3(config)# class-map type inspect match-all IN-NET-CLASS-MAP
- b. R3(config-cmap)# match access-group 101
- c. R3(config-cmap)# exit

Part 4: Specify Firewall Policies

Step 1: Create a policy map to determine what to do with matched traffic.

Use the policy-map type inspect command and create a policy map named IN-2-OUT-PMAP.

R3(config)# policy-map type inspect IN-2-OUT-PMAP

Step 2: Specify a class type of inspect and reference class map IN-NET-CLASS-MAP.

R3(config-pmap)# class type inspect IN-NET-CLASS-MAP

Step 3: Specify the action of inspect for this policy map. The use of the inspect command invokes context-based access control (other options include pass and drop).

- a. R3(config-pmap-c)# inspect
- b. R3(config-pmap-c)# exit
- c. R3(config-pmap)# exit

Part 5: Apply Firewall Policies

Step 1: Create a pair of zones. Using the zone-pair security command, create a zone pair named IN-2-OUT-ZPAIR. Specify the source and destination zones that were created in Task 1.

R3(config)# zone-pair security IN-2-OUT-ZPAIR source IN-ZONE destination OUTZONE

Step 2: Specify the policy map for handling the traffic between the two zones.

- a. R3(config-sec-zone-pair)# service-policy type inspect IN-2-OUT-PMAP
- b. R3(config-sec-zone-pair)# exit

Step 3: Assign interfaces to the appropriate security zones.

Use the zone-member security command in interface configuration mode to assign G0/1 to IN-ZONE and S0/0/1 to OUT-ZONE.

- a. R3(config)# interface g0/1
- b. R3(config-if)# zone-member security IN-ZONE
- c. R3(config-if)# exit
- d. R3(config)# interface s0/0/1
- e. R3(config-if)# zone-member security OUT-ZONE
- f. R3(config-if)# exit

Step 4: Copy the running configuration to the startup configuration.

R3(config)#copy running-config startup-config

Part 6: Test Firewall Functionality from IN-ZONE to OUT-ZONE

Step 1: From internal PC-C, ping the external PC-A server. From the PC-C command prompt, ping PC-A at 192.168.1.3. The ping should succeed.

Step 2: From internal PC-C, SSH to the R2 S0/0/1 interface.

- a. From the PC-C command prompt, SSH to R2 at 10.2.2.2. Use the username Admin and the password Adminpa55 to access R2. The SSH session should succeed.
- b. While the SSH session is active, issue the command show policy-map type inspect zone-pair sessions on R3 to view established sessions.

R3# show policy-map type inspect zone-pair sessions

Step 3: From PC-C, exit the SSH session on R2 and close the command prompt window. Step 4: From internal PC-C, open a web browser to the PC-A server web page.

Enter the server IP address 192.168.1.3 in the browser URL field, and click Go. The HTTP session should succeed. While the HTTP session is active, issue the command show policy-map type inspect zone-pair sessions on R3 to view established sessions.

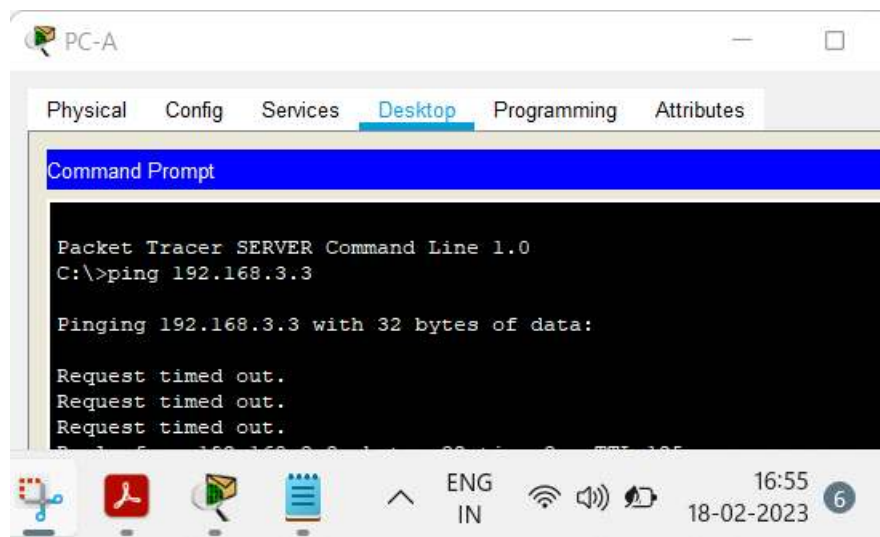
R3# show policy-map type inspect zone-pair sessions

Step 5: Close the browser on PC-C.

Part 7: Test Firewall Functionality from OUT-ZONE to IN-ZONE

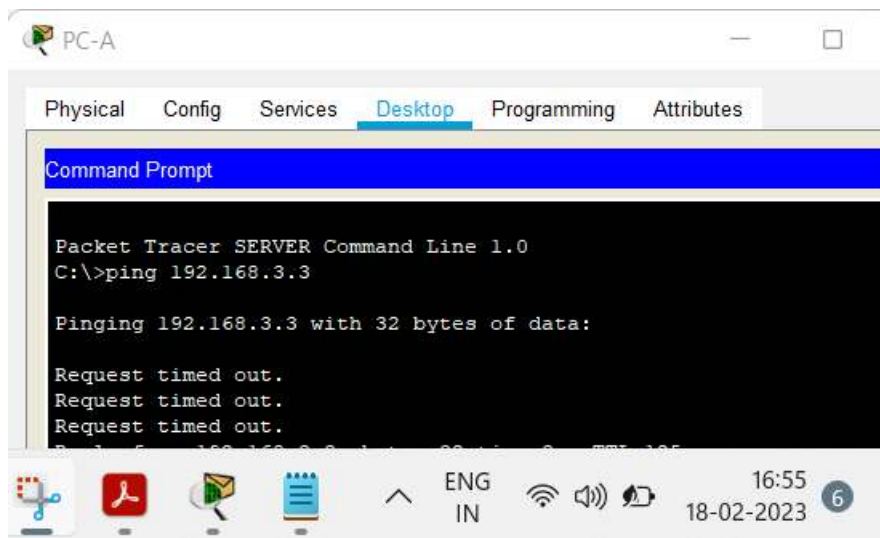
Step 1: From the PC-A server command prompt, ping PC-C.

From the PC-A command prompt, ping PC-C at 192.168.3.3. The ping should fail.



Step 2: From R2, ping PC-C.

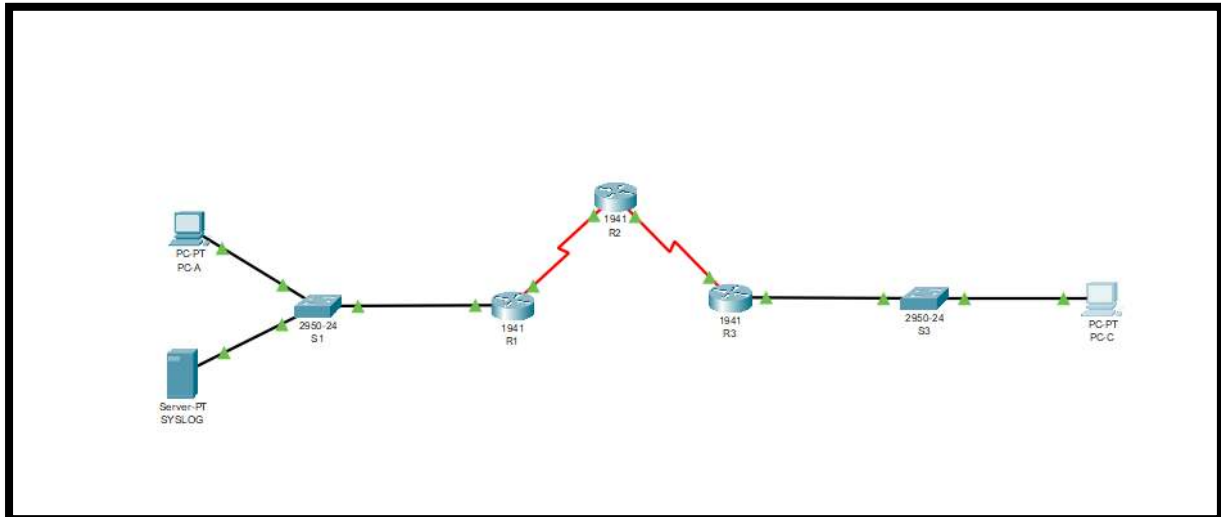
From R2, ping PC-C at 192.168.3.3. The ping should fail.



Practical 8

Q1} Configure IOS Intrusion Prevention System (IPS) Using the CLI

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/1
	S0/0/0	10.2.2.1	255.255.255.252	N/A	N/A
Syslog	NIC	192.168.1.50	255.255.255.0	192.168.1.1	S1 F0/2
PC-A	NIC	192.168.1.2	255.255.255.0	192.168.1.1	S1 F0/3
PC-C	NIC	192.168.3.2	255.255.255.0	192.168.3.1	S3 F0/2

Part 1 : Configure Router

Step 1: Configure secret on router

R(config)# enable secret enpa55

Step 2: Configure console password on router

R(config)# line console 0

R(config-line)# password conpa55

R(config-line)# login

Step 3: Configure SSH login on router

R(config)# ip domain-name ccnasecurity.com

R(config)# username admin secret adminpa55

R(config)# line vty 0 4

R(config-line)# login local

R(config)# crypto key generate rsa

How many bits in the modulus [512]: 1024

Step 4: Configure OSPF on routersExecute command on router 1

R1(config)#router ospf 1

R1(config-router)# network 192.168.1.0 0.0.0.255 area 0

R1(config-router)# network 10.1.1.0 0.0.0.3 area 0

Execute command on router 2

R2(config)#router ospf 1

R2(config-router)# network 10.1.1.0 0.0.0.3 area 0

R2(config-router)# network 10.2.2.0 0.0.0.3 area 0

Execute command on router 3

R3(config)#router ospf 1

R3(config-router)# network 10.2.2.0 0.0.0.3 area 0

R3(config-router)# network 192.168.3.0 0.0.0.255 area 0

Part 2: Enable IOS IPS

Step 1: Enable the Security Technology package

R1# show version

R1(config)# license boot module c1900 technology-package securityk9 (Type yes)

exit

R1# copy run start

R1# reload

R1# show version

```
R1#show version
Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version
15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 23-Feb-11 14:19 by pt_team
```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# license boot module c1900 technology-package securityk9
PLEASE READ THE FOLLOWING TERMS CAREFULLY. INSTALLING THE LICENSE
OR
LICENSE KEY PROVIDED FOR ANY CISCO PRODUCT FEATURE OR USING
SUCH
```

```
ACCEPT? [yes/no]: y
% use 'write' command to make license boot config take effect on next
boot

R1(config)#: %IOS_LICENSE_IMAGE_APPLICATION-6-LICENSE_LEVEL: Module
name = C1900 Next reboot level = securityk9 and License = securityk9
```



```
R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#reload
Proceed with reload? [confirm]
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by Cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC
disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test

Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x2bb1c58
Self decompressing the image :
#####
#### [OK]
Smart Init is enabled
```

```
R1#show version
Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version
15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 23-Feb-11 14:19 by pt_team
```

License Info:

License UDI:

Device#	PID	SN
*0	CISCO1941/K9	FTX1524V69S-

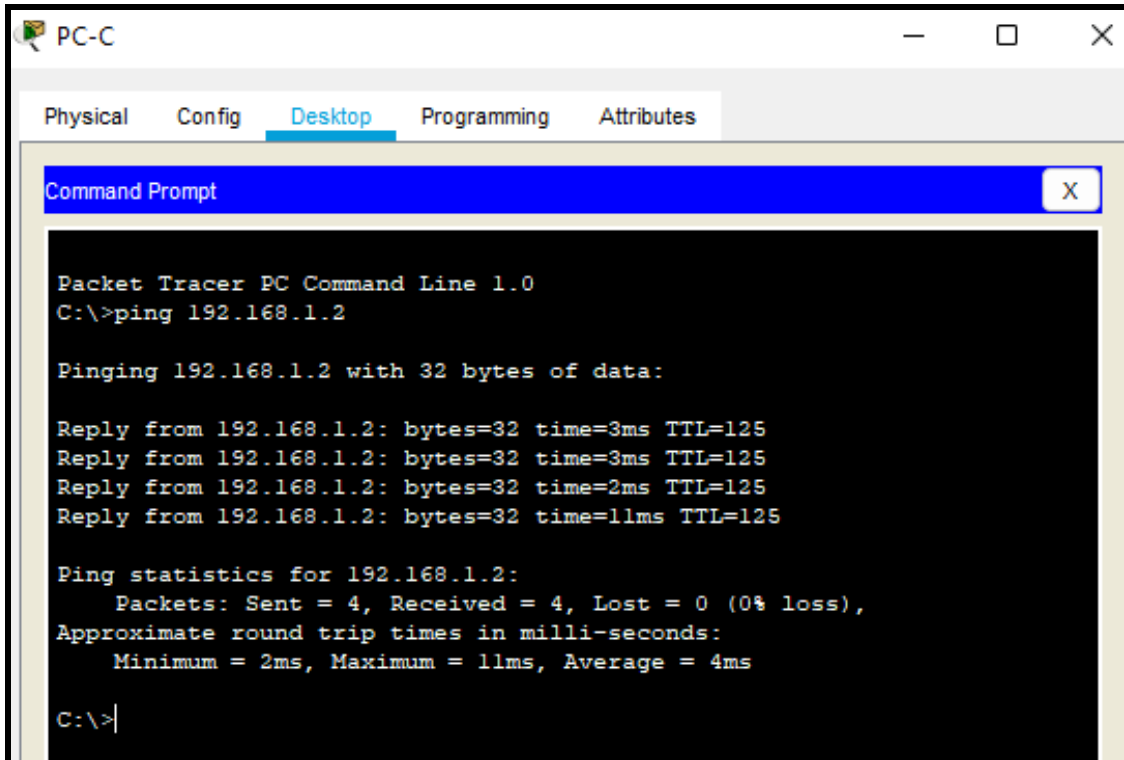
Technology Package License Information for Module:'c1900'

Technology	Technology-package Current	Technology-package Type	Technology-package Next reboot
ipbase	ipbasek9	Permanent	ipbasek9
security	securityk9	Evaluation	securityk9
data	disable	None	None

Configuration register is 0x2102

Step 2: Verify network connectivity

PCA> ping 192.168.3.2 (Successful)



The screenshot shows a Packet Tracer PC window for PC-C. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the execution of the command 'ping 192.168.1.2'. The output indicates that the ping was successful, with 4 packets sent and 4 received, resulting in 0% loss. The round trip times are listed as Minimum = 2ms, Maximum = 11ms, and Average = 4ms.

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

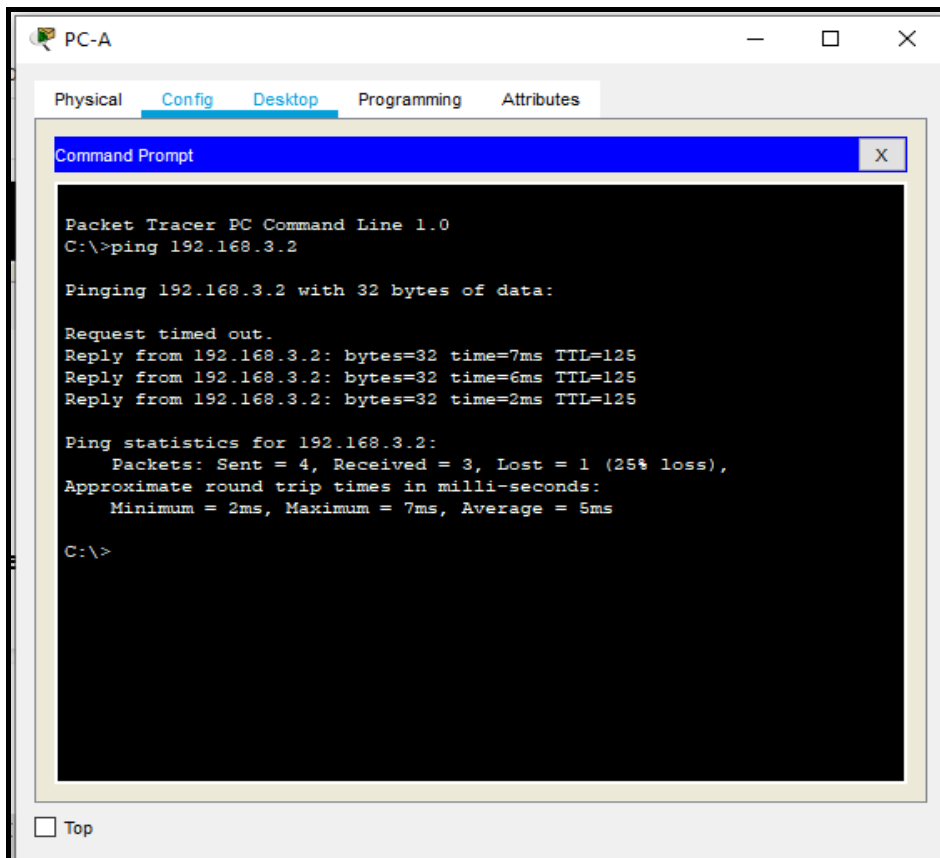
Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=3ms TTL=125
Reply from 192.168.1.2: bytes=32 time=3ms TTL=125
Reply from 192.168.1.2: bytes=32 time=2ms TTL=125
Reply from 192.168.1.2: bytes=32 time=11ms TTL=125

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 11ms, Average = 4ms

C:\>
```

PCC> ping 192.168.1.2 (Successful)



The screenshot shows a Packet Tracer PC window for PC-A. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the execution of the command 'ping 192.168.3.2'. The output indicates that the ping was successful, with 4 packets sent and 3 received, resulting in 25% loss. The round trip times are listed as Minimum = 2ms, Maximum = 7ms, and Average = 5ms.

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.2: bytes=32 time=7ms TTL=125
Reply from 192.168.3.2: bytes=32 time=6ms TTL=125
Reply from 192.168.3.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 7ms, Average = 5ms

C:\>
```

Step 3: Create an IOS IPS configuration directory in flash.**R1# mkdir ipsdir****Create directory filename [ipsdir]? <Enter>**

```
R1#mkdir ipsdir
Create directory filename [ipsdir]?
Created dir flash:ipsdir
```

Step 4: Configure the IPS signature storage location.**R1(config)# ip ips config location flash:ipsdir**

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip ips config location flash:ipsdir
```

Step 5: Create an IPS rule**R1(config)# ip ips name iosips**

```
R1(config)#ip ips name iosips
```

Step 6: Enable logging.**R1(config)# ip ips notify log****R1# clock set hr:min:sec date month year****R1(config)# service timestamps log datetime msec****R1(config)# logging host 192.168.1.50**

```
R1#clock set 09:04:20 25 January 2023
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#service timestamp log datetime msec
R1(config)#exit
R1#
*Jan 25, 09:04:46.044: SYS-5-CONFIG_I: Configured from console by
console
```

```
R1(config)#service timestamp log datetime msec
R1(config)#exit
```

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#logging host 192.168.1.50
```

Step 7: Configure IOS IPS to use the signature categories.

R1(config)# ip ips signature-category

R1(config-ips-category)# category all

R1(config-ips-category-action)# retired true

R1(config-ips-category-action)# exit

R1(config-ips-category)# category ios_ips basic

R1(config-ips-category-action)# retired false

R1(config-ips-category-action)# exit

R1(config-ips-category)# exit

Do you want to accept these changes? [confirm] <Enter>

```
R1(config)#ip ips signature-category
R1(config-ips-category)#category all
R1(config-ips-category-action)#retired true
R1(config-ips-category-action)#ex
R1(config-ips-category)#category ios_ips basic
R1(config-ips-category-action)#retired false
R1(config-ips-category-action)#exit
R1(config-ips-category)#exit
Do you want to accept these changes? [confirm]
Applying Category configuration to signatures ...
%IPS-6-ENGINE_BUILDING: atomic-ip - 288 signatures - 6 of 13 engines
%IPS-6-ENGINE_READY: atomic-ip - build time 30 ms - packets for this
engine will be scanned
```

Step 8: Apply the IPS rule to an interface.

R1(config)# int gig0/1

R1(config-if)# ip ips iosips out

```
R1(config)#ip ips signature-definition
R1(config-sigdef)#signature 2004 0
R1(config-sigdef-sig)#status
R1(config-sigdef-sig-status)#retired false
R1(config-sigdef-sig-status)#enabled true
R1(config-sigdef-sig-status)#exit
R1(config-sigdef-sig)#engine
R1(config-sigdef-sig-engine)#event-action produce-alert
R1(config-sigdef-sig-engine)#event-action deny-packet-inline
R1(config-sigdef-sig-engine)#exit
R1(config-sigdef-sig)#exit
R1(config-sigdef)#exit
Do you want to accept these changes? [confirm]
%IPS-6-ENGINE_BUILDS_STARTED:
%IPS-6-ENGINE_BUILDING: atomic-ip - 303 signatures - 3 of 13 engines
%IPS-6-ENGINE_READY: atomic-ip - build time 480 ms - packets for this
engine will be scanned
%IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 648 ms
```

Step 9: Use show commands to verify IPS.

R1# show ip ips all (Output)

```
R1#show ip ips all
IPS Signature File Configuration Status
  Configured Config Locations: flash:ipsdir
  Last signature default load time:
  Last signature delta load time:
  Last event action (SEAP) load time: -none-

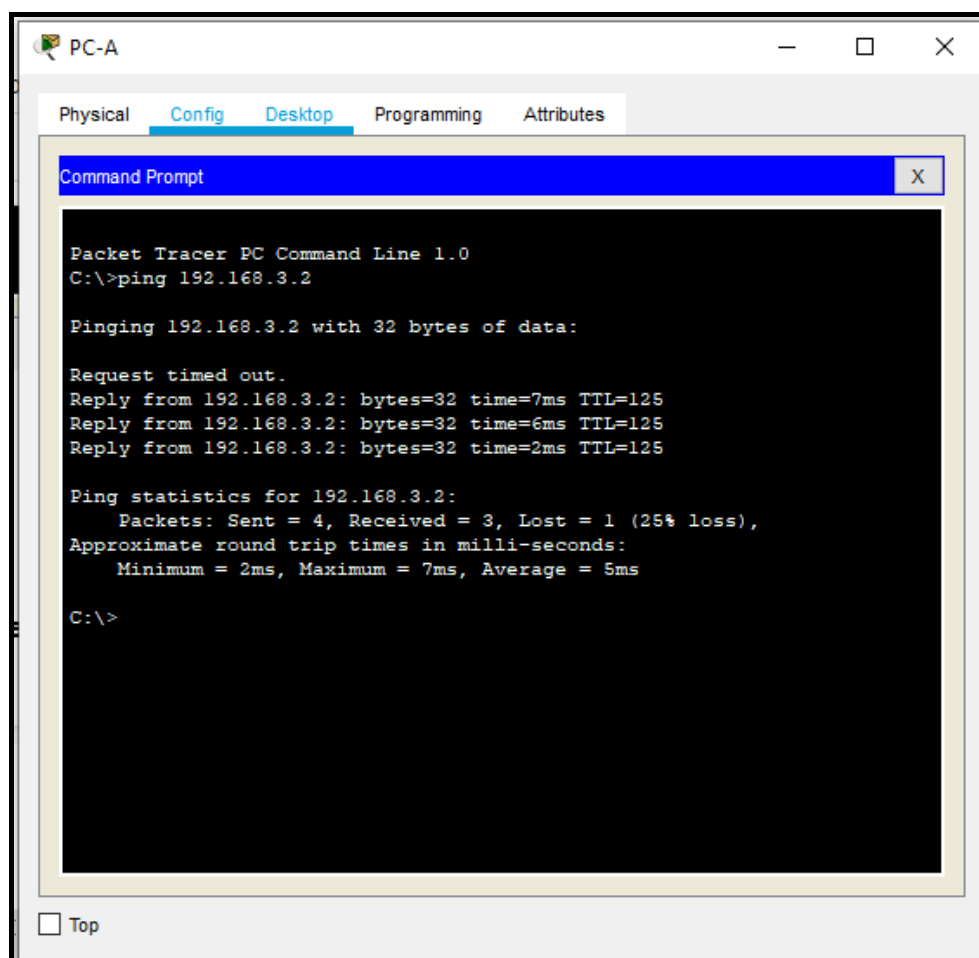
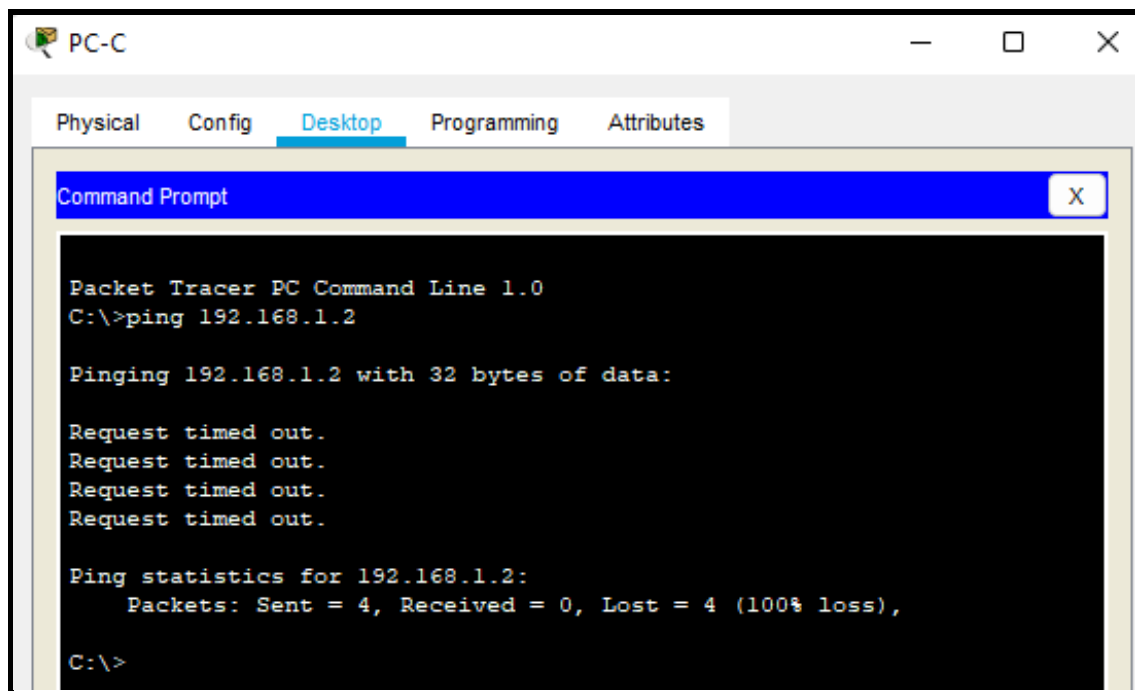
  General SEAP Config:
  Global Deny Timeout: 3600 seconds
  Global Overrides Status: Enabled
  Global Filters Status: Enabled

IPS Auto Update is not currently configured

IPS Syslog and SDEE Notification Status
  Event notification through syslog is enabled
  Event notification through SDEE is enabled

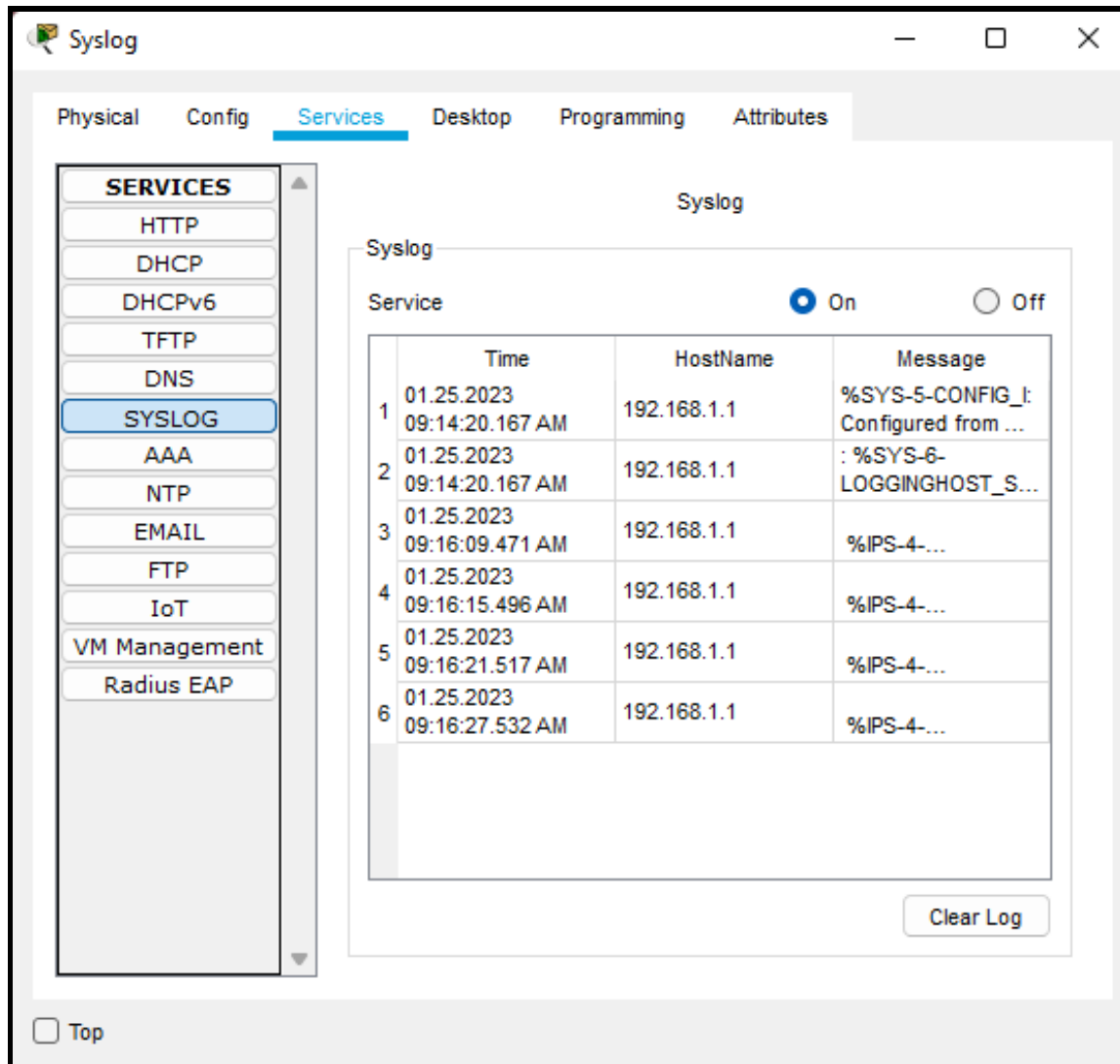
IPS Signature Status
  Total Active Signatures: 1
  Total Inactive Signatures: 0

IPS Packet Scanning and Interface Status
```



Step 10: View the syslog messages.

Click the Syslog server->Services tab-> SYSLOG (Output)



The screenshot shows the Syslog configuration window with the 'Services' tab selected. The 'SYSLOG' service is highlighted in the left-hand menu. The main area displays the Syslog service status as 'On' and a table of log messages.

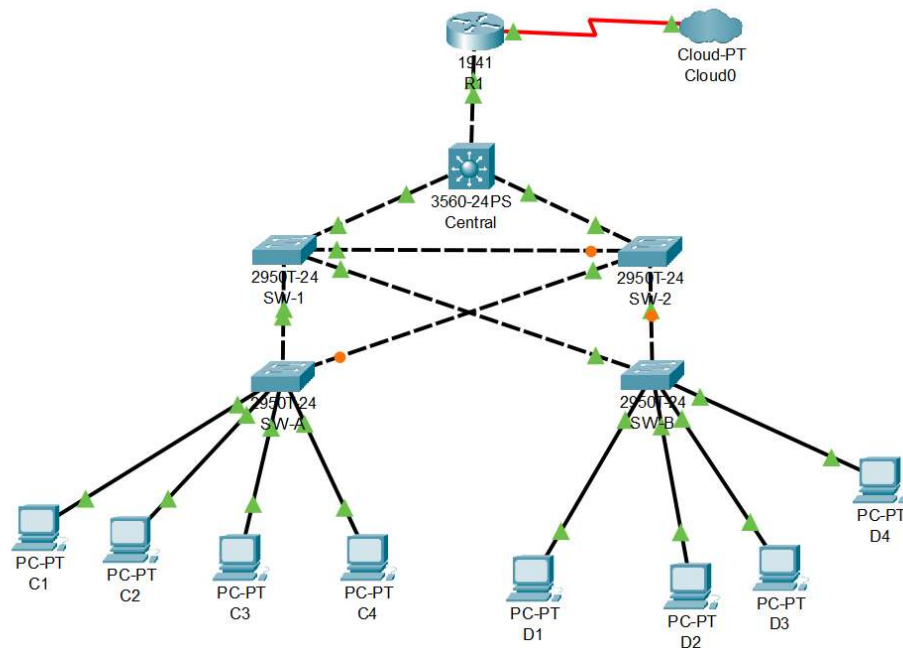
	Time	HostName	Message
1	01.25.2023 09:14:20.167 AM	192.168.1.1	%SYS-5-CONFIG_: Configured from ...
2	01.25.2023 09:14:20.167 AM	192.168.1.1	: %SYS-6- LOGGINGHOST_S...
3	01.25.2023 09:16:09.471 AM	192.168.1.1	%IPS-4-...
4	01.25.2023 09:16:15.496 AM	192.168.1.1	%IPS-4-...
5	01.25.2023 09:16:21.517 AM	192.168.1.1	%IPS-4-...
6	01.25.2023 09:16:27.532 AM	192.168.1.1	%IPS-4-...

Clear Log

Practical 9

Q 1} Layer 2 Security VLAN Security

Topology



Objectives

- Connect a new redundant link between SW-1 and SW-2.
- Enable trunking and configure security on the new trunk link between SW-1 and SW-2.
- Create a new management VLAN (VLAN 20) and attach a management PC to that VLAN.
- Implement an ACL to prevent outside users from accessing the management VLAN.

Part 1: Verify Connectivity

Step 1: Verify connectivity between C2 (VLAN 10) and C3 (VLAN 10).

Step 2: Verify connectivity between C2 (VLAN 10) and D1 (VLAN 5).

Part 2: Create a Redundant Link Between SW-1 and SW-2

Step 1: Connect SW-1 and SW-2.

Step 2: Enable trunking, including all trunk security mechanisms on the link between SW-1 and SW-2.

Part 3: Enable VLAN 20 as a Management VLAN

Step 1: Enable a management VLAN (VLAN 20) on SW-A.

- Enable VLAN 20 on SW-A.
- Create an interface VLAN 20 and assign an IP address within the 192.168.20.0/24 network.

Step 2: Enable the same management VLAN on all other switches.

- Create the management VLAN on all switches: SW-B, SW-1, SW-2, and Central.
- Create an interface VLAN 20 on all switches and assign an IP address within the 192.168.20.0/24 network.

Step 3: Connect and configure the management PC.

Step 4: On SW-A, ensure the management PC is part of VLAN 20. Interface.

Step 5: Verify connectivity of the management PC to all switches.

Part 4: Enable the Management PC to Access Router R1

Step 1: Enable a new sub interface on router R1.

- Create sub interface g0/0.3 and set encapsulation to dot1q 20 to account for VLAN 20.
- Assign an IP address within the 192.168.20.0/24 network.

Step 2: Verify connectivity between the management PC and R1.

Step 3: Enable security.

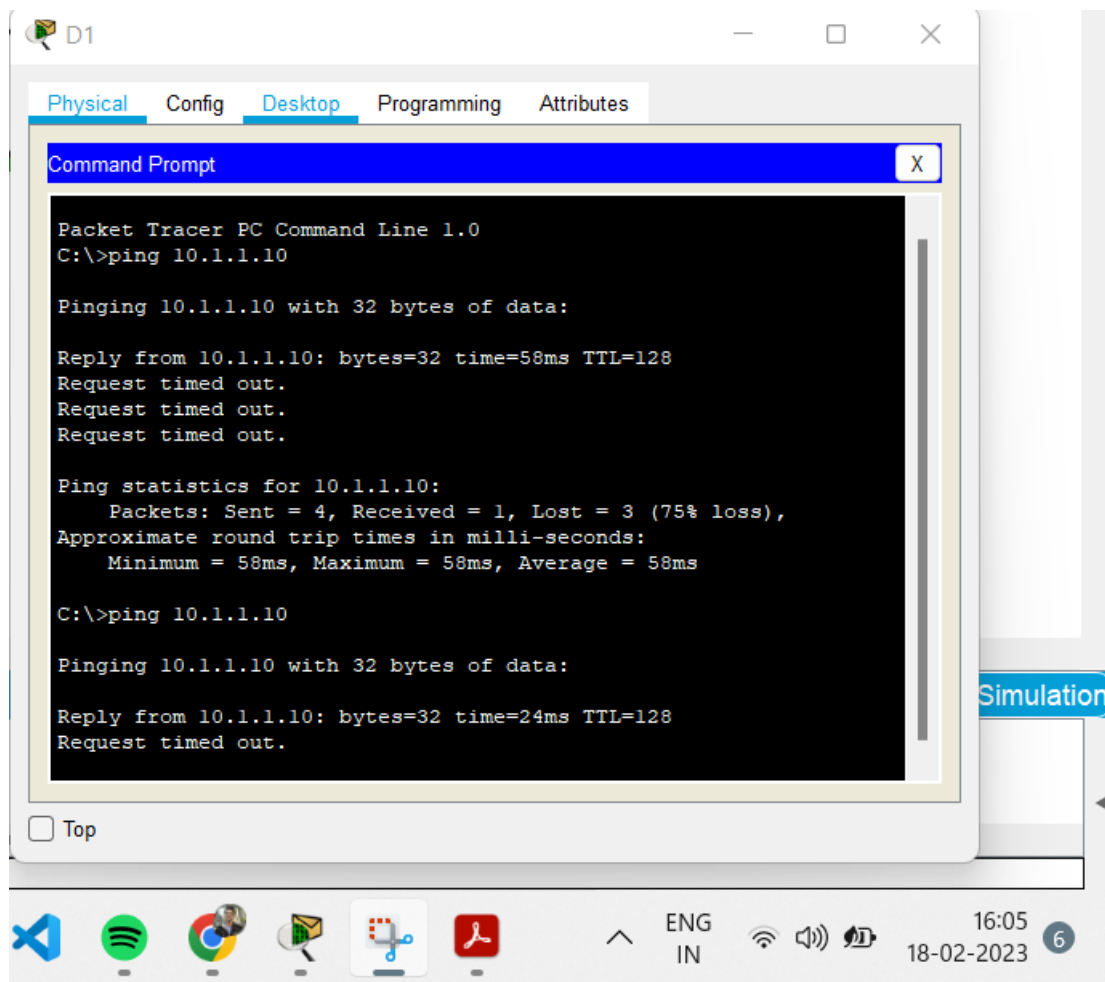
- Create an ACL that allows only the Management PC to access the router.
- Apply the ACL to the proper interface(s).

Step4: Verify Security.

- Verify only the Management PC can access the router. Use SSH to access **R1** with username **SSHadmin** and password **ciscosshpa55**.

```
PC> ssh -l SSHadmin 192.168.20.100
```

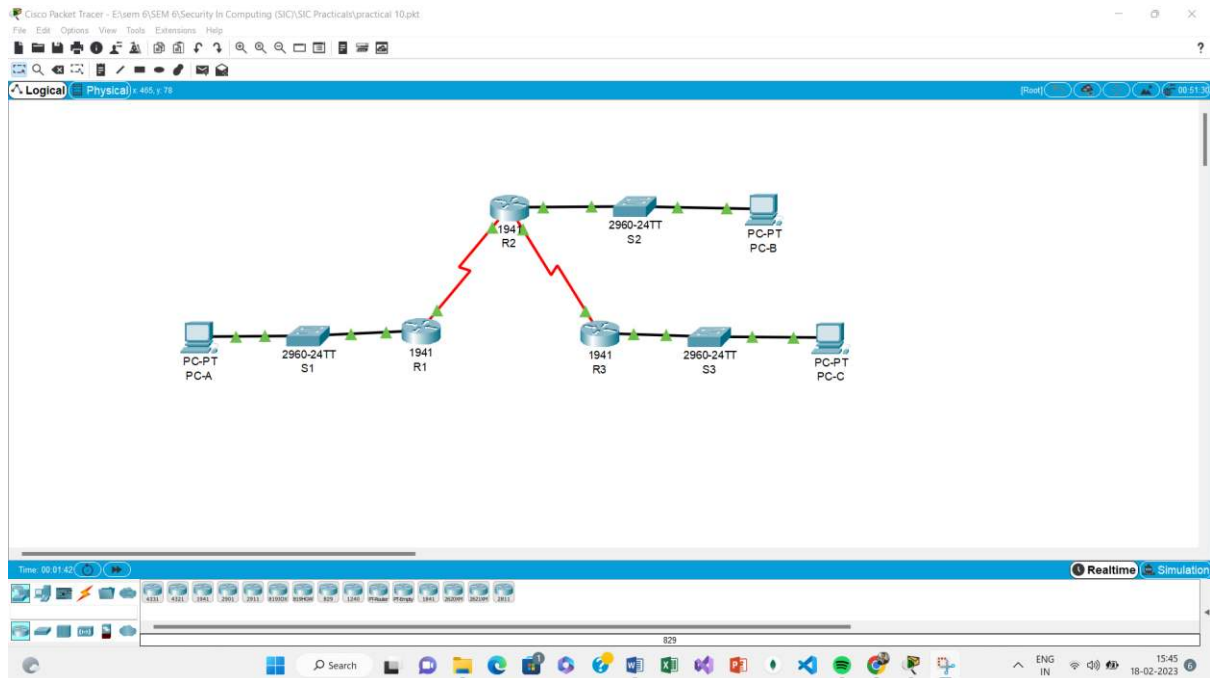
- From the management PC, ping SW-A, SW-B, and R1. Were the pings successful? Explain.
- From D1, ping the management PC. Were the pings successful? Explain.



Practical 10

Q1} Configure and Verify a Site-to-Site IPsec VPN Using

TOPOLOGY:



Addressing Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/0	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A	N/A
R2	G0/0	192.168.2.1	255.255.255.0	N/A	S2 F0/2
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.1	255.255.255.252	N/A	N/A
R3	G0/0	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.2	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/2
PC-B	NIC	192.168.2.3	255.255.255.0	192.168.2.1	S2 F0/1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

Objective

- Verify connectivity throughout the network.
- Configure R1 to support a site-to-site IPsec VPN with R3.

Part 1: Configure IPsec Parameters on R1

Step 1: Test connectivity. Ping from PC-A to PC-C.

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.3: bytes=32 time=3ms TTL=125
Reply from 192.168.3.3: bytes=32 time=2ms TTL=125
Reply from 192.168.3.3: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\>
```

Step 2: Enable the Security Technology package.

- a. On R1, issue the show version command to view the Technology Package license information.

```
R1#show version
Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version
15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 23-Feb-11 14:19 by pt_team
```

- b. If the Security Technology package has not been enabled, use the following command to enable the package.

```
R1(config)# license boot module c1900 technology-package securityk9
```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# license boot module c1900 technology-package securityk9
PLEASE READ THE FOLLOWING TERMS CAREFULLY. INSTALLING THE LICENSE
OR
LICENSE KEY PROVIDED FOR ANY CISCO PRODUCT FEATURE OR USING
SUCH
```

- c. Accept the end user license agreement.

```
ACCEPT? [yes/no]: y
% use 'write' command to make license boot config take effect on next
boot

R1(config)#: %IOS_LICENSE_IMAGE_APPLICATION-6-LICENSE_LEVEL: Module
name = C1900 Next reboot level = securityk9 and License = securityk9
```

- d. Save the running-config and reload the router to enable the security license.

```
R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#reload
Proceed with reload? [confirm]
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC
disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test

Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x2bblc58
Self decompressing the image :
#####
#### [OK]
Smart Init is enabled
```

- e. Verify that the Security Technology package has been enabled by using the show version command.

```
R1#show version
Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version
15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 23-Feb-11 14:19 by pt_team
```

License Info:

License UDI:

```
-----
Device#      PID                      SN
-----
*0           CISCO1941/K9              FTX1524V69S-
```

Technology Package License Information for Module:'c1900'

```
-----
Technology    Technology-package      Technology-package
              Current      Type                Next reboot
-----
ipbase        ipbasek9               Permanent          ipbasek9
security      securityk9             Evaluation         securityk9
data          disable                None                None
```

Configuration register is 0x2102

Step 3: Identify interesting traffic on R1.

```
R1(config)# access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0
0.0.0.25
```

```
R1(config)#access-list 110 permit ip 192.168.1.0 0.0.0.255
192.168.3.0 0.0.0.25
R1(config)#
```

Step 4: Configure the IKE Phase 1 ISAKMP policy on R1.

```
R1(config)# crypto isakmp policy 10
R1(config-isakmp)# encryption aes 256
R1(config-isakmp)# authentication pre-share
R1(config-isakmp)# group 5
R1(config-isakmp)# exit
```


R1(config)# crypto isakmp key vpnpa55 address 10.2.2.2

```
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config-isakmp)#ex
R1(config)#crypto isakmp key vpnpa55 address 10.2.2.2
R1(config)#
R1(config)#
```

Step 5: Configure the IKE Phase 2 IPsec policy on R1.

- a. Create the transform-set VPN-SET to use esp-aes and esp-sha-hmac.

R1(config)# crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac

```
R1(config)#
R1(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R1(config)#
```

- b. Create the crypto map VPN-MAP that binds all of the Phase 2 parameters together. Use sequence number 10 and identify it as an ipsec-isakmp map.

R1(config)# crypto map VPN-MAP 10 ipsec-isakmp

R1(config-crypto-map)# description VPN connection to R3

R1(config-crypto-map)# set peer 10.2.2.2

R1(config-crypto-map)# set transform-set VPN-SET

R1(config-crypto-map)# match address 110

R1(config-cryptomap)# exit

```
R1(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
R1(config-crypto-map)#description VPN connection to R3
R1(config-crypto-map)#set peer 10.2.2.2
R1(config-crypto-map)#set transform-set VPN-SET
R1(config-crypto-map)#match address 110
R1(config-crypto-map)#ex
```

Step 6: Configure the crypto map on the outgoing interface.

R1(config)# interface s0/0/0

R1(config-if)# crypto map VPN-MAP

```
R1(config)#int s0/0/0
R1(config-if)#crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON
R1(config-if)#
```

Part 2: Configure IPsec Parameters on R3

Step 1: Enable the Security Technology package

- a. On R3, issue the show version command to verify that the Security Technology package license information has been enabled.

R3# show version

```
R3#show version
Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version
15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 23-Feb-11 14:19 by pt_team
```

License UDI:

```
-----
Device#      PID                      SN
-----
*0           CISCO1941/K9                  FTX15249T24-
```

Technology Package License Information for Module:'c1900'

```
-----
Technology    Technology-package      Technology-package
              Current       Type                    Next reboot
-----
ipbase        ipbasek9                Permanent            ipbasek9
security      None                    None                  None
data          None                    None                  None
```

Configuration register is 0x2102

- b. If the Security Technology package has not been enabled, enable the package and reload R3.

```

R3#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R3#reload
Proceed with reload? [confirm]
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC
disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test

Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x2bb1c58
Self decompressing the image :
#####

```

License Info:

License UDI:

Device#	PID	SN
*0	CISCO1941/K9	FTX15249T24-

Technology Package License Information for Module:'c1900'

Technology	Technology-package Current	Technology-package Type	Technology-package Next reboot
ipbase	ipbasek9	Permanent	ipbasek9
security	securityk9	Evaluation	securityk9
data	disable	None	None

Configuration register is 0x2102

Step 2: Configure router R3 to support a site-to-site VPN with R1.

```
R3(config)# access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0  
0.0.0.255
```

```
R3(config)#access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0  
0.0.0.255  
R3(config)#
```

Step 3: Configure the IKE Phase 1 ISAKMP properties on R3.

```
R3(config)# crypto isakmp policy 10  
R3(config-isakmp)# encryption aes 256  
R3(config-isakmp)# authentication pre-share  
R3(config-isakmp)# group 5  
R3(config-isakmp)# exit  
R3(config)# crypto isakmp key vpnpa55 address 10.1.1.2
```

```
R3(config)#crypto isakmp policy 10  
R3(config-isakmp)#encryption aes 256  
R3(config-isakmp)#authentication pre-share  
R3(config-isakmp)#group 5  
R3(config-isakmp)#ex  
R3(config)#crypto isakmp key vpnpa55 address 10.1.1.2
```

Step 4: Configure the IKE Phase 2 IPsec policy on R3.

- a. Create the transform-set VPN-SET to use esp-aes and esp-sha-hmac.

```
R3(config)# crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
```

```
R3(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R3(config)#
```

- b. Create the crypto map VPN-MAP that binds all of the Phase 2 parameters together. Use sequence number 10 and identify it as an ipsec-isakmp map.

```
R3(config)# crypto map VPN-MAP 10 ipsec-isakmp
```

```
R3(config-crypto-map)# description VPN connection to R1
```

```
R3(config-crypto-map)# set peer 10.1.1.2
```

```
R3(config-crypto-map)# set transform-set VPN-SET
```

```
R3(config-crypto-map)# match address 110
```

```
R3(config-crypto-map)# exit
```

```
R3(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R3(config-crypto-map)#description VPN connection to R1
R3(config-crypto-map)#set peer 10.1.1.2
R3(config-crypto-map)#set transform-set VPN-SET
R3(config-crypto-map)#match address 110
R3(config-crypto-map)#ex
```

Step 5: Configure the crypto map on the outgoing interface.

```
R3(config)# interface s0/0/1
```

```
R3(config-if)# crypto map VPN-MAP
```

```
R3(config)#int s0/0/1
R3(config-if)#crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
```

Part 3: Verify the IPsec VPN

Step 1: Verify the tunnel prior to interesting traffic. I

R1# show crypto ipsec sa

```
R1#show crypto ipsec sa

interface: Serial0/0/0
  Crypto map tag: VPN-MAP, local addr 10.1.1.2

protected vrf: (none)
local  ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.230/0/0)
current_peer 10.2.2.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
#pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0

  local crypto endpt.: 10.1.1.2, remote crypto endpt.:10.2.2.2
  path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
  current outbound spi: 0x0(0)

inbound esp sas:
```

Step 2: Create interesting traffic.

Ping PC-C from PC-A.


```
C:\>ping 192.168.3.3
```

```
Pinging 192.168.3.3 with 32 bytes of data:
```

```
Reply from 192.168.3.3: bytes=32 time=3ms TTL=126
```

```
Reply from 192.168.3.3: bytes=32 time=2ms TTL=126
```

```
Reply from 192.168.3.3: bytes=32 time=3ms TTL=126
```

```
Reply from 192.168.3.3: bytes=32 time=3ms TTL=126
```

```
Ping statistics for 192.168.3.3:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:
```

```
    Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

Step 3: Verify the tunnel after interesting traffic.

R1# show crypto ipsec sa

```
R1#show crypto ipsec sa
```

```
interface: Serial0/0/0
```

```
    Crypto map tag: VPN-MAP, local addr 10.1.1.2
```

```
    protected vrf: (none)
```

```
    local ident (addr/mask/prot/port):
```

```
(192.168.1.0/255.255.255.0/0/0)
```

```
    remote ident (addr/mask/prot/port):
```

```
(192.168.3.0/255.255.255.0/0/0)
```

```
    current_peer 10.2.2.2 port 500
```

```
    PERMIT, flags={origin_is_acl,}
```

```
    #pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 0
```

```
    #pkts decaps: 6, #pkts decrypt: 6, #pkts verify: 0
```

```
    #pkts compressed: 0, #pkts decompressed: 0
```

```
    #pkts not compressed: 0, #pkts compr. failed: 0
```

```
    #pkts not decompressed: 0, #pkts decompress failed: 0
```

```
    #send errors 1, #recv errors 0
```

```
    local crypto endpt.: 10.1.1.2, remote crypto endpt.:10.2.2.2
```

```
    path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
```

```
    current outbound spi: 0x5A2F065B(1513031259)
```

```
    inbound esp sas:
```

```
        spi: 0x2FAD4AE7(799886055)
```

```
--More--
```